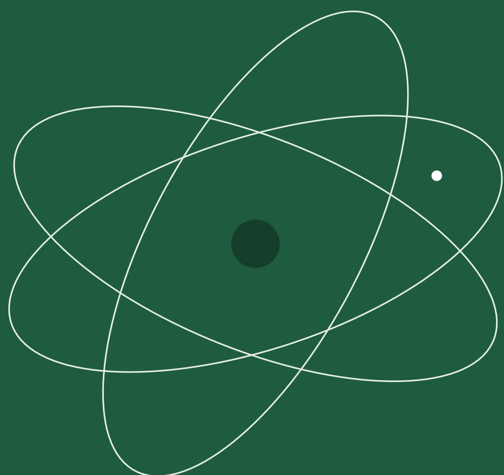
CardSat

The Complete Operating Manual



Firmware v0.9.64

Paul Stoetzer · N8HM

A complete guide to operating **CardSat**, the amateur-radio satellite tracker and multi-radio CAT Doppler controller for the M5Stack Cardputer ADV (Icom, Yaesu, Kenwood).

> **Status.** CardSat runs on the Cardputer ADV, with its core features (display, keyboard, GPS, pass prediction, tracking, alarms, logging, and the on-device data caches) confirmed on hardware. Two CAT paths are hardware-confirmed on an **IC-821**: **single-pin CI-V** (full bidirectional exchange over one wire) and **CAT over a USB↔serial adapter** (FTDI; engage, disengage, re-engage, and Doppler tracking over many cycles). The **Icom LAN (RS-BA1)** transport is confirmed able to control a real Icom (an IC-705), but has not yet been tested against the **IC-9700** it is intended for. The remaining CAT paths — separate-pin CI-V, the Yaesu and Kenwood encoders — and **all** of the rotator backends (GS-232, Easycomm I/II/III, and SPID over the I²C→UART bridge, the Grove port, or a USB adapter; the rotctld TCP client; PstRotator UDP; and the direct-Yaesu I²C interface) are host-tested but have not yet driven that specific hardware, and running a radio *and* a rotator on USB at once is untested. Keep the serial monitor open and verify on the air before trusting CAT for a contact.

Contents

01

Overview

CardSat turns a pocket-sized Cardputer ADV into a standalone satellite station controller. It:

- downloads GP (General Perturbations / OMM) orbital elements from AMSAT and transponder frequencies from the SatNOGS database, and caches both to flash for **fully offline** use;
- predicts passes with **SGP4** and shows live az/el/range, a polar sky plot, and per-pass elevation curves;
- drives an **Icom, Yaesu, or Kenwood** transceiver over **CAT** with continuous **Doppler correction**, keeping a constant frequency *at the satellite* (the AMSAT "One True Rule") so your signal stays put in a linear transponder's passband;
- lets you tune the passband from the device **or from the radio's own knob**;
- maintains a **favorites** list, a unified **Next Passes** schedule, an **AOS alarm**, a **deep-sleep-until-next-pass** power saver, and **sun/eclipse** status;
- helps you **observe** as well as work the birds — it flags **visually observable passes** (sunlit satellite, dark sky), predicts **Sun/Moon transits** (when a bird crosses the solar or lunar disc from your location), shows **decay/reentry** watch flags on declining orbits, and keeps a **per-satellite operating note** for each bird;
- points an **az/el antenna rotator** — a wired **GS-232**, **Easycomm**, or **SPID** controller, a **Yaesu rotator driven directly** (no GS-232 box, via an I²C ADC + output expander), or a networked **Hamlib** `rotctld` / **PstRotator** server;
- **logs QSOs** on the device and **exports ADIF** for LoTW / eQSL or your main logger.

Everything runs on the device. WiFi is needed only to refresh GP/transponders and to set the clock.

02

What you need

- **M5Stack Cardputer ADV** (ESP32-S3, 8 MB flash, 240×135 screen, 56-key keyboard, speaker, microSD, Grove port, 2×7 header).
- **A supported transceiver** in one of three CAT families:
  - **Icom CI-V** — IC-820, IC-821, IC-910, IC-970, IC-9100, IC-9700;
  - **Yaesu** — FT-847, FT-736R;
  - **Kenwood** — TS-790, TS-2000;
  - **None** — no radio. Select this to run CardSat as a pure tracker / rotator controller; all CAT features become inactive and the radio can't be turned on from the Track screen.
- **A CAT interface suited to that radio.** The ESP32-S3 pins are **not** 5 V tolerant — never connect CAT directly. **Icom:** a 3.3 V-safe CI-V interface (one-transistor circuit or a ready-made board). **Kenwood:** a **MAX3232** RS-232 level shifter on the DB-9 COM port. **Yaesu:** a serial CAT interface (verify TTL vs RS-232 per the CAT manual).
- (*Optional*) a **GPS**: a NMEA receiver on the Grove port, or an **M5Stack Cap LoRa** (868 or 1262) module with onboard GNSS.
- (*Optional*) **WiFi** to fetch GP/transponders and sync the clock.

03

Connecting your radio

All three CAT families share **UART1**, **RX = G1 / TX = G2** by default, but the **interface hardware differs** (see §2 and §16).

1. Wire the appropriate CAT interface between the radio's control jack (Icom **REMOTE**, Kenwood **COM** DB-9, Yaesu **CAT**) and **G1/G2**. **Never connect the radio directly to the GPIO pins** — the ESP32-S3 is not 5 V tolerant.
2. Note the radio's CAT settings: **baud** (all families) and, for **Icom**, the **CI-V address** (fixed on older rigs).
3. In CardSat, open **Settings**, choose your **radio model** (this auto-fills the defaults), then adjust **CAT baud** — and, for Icom, the **CI-V address** — to match.

CardSat drives two independent VFOs. By default **downlink = Sub/RX, uplink = Main/TX**, but the **VFO Type** setting can swap the roles (*Main Dn/Sub Up*). On **Icom** it manages MAIN/SUB and, by default, leaves the rig's satellite mode **off**; the **Sat mode** setting commands it on/off when you engage radio control (a no-op on rigs without one, such as the IC-820/821/970). On **Yaesu and Kenwood** the rig's own satellite / full-duplex mode and the band pair are set up **by you on the radio** — CAT can't switch bands on those rigs — and CardSat Doppler-tunes within them. (See §16.)

Icom CI-V: separate-pin or single-wire

Icom's CI-V bus is electrically a **single half-duplex wire**, so CardSat offers two ways to connect it, chosen with **CI-V wiring** in *Settings → Radio / CAT*:

- **Separate TX/RX (G2/G1)** — the default and most reliable path: G2 carries transmit, G1 carries receive, through your interface to the radio's single CI-V line. This is the recommended option.
- **Single-pin (1-pin G2** or **1-pin G1)** — the entire CI-V bus runs on **one shared GPIO**, matching CI-V's native single-wire wiring. This is **confirmed working on an IC-821** (full bidirectional exchange — frequency reads and ACKs — over a single wire), but because both directions share one pin it needs a proper **open-drain, level-correct** interface between the 3.3 V GPIO and the radio's CI-V line; get the 5 V/3.3 V interfacing right before using it. See **docs/interfaces/CIV_SINGLE_PIN.md** for the interface detail and cautions. If in doubt, use separate TX/RX.

This choice applies to **wired Icom CI-V only**; Yaesu and Kenwood use their own serial wiring, and Icom-over-LAN (below) needs no CI-V wire at all.

Icom over the network (no CI-V wiring)

The **IC-9700** can be controlled over **WiFi/Ethernet** instead of the wired CI-V bus — CardSat speaks the same RS-BA1 UDP protocol as Icom's own remote software. No level shifter, no UART: the radio's CI-V/G1/G2 pins stay free.

> **Scope.** Icom LAN in CardSat is **intended for the IC-9700** — the one network-capable Icom in CardSat's radio list that is a full-duplex satellite rig. The LAN

transport itself is **confirmed working**: CardSat has successfully controlled an **IC-705** over the network (with the correct CI-V address set), which proves the RS-BA1 connect/auth/CI-V path drives a real Icom radio. The IC-705 isn't usable for live satellite work, though — it's a single-receiver radio, so its two VFOs just swap back and forth rather than offering the simultaneous MAIN/SUB downlink+uplink the IC-9700 provides. Other networked Icoms (IC-7610, IC-785x) speak the same protocol but likewise lack the satellite architecture. **The IC-9700 itself has not yet been tested** over LAN — the path should work given the IC-705 result, but confirm it before relying on it. Select **Icom LAN** with the **IC-9700** as the radio model.

On the radio (IC-9700: **MENU > SET > Network**): set **Network Control = ON**, set a **Network User1** id + password, leave the **Control** port at **50001** (Serial = 50002, Audio = 50003 follow it), and turn **CI-V Transceive ON** so the radio reports changes. Give the radio a stable IP (DHCP reservation or static).

In CardSat **Settings**: set **CAT type → Icom LAN**, **LAN host** to the radio's IP, **LAN port** to 50001 (unless you changed it), and **LAN user / LAN pass** to the Network User1 credentials. CardSat connects in the background; once the link is up the radio behaves exactly like a wired IC-9700 (MAIN/SUB, Doppler, sat mode, CTCSS all work the same). Only **CAT** is carried — CardSat does not open the audio stream. Keep the credentials on a trusted LAN. (See §16.)

Driving a radio through Hamlib (rigctl over the network)

As a third option, CardSat can control **any radio Hamlib supports** by talking to a **`rigctld`** server over the network instead of wiring the radio to CardSat. Set **CAT type → rigctl (net)** and point **LAN host / LAN port** at the machine running `rigctld` — set the port to the one your `rigctld` listens on (Hamlib's conventional default is **4532**). CardSat is the client — the PC's `rigctld` holds the actual serial/USB connection to the rig and CardSat sends it set/read commands, mapping VFO-A to the downlink and VFO-B to the uplink. It drives the two legs the way Gpredict does — **VFO mode**: it selects the VFO (`set_vfo VFOA`/`VFOB`) and then sends plain `set_freq`/`set_mode` on it, rather than the older `set_split_freq` convention that makes Hamlib retune the wrong VFO on Icom sat rigs. On connect it probes `chk_vfo`, so a `rigctld` started with `--vfo` (which requires the VFO inline on every command) is handled automatically. This works with mainstream Hamlib backends and with CardSat's own built-in `rigctld` server. This is handy when the radio is already cabled to a shack computer, or for a rig CardSat has no native backend for. It is **host-tested only** — verify against your `rigctld` before a pass. If your rig is an Icom in sat mode and you see the active VFO flicker while tracking, start `rigctld` with `-x 1` (the Hamlib `--uplink` switch) to suppress uplink read-back, which reduces VFO switching.

Two half-duplex radios on one pass — the CardSatDualRig companion

A full-duplex rig (IC-9700, FT-847, TS-2000) transmits and receives at once, so CardSat drives it directly. **Half-duplex and receive-only** radios (IC-705, FT-817, IC-R8600, TH-D74, …) cannot, so a proper linear-transponder station needs *two* of them — one on the downlink, one on the uplink. The **CardSatDualRig** companion firmware (in `companion/CardSatDualRig/`, for the M5StickS3) makes that work: it hosts the two radios on its own USB port, speaks each one's native CAT, and presents CardSat a single `rigctld` server. CardSat steers two VFOs (VFOA = downlink, VFOB = uplink) and the Stick fans them out to the two radios. PTT is manual — you key your transmit radio by hand. Point CardSat at the Stick with either **rigctl** (**net**) over Wi-Fi or **rigctl** (**Grove**) below.

Driving the companion (or any rigctld) over a Grove cable — no Wi-Fi

Set **CAT type → rigctl** (**Grove**) to run the exact same VFO-mode `rigctld` protocol over the Cardputer's **Grove port** (**G1/G2**) instead of TCP. This is the no-network field mode: a single Grove cable from the Cardputer to the Stick carries all radio control. Wiring is Cardputer Grove **TX → Stick RX**, **RX ← Stick TX**, and GND↔GND; both ends are 3.3 V, so no level shifter. The **Grove baud** (the row that reads "LAN port" for the network options) must match the Stick's Grove baud — default **115200**. Because rigctl (Grove) claims the Grove UART, it follows the same rule as wired CI-V: a **Grove rotator** must move to USB/LAN or the I²C bridge, and it can't share the port with a **Grove GPS**. The link is bidirectional, so CardSat can also configure the Stick over it (see the companion's `\csdr_*` escape) — no phone or portal needed.

Configuring the companion from CardSat — Dual-Rig setup screen

Once CAT type is **rigctl** (**net**) or **rigctl** (**Grove**) and CardSat can reach the Stick, open **Settings → Radio → Dual-Rig setup** (**Stick**) to set the companion up from the Cardputer — no phone, portal, or second screen. On entry CardSat queries the Stick and shows two things: the two legs (**DN** (**RX**) = downlink, **UP** (**TX**) = uplink), each with its radio model, bound USB device, and CI-V address; and the **live USB enumeration** — every serial adapter the Stick currently sees, with product name and VID:PID. That enumeration is the whole point: with two identical adapters plugged in, the only reliable way to say *which* radio is the downlink is to pick it from what the Stick actually enumerated, so you bind by the device's own USB serial rather than by a guessable port order. Move the cursor with `;`/`.`, change a field with `,`/`/`, and press a digit **1–8** to bind that enumerated device to the selected leg. `q` re-queries the Stick, `s` saves the whole configuration back to it (persisted to the Stick's flash), and `` ` `` returns. The model list is fetched from the Stick, so it always matches whatever radios that companion build supports.

The **Dual-Rig setup** (**Stick**) row in the Radio settings list carries a small **link dot** on its right edge — green when the Stick answers, red when it doesn't, gray when the check hasn't run or CAT isn't a rigctl type — plus a "linked" / "no link" word. It updates while the row is highlighted, so during bring-up you can see at a glance whether the Grove (or Wi-Fi) link is alive before opening the screen; a green dot means the cable, baud, and `\csdr_*` handshake are all working.

CAT over a USB-to-serial adapter (any radio, no level shifter)

Another option: plug a **USB↔serial adapter** into the Cardputer's **USB-C port** and set **CAT type → USB serial**. This swaps only the transport — the CI-V, Yaesu, and Kenwood protocol encoders are unchanged — so it works for **every radio in the table**, and for the pre-USB rigs (IC-821, TS-790, FT-847…) it replaces the MAX3232/CI-V interface with an off-the-shelf adapter. **Bench-proven on an IC-821 with an FTDI adapter** (engage, disengage, re-engage, and Doppler tracking over many cycles).

Which adapters work: any **CDC-ACM** device, plus the vendor bridges EspUsbHost knows by VID:PID — **FTDI** (FT232R/2232/4232/232H/230X), **CP210x**, **CH34x**, and **PL2303**. A clone with an unlisted product ID enumerates but is not recognized; the status line distinguishes `No`

`USB device detected` (nothing there) from `Not a known serial adapter: <n> <vid:pid>` (there, but unlisted).

Two things to know before engaging:

1. **The serial console goes away while USB is engaged.** The ESP32-S3 has one USB PHY; engaging USB CAT (or a USB rotator) claims it, and the console drops for as long as the USB device is engaged. Since v0.9.64 the console is reinitialized when you disengage (the host releases the PHY), so turning USB CAT off should bring a serial terminal back — but while it's engaged there's no console. Diagnostics move to disk regardless: `/CardSat/Logs/usb.log` records every engage/bind/failure automatically, and **Settings → Station / logging →** **`Console to file`** can mirror the whole console to `console.log` (see §16). If you want CAT *and* a live console at the same time, run the radio on Grove or LAN instead.
2. **Build availability.** USB CAT is **on by default** (since 0.9.59) — in the prebuilt release binaries and in source builds alike. Building from source therefore requires the **EspUsbHost** library (**v2.4.1** is what the current release is built against and is recommended; v2.3.1+ also compiles clean against arduino-esp32 3.2.1); see **docs/BUILD_AND_FLASH.md**, which also covers compiling it out (`CARDSAT_HAS_USBCAT 0` — no library needed then). A build without it simply doesn't offer *USB serial* in the CAT type row.

Settings: **Scan USB adapters** (under *Radio / CAT*) enumerates what's plugged in — a deliberate keypress, because it closes the console. Every device string starts with **`#N`**, the USB device address (0.9.59): two identical adapters — the classic pair of Prolifics — are otherwise byte-for-byte the same, and it's the distinguishing VID:PID *tail* that truncates off a narrow row, so the one token that tells them apart now leads. It's also exactly the address explicit binding stores, so what you read is what the firmware binds. **Radio USB** picks the adapter when more than one is present. Running a radio **and** a rotator on USB at the same time is supported but experimental; see §17 → Two USB adapters.

04

Connecting a GPS (optional)

GPS runs on **UART2**, independent of CAT. On the **Location** screen press `s` to cycle the source:

Source	Pins	Baud	Notes
Grove 9600	G1/G2	9600	**!** same pins as CAT — don't use both at once
Grove 115200	G1/G2	115200	**!** same pins as CAT — don't use both at once
Cap LoRa868	G15/G13	115200	M5Stack Cap LoRa (868) onboard AT6668 GNSS
Cap LoRa1262	G15/G13	115200	M5Stack Cap LoRa (1262) onboard AT6668 GNSS

Both Cap LoRa modules use identical GPS settings (115200 8N1 on G15/G13). For a Grove-port receiver, pick the baud that matches it.

Press `p` on the Location screen to enable/disable GPS use. With a fix, your latitude, longitude, altitude, and grid update automatically, and the clock is set from GPS time. The Location screen also refreshes on its own the moment a fix is gained or lost or the satellite count changes.

> If you use the **Grove** GPS option you cannot use CAT on G1/G2 at the same time. The two **Cap LoRa** options use G15/G13 and coexist with CAT.

05

Installing the firmware

There are two ways to get CardSat onto the device: flash a **prebuilt binary** (no toolchain needed — easiest), or **build from source** with the Arduino IDE or PlatformIO. Two binaries are published with each release:

File	Use it with	Notes
`CardSat-app.bin`	**Launcher** (bmorcelli/Launcher)	App-only image; Launcher writes the partition table and bootloader, so this file is **only** usable through Launcher — it cannot be flashed on its own.
`CardSat-merged.bin`	**M5Burner**, or a **direct flash** (esptool / web flasher)	Complete standalone image (bootloader + partition table + app + empty filesystem), written at offset `0x0`.

Both prebuilt binaries are built **with USB CAT included** — *USB serial* appears in the CAT type row out of the box. (Source builds have it on by default too; see below.)

Install with Launcher (`CardSat-app.bin`)

Launcher by bmorcelli is a firmware launcher for the Cardputer (and many other ESP32 boards) that installs and runs binaries from a microSD card, over-the-air, or through its WebUI — and builds the right partition layout for each app automatically. It must already be installed on the device (see its repository for how to flash Launcher itself).

1. Copy **`CardSat-app.bin`** to the Launcher binaries folder on the device's microSD card (or push it via Launcher's WebUI / OTA favorites).
2. On the device, start **Launcher**, browse to **CardSat**, and install/run it. Launcher writes the appropriate partition scheme and the app for you.
3. `CardSat-app.bin` is an app-only image with no standalone bootloader or partition table, so it works **only** through Launcher. To flash without Launcher, use the merged image below.

Install with M5Burner, or direct flash (`CardSat-merged.bin`)

`CardSat-merged.bin` is a **complete image** — bootloader, partition table, application, and the empty LittleFS filesystem combined, written at offset `0x0`. Use it with **M5Burner** (add it as a custom firmware and burn), for a fresh device, to recover one in an unknown state, or whenever you are not using Launcher.

To flash it directly with **esptool** (replace the port with yours):

```
esptool.py --chip esp32s3 --port /dev/ttyACM0 --baud 921600 \
  write_flash 0x0 CardSat-merged.bin
```

Or with the web **ESP Tool** (https://espressif.github.io/esptool-js/): connect the Cardputer over USB, set chip to **ESP32-S3**, add `CardSat-merged.bin` at address **`0x0`**, and flash.

Hold the device's reset/boot as your tool instructs if it does not enter download mode automatically. After flashing, the device reboots into CardSat. (The merged image starts with an **empty** filesystem; run **Update** once on first boot to download GP elements — see §7.)

Build from source — Arduino IDE (single file)

1. Install libraries: **M5Cardputer**, **ArduinoJson** (v7), **TinyGPSPlus**, **RadioLib** (by Jan Gromes), **ESP_SSLClient** (by Mobizt), and **EspUsbHost** (by tanakamasayuki) from the Library Manager, and the Hopperpop **Sgp4** library via *Sketch → Include Library → Add .ZIP Library* (from https://github.com/Hopperpop/Sgp4-Library). **ESP_SSLClient is required** — all HTTPS runs on its BearSSL stack and the build will not compile without it. **EspUsbHost is required by the default build** (USB CAT is on by default) — install **v2.3.1 or later**, which compiles clean against arduino-esp32 3.2.1 (only stale ≤2.3.0 copies need the one-line patch kept in **docs/BUILD_AND_FLASH.md**, which also covers building without USB CAT instead).
2. Open `CardSat.ino`. Under **Tools**, set:
   - **Board:** ESP32S3 Dev Module (or M5StampS3)
   - **Flash Size:** 8MB
   - **Partition Scheme: Huge APP (3MB No OTA/1MB SPIFFS)** — *required*
   - **PSRAM:** Disabled · **USB CDC On Boot:** Enabled
3. Upload. Open the Serial Monitor at **115200** baud for diagnostics.

USB CAT is part of the default build. To build **without** it — dropping the EspUsbHost dependency and the *USB serial* CAT type — set `CARDSAT_HAS_USBCAT` to `0` in `CardSat.ino`; **docs/BUILD_AND_FLASH.md** has the detail, plus a linker note specific to the single-file `.ino`.

Build from source — PlatformIO

`pio run` to build, `pio run -t upload` to flash, `pio device monitor` to watch the log. The `cardputer_adv` env pins the libraries and the 8 MB partition layout.

Upgrading an existing install

Updating to a newer CardSat version uses the same tools as a first install. The one thing to understand is **what each method does to your saved data** — your station settings, radio/rotator calibration, per-satellite operating notes and CTCSS overrides, favorites, and the cached GP/AMSAT data.

Where your data lives. CardSat **prefers a microSD card** for all storage. When a FAT32 card is inserted, everything is saved to the card (under `/CardSat`), and the device's internal flash (LittleFS) is used only as a fallback when no card is present. Flashing firmware only writes the **internal flash** and never touches the SD card — so **if you keep an SD card in the device, your configuration survives any upgrade, including a full merged flash.** A factory reset likewise leaves the SD card's contents in place. The table below describes the **no-SD-card** case, where data is stored internally in LittleFS.

Method (no SD card)	File	Effect on internal saved data
Launcher (recommended for upgrades)	`CardSat-app.bin`	**Preserved.** Launcher replaces only the application partition, leaving the LittleFS filesystem intact, so settings, notes, favorites, and cached elements survive the update.
M5Burner / direct flash at `0x0`	`CardSat-merged.bin`	**Erased.** The merged image carries a fresh empty filesystem, so flashing it at `0x0` overwrites internally-stored data. You'll reconfigure and run **Update** again.

In short, your settings carry over if **either** an **SD card** is present (regardless of how you flash) **or** you upgrade with **Launcher** (`CardSat-app.bin`). Data is only lost when it was stored **internally** *and* you do a full `CardSat-merged.bin` flash.

If you do land on a clean slate, plan to set up again: re-enter your **location/grid**, your **radio (CAT) and rotator** settings, any **calibration**, then run **Update** once to refetch GP elements (see §7). Advanced users without an SD card who still want to keep internal data across a binary update can flash only the application partition (point esptool at the app offset instead of `0x0`, leaving the filesystem partition untouched), but Launcher does this automatically and is less error-prone.

After upgrading, confirm the running version on the **About** screen (Main menu → **About**). A power cycle after flashing is good practice to clear any stale state.

06

The keyboard: how to navigate

CardSat uses the arrow legends printed on the Cardputer keys:

Key	Action
`;`	up
`.`	down
`,`	left
`/`	right
ENTER	select / confirm
` or **DEL**	back / cancel
`{ }`	page up / page down (lists)
Fn + key	a modifier used by the Notes editor for cursor movement and save, so the `; . , /` keys stay typeable as punctuation (see §8 → Notes)
`h`	open **Help** (global; on text-entry screens use **`Fn+h`** — see below)
`b`	save a screenshot to the SD card (on letter-consuming screens use **`Fn+b`**) — see §20

Other letter keys are screen-specific actions and are shown in the **footer** at the bottom of each screen. When in doubt, read the footer. Two global exceptions are deliberately not shown in footers: **`h`** opens Help and **`b`** saves a screenshot. Both work on any screen *except* where letters are themselves input — the text editors, the **Tiny BASIC editor**, the **scientific calculator's** expression entry, the LoTW passphrase prompt, the Tools list's first-letter jump, and the DXCC / character lookups' type-to-search.

On every screen that claims bare letters, the two globals stay reachable as **`Fn+h`** (Help) and **`Fn+b`** (screenshot), so a plain `h` or `b` still does the screen's own job — typing the letter in the editors and the calculator, or jumping/searching in the Tools list and the DXCC / character lookups. This matches the `Fn`-is-for-commands convention those screens already use for save, run, print and cursor movement, and it means **every screen can be screenshotted**. (`Fn+shift+b` still types a capital B.)

07

First-time setup

1. **Settings** (`;`/`.` to move, `,`/`/` to change a value, ENTER to edit text):
   - **Radio** — select your model; CAT baud (and, for Icom, the CI-V address) auto-fill.
   - **CAT baud** — set to match the radio's CAT menu (applies to every CAT family).
   - **CI-V addr** — set to match the radio; **Icom only**.
   - **CI-V wiring** — (*Icom wired CI-V only*) choose **TX/RX (G2/G1)** — the normal, recommended separate-wire path — or a **single-pin** mode (**1-pin G2** or **1-pin G1**) that drives the whole CI-V bus over one shared open-drain GPIO. Single-pin is confirmed working on hardware (verified on an IC-821) but still needs correct 5 V/3.3 V level interfacing; the separate TX/RX path is simplest. See `docs/interfaces/CIV_SINGLE_PIN.md` and mind the level cautions before wiring.
   - **CAT serial monitor** — opens a live diagnostic that shows raw CAT traffic (TX in cyan, RX in green) as hex, for any wired protocol (CI-V, Yaesu, Kenwood). For a **USB CAT** radio it brings the USB transport up on entry so there's a live link to watch, and drops it again on exit if the monitor was what engaged it. While open it **actively polls the rig** (~once per 700 ms) by reading the downlink frequency, so you always see live traffic — your read request as TX and the radio's reply as RX — making it a real bench check of the CAT link with no satellite pass required. Press `p` to toggle polling off for a purely passive view; `s` to type a raw frame in hex (e.g. `FE FE 4C E0 03 FD`) and transmit it; `;`/`.` scroll back through the log; `` ` `` exits. Polling pauses automatically while Doppler tracking is engaged so the two don't collide. If you see "(no traffic yet)" with polling on, check the CAT wiring, baud, and (for Icom) the CI-V address. **Caution:** sending raw frames transmits arbitrary bytes to the radio — a malformed or wrong-address frame can mis-set the rig.
   - **Min pass el** — passes whose **peak elevation** never reaches this value are hidden from the pass lists and schedule (default 5°).
   - **Visible passes** — when on, the schedule flags passes that are **visually observable** (satellite sunlit, sky dark at your location, and the bird high enough) with a yellow `*`, and the pass-detail screen shows a verdict ("Visible: YES", or the reason it isn't — daylight, satellite in shadow, or too low). **Sky-dark gate** chooses how dark the sky must be — civil (Sun −6°), nautical (−12°), or astronomical (−18°) — and **Visible min el** is the minimum peak elevation to count (default 10°). This turns the schedule into a "what can I go outside and watch" tool; the ISS and other bright objects light up `*` on a good evening pass.
   - **WiFi SSID / WiFi pass** — enter your network, then **Save & test WiFi**. On the **WiFi SSID** row you can instead press `s` to **scan** for nearby networks, pick one from the list (strongest first; `*` = secured), and then enter its password (open networks skip the password step).
   - **WiFi 2 SSID / WiFi 2 pass** — an **optional second network**. If the primary network can't be joined, CardSat automatically falls back to this one. Handy in the field: keep your home router as the primary and a **phone hotspot** (or a portable travel router) as the second, and the device connects to whichever is present. Leave it blank to disable.

- **AOS alarm** — on/off for the pre-pass beeps.
- **VFO Type** — which physical VFO carries each leg: *Main Up/Sub Dn* (default) or *Main Dn/Sub Up*.
- **Beacon/RX-only DL** — which VFO carries the downlink for receive-only entries (beacons, telemetry, SSTV, CW — anything with a downlink but no uplink). *Follow VFO* (default) puts it on the same VFO as a normal transponder downlink, per VFO Type; *Main* or *Sub* force it to that VFO. Use *Follow VFO* if swapping to a beacon mid-pass moves your receive to the wrong VFO; force *Main* only if your rig reads that VFO back more reliably for receive-only.
- **Sat mode** — command the rig's own satellite mode on/off when you engage CAT (a no-op on rigs without CAT sat mode, including the IC-820/821/970 — engage SAT on those from the front panel).
- **CAT rate** — how often Doppler/CAT updates are sent to the radio (default **500 ms**, adjustable in 10 ms steps). A soft floor keeps the *effective* rate no faster than the CAT baud can service one update; the row shows `(min N)` when your setting is being clamped.
- **CAT delay** — how long to pause *after each CAT command* before sending the next one to the radio (default **70 ms**, 0–200 ms in 2 ms steps). Raise it if an older or slow radio drops or mis-handles back-to-back commands; lower it toward 0 for the fastest tuning on a radio that keeps up. CI-V (Icom) only at present.

2. **Location** — set your position one of three ways:
   - `e` latitude, `o` longitude, `a` altitude; or
   - `g` Maidenhead grid (uppercased as you type — hold shift for lowercase sub-squares — as on the log and mutual-pass grid fields); or
   - `p` enable GPS (and `s` to pick the GPS source).
   - If you have no network or GPS, press `c` to set the **UTC clock** manually (`YYYY-MM-DD HH:MM:SS`).
3. **Update** — press `k` (or ENTER) to download GP data; this also syncs the clock over NTP if WiFi is configured. Optionally press `a` to cache **all** transponders for offline use.
4. You're ready: **Satellites** to pick birds, **Next Passes** to see what's up.

At every power-on, if a WiFi network is configured CardSat connects automatically and sets the clock over **NTP** before loading data — so a network-connected unit boots with the correct time and no key presses. If WiFi is unavailable it carries on with GPS or the cached/manual clock; the attempt is best-effort and non-fatal (it can add a few seconds to boot while it tries). If the clock is set and even the freshest cached element set is **over a week old**, CardSat also refreshes GP automatically at boot. The display blanks its backlight after the **Screen sleep** idle time (never while tracking or alarming); any key wakes it.

08

Screen map and navigation

CardSat is a simple state machine. From **Home** you reach every screen; ` or **DEL** always steps back. Backing out **returns you to the screen you came from** — so opening **Passes** from the Home menu backs to Home, while opening it from the **Satellites** list backs to the list; the same applies to **Track**, which returns to Passes, Next Passes, or Home depending on where you launched it.

One deliberate exception: leaving the **Track** screen does **not** stop tracking. If the radio and/or rotator are engaged they keep running in the background (CAT Doppler and rotator pointing are serviced regardless of the current screen), and a green **RAD** / **ROT** / **R+R** tag appears in the header so you know the rig may still be transmitting. Stop with `r` (radio) or `o` (rotator) — see §8 → Track.

Home

A menu of twenty items: **Satellites · Next Passes (favs) · Passes (sel) · Track (sel) · World Map · Sun / Moon · Space Wx · Weather · Activations · AMSAT status · Overhead now · Grid dist/bearing · QRZ Lookup · Location · Update · Settings · Log · Messages · About · Charge / Sleep.** The currently selected satellite is shown at the bottom right. `;`/`.` move, ENTER selects, and any letter jumps to the next item starting with it.

About

Author credit (**Paul Stoetzer, N8HM**), firmware version and build date, storage backend (microSD or internal flash), GP catalog size and freshest element age, WiFi/IP, battery level, free heap, and uptime. The heap line shows two numbers — total free memory and the **largest single free block** (“max blk”). On this no-PSRAM board the second is the one that matters for network uploads: a TLS connection needs one big *contiguous* chunk, not just scattered free bytes, so if uploads ever misbehave this is the number to glance at first. Both figures dip while features run and recover afterward; that’s normal. Press `l` for a **License & credits** screen — license pointer, no-warranty and hardware disclaimers, credit for the outside data sources, and a recommendation to support AMSAT. ` or ENTER returns home.

Press `k` for **Calendar export**, which writes standard **iCalendar** (`.ics`) files into `/CardSat/Calendars/` for import into a phone or desktop calendar. Pick a source — **favorite passes**, the **selected pass** (active satellite’s next pass), **activations + sked reminders** (from the hams.at feed and your manual entries), **EME good days** (the next 90 days’ best moonbounce days), or **visible passes + solar transits** — and CardSat writes a timestamped file and reports its name. Each event’s description carries the satellite, frequencies, mode, maximum elevation, rise direction, and your station grid and coordinates, so the event is self-explanatory in any calendar app. Download the files over WiFi from the web **Files** page (§18), then open or email them to import. All times are written in UTC, which every calendar app converts to the viewer’s local zone.

Figure 1: Charge / Sleep

Charge / Sleep

The last item on the home menu is a minimal low-power mode for charging or parking the device. Selecting it **blanks the backlight and stops radio/rotator output**, and goes further to cut power: it **powers the WiFi radio fully off** (WiFi is the single largest consumer) and **drops the CPU clock from 240 MHz to 80 MHz**, and the main loop skips all network and radio services while parked — so the device idles at a fraction of its normal draw. **Any key wakes the screen** to show battery status — a large gauge, charging state, and the cell voltage — then it auto-blanks again after about 5 seconds; the keyboard is still polled even in the low-power state, so a keypress responds instantly. **ESC/back** returns to the home menu and **restores full speed and reconnects WiFi**.

The battery percentage here is derived from the cell **voltage** against a LiPo discharge curve (3.30 V ≈ 0 %, 4.20 V ≈ 100 %), which is more accurate in the flat middle of the discharge than the PMIC's raw level; it falls back to the raw level if voltage isn't available. Charging state comes from the Cardputer ADV's PMIC. The favorites **AOS alarm keeps running** while parked (it's pure computation, no WiFi), so you still get pass-countdown beeps on the charger.

Pressing **H** in this mode triggers an **on-demand heap reset**: on a long session, repeated HTTPS/TLS handshakes can fragment the (no-PSRAM) heap until the largest free block is too small for a new TLS context and fetches start failing, even though total free heap still looks healthy. The reset restores full speed, reconnects WiFi/TLS to free and coalesce those transient allocations, and reports the before/after largest contiguous block.

This manual reset is rarely needed: CardSat does the same coalescing **automatically** just before a handshake that would otherwise fail (it briefly cycles WiFi to return the fragmented buffers, then proceeds or declines gracefully), and the firmware keeps more contiguous heap free in the first place. The `H` key remains as a manual override.

Logging QSOs (Log)

There are two ways to start an entry:

- **From a pass** — press `l` on the **Track** or **Polar** screen to log the contact you're working; radio control keeps running while you type. CardSat snapshots the UTC time, satellite, mode and the live **Doppler-corrected** uplink/downlink frequencies, plus your grid and callsign (from **My callsign** in Settings). The form opens with the **Call** field already selected, so you can press ENTER and type the callsign straight away.

- **After the fact** — choose **New QSO entry** from the **Log** menu when you're *not* working a pass. The entry opens with the time set to now and everything ready to edit, so you can log a contact you made earlier or on another rig.

Every field is editable on the entry screen — `;`/`.` move, ENTER edits the selected field:

- **Date / Time** — UTC, entered as `YYYY-MM-DD` and `HH:MM:SS`.
- **Sat** — ENTER opens the satellite list; pick a bird and CardSat fills the **Sat**, **Mode** and frequency fields with its defaults — the **non-Doppler center** of a linear transponder's passband, or the **nominal** downlink/uplink for an FM or single-channel satellite.
- **DL MHz / UL MHz** — downlink and uplink in MHz (e.g. `145.960`).
- **Call** — the station worked (required to save). Letters default to **uppercase** (hold shift for lowercase).
- **Mode** — on a **linear** transponder, ENTER toggles **SSB ↔ CW** for this contact; on an FM bird the mode is fixed.
- **RST S / RST R** — report sent and received (default `59`).
- **Grid** — the other station's grid, **uppercase** by default (optional, but grids are the point of sat ops).
- **Notes** — free text.

When you are **editing an existing logged QSO** (via **View / edit log**), two more rows appear at the bottom:

- **LoTW / Cloudlog** — whether this QSO is marked as already uploaded to each service. ENTER toggles the flag. Editing any of the fields above **automatically clears both flags** so the corrected QSO is re-sent on your next upload; these rows let you override that — for example, after fixing a typo in **Notes** you can mark the QSO back as already uploaded so it isn't sent again — or simply flip a flag on its own. The rows always show the state that will be saved. (They don't appear when logging a brand-new QSO, which starts as not-yet-uploaded.)

`s` saves, `` ` `` cancels. QSOs are appended to **`/CardSat/qso_log.csv`** (one row each; `notes` is the last column, so commas there are fine).

The **Log** item on the main menu offers **New QSO entry**, **View / edit log**, **Export to ADIF** (writes `/CardSat/qso_log.adi`, including `STATION_CALLSIGN` from My callsign, for upload to LoTW/eQSL or import into your main logger), **Sign & upload to LoTW** (see LoTW upload below), **Upload to Cloudlog** (see Cloudlog upload below), **Voice Memos** (the on-device voice-memo browser described above), and **Notes** (a free-form text editor, see Notes below), and **Awards** (a summary of your worked totals, see Awards below). The two upload options sit together: uploading to Cloudlog can itself forward your QSOs to LoTW (if your Cloudlog instance is configured for that), so for most operators they're alternatives rather than things to do separately. The core logging functions are listed first and the menu scrolls; the non-core **Voice Memos** and **Notes** tools are grouped together at the bottom.

LoTW limits the `SAT_NAME` field to six characters and uses its own names, so on export CardSat translates each satellite via **`/CardSat/lotw_sats.csv`** (rows of `SAT_NAME,AMSAT_NAME`; a built-in table covers the common cases if the file is absent). If a logged satellite has no match, CardSat prompts you for its `SAT_NAME` (entry capped at six characters, uppercased) and saves it back to the CSV, so it's only asked once. Update that CSV on the card as new satellites appear.

Logbook of the World (LoTW) direct upload

Sign & upload to LoTW (on the **Log** menu) signs your un-uploaded satellite QSOs into a `.tq8` file and sends them straight to ARRL's Logbook of the World over WiFi — no PC, no TQSL, no separate upload step. CardSat builds the same cryptographically-signed file TQSL would, signs each QSO with your callsign certificate's private key, and posts it to LoTW's self-authenticating upload service (the signed payload authenticates itself, so no login is needed).

> **! This feature requires a microSD card, and your LoTW private key lives on it.** Read this whole section before using it. The key is your amateur-radio identity credential. Treat the card like you'd treat that key: keep it in your possession, and don't use a shared or disposable card. CardSat loads the key only when you run an upload, keeps it in RAM only, and never copies or transmits it anywhere except as the signature inside the `.tq8`. **Use this at your own risk.**

This is an upload feature, not enrollment. ARRL issues first-time certificates only through TQSL plus a postcard mailed to your FCC address — that can't happen on a handheld. You enroll once on a computer the normal way, then copy your existing credential to the card.

One-time setup (on your computer):

1. In TQSL, you already have your callsign certificate. Export it to a `.p12` file (TQSL: *Callsign Certificates → right-click your call → Save Callsign Certificate*, choosing the `.p12`/PKCS#12 format with a password).
2. Convert the `.p12` into the two PEM files CardSat reads. The easy way is the **bundled converter**: open **`tools/lotw_cert_converter.html`** in any web browser, choose your `.p12` file, enter the export password, and click **Convert** — it produces **`lotw_key.pem`** and **`lotw_cert.pem`** for you. It runs entirely in the browser (it even works with your network disconnected), and your private key and password never leave your computer.

   If you'd rather use the command line, OpenSSL does the same thing:

   ```
   openssl pkcs12 -in CALL.p12 -nocerts -nodes | openssl rsa -out lotw_key.pem
   openssl pkcs12 -in CALL.p12 -clcerts -nokeys -out lotw_cert.pem
   ```

   (The converter writes an unencrypted key, which is what CardSat expects. With the OpenSSL route you can instead keep the key encrypted on the card by dropping `-nodes` and setting a password; CardSat will ask for it at upload time.)
3. Copy **`lotw_key.pem`** and **`lotw_cert.pem`** into the **`/CardSat/`** folder on the microSD card.
4. In **Settings → Station / logging**, set your LoTW station location. The fields form a chain — each one narrows the next:
   - **LoTW DXCC** — press ENTER to pick your entity from the full DXCC list. The entities that have a subdivision (United States, Canada, Japan, the Russias, China, Australia, Finland/Åland/Market Reef) are grouped at the **top** of the list, marked with a >; everything else follows alphabetically. Start typing to filter the list (e.g. "japan" or "unit"). The list is long, so the typeahead is the fast way in.
   - **LoTW CQ zone** and **LoTW ITU zone** — your zones.
   - **LoTW (primary)** — the row is labeled with your entity's actual subdivision name: **state** (US), **province** (Canada/China), **oblast** (Russia), **prefecture** (Japan), **state** (Australia), or **kunta** (Finland). Press ENTER to pick from the valid list for your

DXCC; type to filter. If your entity has no subdivision the row shows **(n/a)** and you can skip it. (US/AK/HI **must** set a state — LoTW rejects uploads from those entities without one.)
- **LoTW (secondary)** — only two entities have a second level: the **United States** (county, which lets your contacts earn county awards) and **Japan** (city/gun/ku). The row is gated by your primary choice — pick the state first and the county list follows; pick the prefecture first and the city list follows. Other entities show **(n/a)**.
- **LoTW IOTA** — optional, any entity: enter your island reference (e.g. `NA-005`) if you're operating from a qualifying island.

This is the same flow for everyone — the US is just the DXCC "United States" with "state" and "county" as its two levels, exactly like Japan's "prefecture" and "city". Your callsign and grid come from **My callsign** and your location, which CardSat already has. (Inside any picker: `;/.` move, ENTER selects, typing filters, and ` clears the filter or backs out.)

Uploading: open **Log → Sign & upload to LoTW**. The screen shows whether the card, the key/cert, and the station fields are present, and how many QSOs haven't been uploaded yet. Press **u** to sign and send; if your key is password-protected you'll be asked for the password (leave it blank if you exported with `-nodes`). CardSat signs each pending QSO, posts the `.tq8`, shows how many were accepted, and flags those QSOs so they're never sent twice. (LoTW rejects duplicate uploads, and CardSat tracks what it has sent in a new `uploaded` column in `qso_log.csv`.)

Large uploads are automatic. A full log is uploaded in **batches of 6 QSOs**, each small enough to clear a fixed limit in the ESP32's network stack (a larger single upload would stall part-way through — see docs/design/UPLOAD_AND_AUDIO_TLS_POSTMORTEM.md). When more QSOs remain after a batch, CardSat **continues on its own in the same session**, and you **enter your key password only once** for the whole run. The upload screen shows the progress ("Batch sent; N left..."). This works for both **u upload (un-uploaded only)** and **upload ALL** (re-send) — a 14-QSO upload, for instance, goes up as 6 + 6 + 2 across three batches and stops by itself, with no interruption. The log is split into small batches only to keep each `.tq8` file small; earlier firmware rebooted between batches, but as of v0.9.43 the whole run completes in one session on the BearSSL TLS stack. Your key password is held only in volatile memory during the run and is erased when it finishes or fails; it is never written to the card or flash.

If a connection is refused outright (rare now — a genuine transport failure, e.g. very weak WiFi), CardSat offers a **reboot-and-upload** recovery — **ENTER** to reboot and finish from a clean start, ` to cancel. This is a user-confirmed failsafe, not part of the normal flow.

The satellite `SAT_NAME` translation described above applies here too — the same `/CardSat/lotw_sats.csv` map is used, so make sure each bird you've worked has a LoTW name before uploading.

View / edit log is a scrollable list of recent contacts (`;/.` to move), **ordered newest-first** — the most recent QSO is at the top, and the list opens there. Open one with ENTER to correct **any field — including the date, time, satellite and frequencies** — then `s` to save, or press `x` twice to delete it; changes are written straight back to the CSV. The most recent **60** QSOs are available on the device — the complete log always lives in `/CardSat/qso_log.csv`, which you can also read or edit on a computer.

Cloudlog / Wavelog upload

CardSat can upload your satellite QSOs to a self-hosted **Cloudlog** (or the compatible **Wavelog**) online logbook over WiFi. Because a Cloudlog instance can itself be set up to forward QSOs on to LoTW, eQSL, and others, for most operators this is an **alternative** to the on-device LoTW upload, not something to do in addition. CardSat tracks the two independently, so a QSO uploaded to Cloudlog is not marked as sent to LoTW, and vice-versa.

Setup (all under *Settings → Station / logging*):

- **Cloudlog URL** — the base address of your Cloudlog instance, e.g. `https://log.example.com`. Both `https://` and `http://` (for a LAN instance) are accepted. A trailing slash is trimmed automatically; CardSat appends the API path for you.
- **Cloudlog API key** — generate a **read-write** key in your Cloudlog account's API page and enter it here. (A read-only key is rejected for uploads.)
- **Cloudlog station ID** — the numeric *station profile id* the QSOs should be filed under, shown in your Cloudlog station-profile settings. Cloudlog verifies that this profile belongs to your key and that its callsign matches each QSO's station callsign, so make sure your **My callsign** setting matches the profile.

Uploading: open **Log → Upload to Cloudlog**. The screen shows your server, whether the key and station ID are set, and the count of QSOs not yet sent to Cloudlog. Press `u` to upload (CardSat connects to WiFi if needed). To re-send contacts that are already marked uploaded — for instance after fixing a setting — press `a` to toggle re-send mode, then `u`. On success the screen reports how many QSOs Cloudlog imported. If the **server** rejects the upload it shows the reason (a wrong key, insufficient key rights, or a station-ID/callsign mismatch are the usual causes). Your API key is treated as a secret and is never written to the serial log.

Large uploads are batched automatically, the same way as LoTW: Cloudlog uploads in **15-QSO batches** (Cloudlog's records are smaller than a signed `.tq8`), all **in one session**. Cloudlog authenticates with your API key rather than a passphrase, so the continuation is fully automatic — nothing to re-enter.

If instead the upload can't even **connect** (rare now — a genuine transport failure such as very weak WiFi), CardSat offers a failsafe: it asks whether to **reboot and upload** — press **ENTER** to reboot and complete the upload from a clean start, or ` to cancel and leave the QSOs pending. (This only appears for a true connection failure; a server rejection, which a reboot wouldn't fix, is just reported.)

Notes (free-form text editor)

Notes on the **Log** menu is a simple multi-page text editor for jotting things down on the device — a sked frequency, a grid you still need, antenna settings, a reminder for the next pass. Each note is a plain `.txt` file stored under **`/CardSat/notes/`** (on the microSD card if one is fitted, otherwise in internal flash, so notes work with or without a card), which means you can also read or edit them on a computer.

The **browser** lists your notes newest-first, each with the date and time it was last saved (UTC, like every other time on the device). Press **`n`** to start a new note, **ENTER** to open the highlighted one, and **`d`** then **ENTER** to delete it (a confirmation step guards against an accidental press). ` returns to the Log menu.

The **editor** is a full multi-line editor: just type, with **ENTER** for a new line and the **delete** key for backspace. Because the Cardputer's `; . , /` keys are needed as ordinary punctuation in your text, the editor's commands all use the **Fn** modifier, so every plain key types literally:

- **Fn +** `,` and **Fn +** `/` move the cursor **left** and **right**.
- **Fn +** `;` and **Fn +** `.` move the cursor **up** and **down**.
- **Fn +** `s` **saves** (the first time, you're asked for a name; allowed characters are letters, numbers, spaces, `-` and `_`, capped at 31 characters).
- ``` ` ``` **exits** to the browser. If you have unsaved changes, a named note is saved automatically; an unnamed note prompts you for a name first (or press ` again at that prompt to discard it).

Notes are limited to 4 KB each, which is several pages of text — ample for operating notes.

Awards (Log → Awards)

Figure 2: Awards

Awards on the **Log** menu summarises what you've worked, read straight from `/CardSat/qso_log.csv`. The top of the screen shows the **all-satellites totals**: total **QSOs**, unique **grid squares**, the number of distinct **satellites** in your log, unique US **states** (out of 51, counting DC), and unique **DXCC entities**. Below that is a scrollable list of every satellite you've worked and how many contacts you've made through each.

From either the all-satellites view or a per-satellite view you can open a **scrollable list of the actual worked grids, states, or DXCC entities**: press `g` for grids, `s` for states, or `d` for DXCC. The list shows the decoded entries in pages (`;`/`.` scroll a line, `{`/`}` page); press `g`/`s`/`d` again to switch between the three lists, and ` to step back. Press **ENTER** on a satellite in the all-sats list first to scope these lists to **that satellite's** worked grids/states/entities; the per-satellite screen also shows that bird's own QSO, grid, state and DXCC counts.

> **How states and DXCC are counted.** CardSat does **not** store a state or DXCC field per QSO. It **derives** both from each contact's **grid square** by the same geographic point-in-polygon calculation the workable-states and workable-DXCC footprint screens use (with the island/micro-entity entities matched to the nearest reference point from a bundled `cty.dat`-derived table). This has two consequences: a QSO logged **without a grid** counts toward your QSO, satellite and band totals but **cannot** be placed in a state or entity; and a grid that sits very close to a border, or a rare entity not in the built-in table, may occasionally be attributed differently than an official award check would. Treat the state and DXCC numbers as a **good working estimate for your own tracking**, not as an authoritative award credit. The QSO, grid, satellite and band counts are exact (grids are taken directly from the log).

Awards are computed each time you open the screen, so they always reflect the current log, including QSOs you've just added or imported.

Importing an existing ADIF log (`tools/adif2csv.py`)

If you already have a satellite log in another program, you can seed CardSat's log with it. The bundled **`tools/adif2csv.py`** converts a standard **ADIF** export into the CSV format CardSat stores on the device:

```
python3 tools/adif2csv.py mylog.adi          # writes qso_log.csv next to it
python3 tools/adif2csv.py mylog.adi out.csv  # explicit output name
```

It keeps **only satellite QSOs** — records whose ADIF `PROP_MODE` is `SAT` (if a record has no `PROP_MODE` but does carry a `SAT_NAME`, it's treated as a satellite contact) — and **drops everything else**, so your terrestrial HF/VHF contacts are left out. Only the fields CardSat uses are carried across (callsign, satellite, mode, up/downlink frequency, RST sent/received, your grid and the worked grid, your callsign, and a comment); anything else in the ADIF is ignored. Callsigns and grids are upper-cased to match CardSat's own entry rule, and frequencies are converted from MHz to Hz. Copy the resulting **`qso_log.csv`** onto the card at **`/CardSat/qso_log.csv`** (see below for transferring files).

> The same caveat as the Awards screen applies: the converter does not read or write any state or DXCC field — CardSat derives those from the grid square — so make sure your ADIF records include `GRIDSQUARE` if you want imported QSOs to count toward states and DXCC. Records without a grid still import fine; they just won't be placed in a state or entity.

Satellites

The catalog (up to 150 sats held from the GP data, plus any you add manually; if a source carries more than 150 — some CelesTrak groups do — your **favorites are loaded first**, the rest fill in file order, and the status line reports "Loaded X of Y").

- `;`/`.` move, `{`/`}` jump 10 rows.
- `f` — toggle **favorite** (favorites are marked with `*`).
- `v` — toggle **favorites-only** filter.
- `/` — **search the entire public catalog on CelesTrak** by name or NORAD number and add any object as a favorite — independent of whatever your primary GP source is. Type a name fragment (`NOAA`, `FUNCUBE`, `COSMOS 2251`) or an all-digit NORAD ID; results stream in showing name, NORAD, perigee–apogee band, and inclination. Rows shown **green** are already in your loaded catalog. **ENTER** on a result:
  - if the object is already covered by your primary source, it's simply marked favorite;
  - otherwise it's saved to `/CardSat/ctx.json` as a **CelesTrak extra**, merged into the catalog, and marked favorite — and, crucially, **every GP update re-fetches its elements from CelesTrak** so it never goes stale, no matter what your primary source is. (Hand-entered `n` satellites are deliberately *never* auto-fetched — pre-launch fits and state-vector saves usually aren't in the public catalog at all.)

  CelesTrak courtesy limits are enforced so a shared IP never gets banned: at least **10 s between searches**, an **identical query within 2 h reuses the cached result** instead of re-fetching, the extras refresh runs at most **once per 2 h** (the timestamp is persisted, so reboots can't bypass it), individual refresh fetches are spaced 2 s apart, and at most 25 extras refresh per update. The screen and the status line say when a limit is in effect.

- `n` — add a **manual GP satellite**: enter the name, NORAD ID, epoch, then each orbital element in turn (see **Edit** under §8). If your elements are only in TLE form, convert them first with `tools/tle2gp.py` (see §8). For an object that's in the public catalog, prefer `/` — it types less and keeps itself updated.
- `x` — **delete** the selected satellite, if it was added by you — either a manual `n` satellite (removed from `/CardSat/mgp.json`) or a `/` CelesTrak extra (removed from `/CardSat/ctx.json`, which also stops its per-update refresh). Press `x` once to arm, `x` again to confirm; the favorite mark is dropped too. Cached GP satellites from the network can't be deleted this way (they'd just return on the next Update). To **edit** a manual satellite, delete it and re-add it with `n`.
- `e` — open the **EQX table** (equatorial crossings) for the selected satellite, for use with an OSCARLOCATOR (see Transponder/EQX screens in §8).
- `2` — open the **sat-to-sat visibility finder**: the windows when this satellite and a second one (chosen from your favorites) are **both above your horizon at once** over the next few days (see below).
- **ENTER** — load the satellite's transponders and open its **Passes**.
- ` — back to Home.

At the right edge each row may show an **AMSAT activity mark**: a **filled dot** means the satellite has been reported **heard** (active) recently, a **filled square** means it has only been reported **telemetry only** (a beacon/telemetry is alive but no transponder or voice contact was logged), a **hollow ring** means it has only been reported **not heard** recently, and **no mark** means there are no recent reports. When a satellite has a mix of reports the strongest wins (heard > telemetry > not heard). The marks come from the AMSAT OSCAR Status page and are refreshed whenever you run **Update** (see §GP source). Matching is by base designator, so it works for AMSAT-named birds; satellites loaded from CelesTrak categories usually won't match and simply show no mark.

"Recently" is a setting. The status window defaults to the last **24 hours** and can be changed to **3, 6, 12, 24, 48, or 72 h** in *Settings → AMSAT status window* (re-run Update to re-fetch with the new window). A shorter window is a sharper "is it workable right now" signal; a longer one catches rarely-reported birds.

AMSAT status screen. For a full view of what's on the air, open the dedicated **AMSAT status** screen — press **s** from the Satellites list, or pick **AMSAT status** from the main menu. It lists every bird with a report in the window, **sorted most-active-and-most-recent first**, and for each shows the status word (colored), **how long ago** the most recent report came in (e.g. "45m", "3h", "2d" — for heard, telemetry, *and* not-heard alike, since "not heard 1h ago" tells you someone tried recently), and how many stations reported. **ENTER** adopts a bird as the active satellite and jumps to Track; **u** re-fetches in place; ` goes back. (The "N ago" ages need the clock set; the feed buckets to about half-hour resolution, so they're coarse beyond the first hour.)

Each row may also show a small **down-arrow** to the left of the activity mark — a **decay flag** for orbits that are dropping. It's colored by severity: **yellow** (watch), **orange** (decaying), or **red** (reentry imminent), derived from the satellite's perigee altitude and the lifetime estimate (see the **Perigee** line on the Orbital-analysis screen). It's an at-a-glance "this bird is on its way down" cue — useful for working a satellite before it's gone, and for spotting objects nearing reentry. The level is an order-of-magnitude estimate from the elements, **not** a precise reentry prediction; for that, consult CelesTrak or Space-Track.

Orbital analysis (o)

Opened with `o` from **Satellites**. Eleven pages, paged with `,`/`/`, `r` to recompute, ` to leave.

8.0.11.1 Theory of operation — how these numbers are produced A short primer, because the pages below assume some of this:

The elements. Each satellite is described by a set of **mean orbital elements** (the modern GP/OMM form of the classic two-line element set, TLE). The six that fix the orbit's size, shape and orientation are: **semi-major axis** a (orbit size, here derived from mean motion), **eccentricity** e (how elliptical, 0 = circle), **inclination** i (tilt of the orbit plane to the equator), **right ascension of the ascending node** Ω/RAAN (where the orbit plane cuts the equator going north, measured around the equator), **argument of perigee** ω (where the low point sits within the orbit plane), and **mean anomaly** M (where the satellite is along the orbit at the element epoch). **Mean motion** n (revolutions/day) stands in for a, and **B*** carries atmospheric drag. "Mean" means small periodic wobbles have been averaged out; the propagator puts them back.

The propagator. CardSat predicts position with **SGP4** — the same Simplified General Perturbations model the elements were built for. SGP4 is not a plain Kepler solver: it folds in the dominant perturbations (Earth's oblateness through the **J2** harmonic, plus atmospheric drag via B*) so that a near-Earth satellite's real motion is reproduced to roughly a kilometer near epoch. Every live read-out — az/el, range, range-rate, sub-point — comes from propagating the elements to the current instant and converting the resulting position/velocity vectors. **Accuracy decays with element age**, which is why the Info and Orbit-position pages show how old the set is and CardSat nudges you to refresh GP. For the full theory of GP/OMM elements, what SGP4 is and isn't, the coordinate frames involved, and where prediction error comes from, see §14.

Range rate and Doppler. The radial velocity (range-rate) is taken straight from the SGP4 velocity vector — the component along the station→satellite line. Doppler shift is then `Δf = −(range-rate ÷ c) · f`: positive range-rate (receding) lowers the received frequency, and the shift scales with frequency, so a 70 cm downlink swings roughly three times as far as a 2 m one. This is the same quantity the Track screen uses to tune the radio (see §9).

J2 and why pass times drift. A spherical Earth would hold the orbit plane fixed in space; the real Earth's equatorial bulge (the J2 term) makes the plane **precess**. The two secular rates CardSat reports come directly from J2: the node regresses at `Ω̇ = −1.5 n J2 (Re/p)² cos i` (°/day) and perigee rotates at `ω̇ = 0.75 n J2 (Re/p)² (5cos²i − 1)` (°/day), where $p = a(1-e^2)$. Node regression is what walks a Sun-relative orbit's pass times earlier each day, and when it happens to equal the Earth's ~0.986°/day motion around the Sun the orbit is **sun-synchronous** (it keeps a fixed local solar time — the **LTAN**). Apsidal precession turns ω, moving where perigee sits; it goes to zero at the **critical inclination** of 63.4° (where $5\cos^2 i-1 = 0$), the trick Molniya orbits use to keep perigee parked.

Beta angle and eclipses. The **solar beta angle** β is the angle between the Sun and the orbit *plane* (CardSat computes it as 90° minus the angle between the orbit- normal vector and the Sun vector). It governs lighting: at low β the satellite dips into Earth's shadow once per revolution; past a threshold **β*** = arccos(Re/(Re+h)) the orbit rides above the shadow and is in **continuous sunlight**. CardSat tests shadow with a **cylindrical-umbra** model (Earth casts a parallel-sided shadow tube, no penumbra) — simple, and good to about a minute for power-budget purposes. As the node precesses (above), β cycles over weeks, producing

eclipse seasons and full-sun seasons; the Sun/Beta and Illumination screens visualize this.

Footprint geometry. The ground footprint is the cap of Earth from which the satellite is above the horizon. Its diameter is `2·Re·acos(Re/(Re+h))`, set purely by altitude *h* — higher means a wider circle and a longer maximum possible contact. For an elliptical orbit the footprint breathes between its apogee (largest) and perigee (smallest) values, both of which the Info page lists.

The pages:

- **Info** — NORAD, current altitude, **footprint** diameter, period, apogee/ perigee altitudes, the **footprint diameters at apogee and perigee**, inclination/eccentricity, semi-major axis, B* with a rough drag-decay estimate, element age/revolution, and the next ascending-node longitude/time. The decay estimate integrates B* through an exponential-atmosphere model with a **King-Hele** treatment that lets an eccentric orbit circularize (drag at perigee lowers apogee fastest), down to a ~120 km reentry. The physics: drag acts hardest where the air is densest, which is at **perigee**, and a velocity loss there pulls down the *opposite* side of the orbit — so an elliptical orbit rounds off (apogee drops toward perigee) before the whole thing spirals in. B* scales how much atmosphere the satellite "feels" (ballistic coefficient × a reference density), and the model decrements energy each revolution until the perigee reaches the ~120 km reentry floor. Because thermospheric density swings about a factor of ten over the solar cycle - the single biggest driver of lifetime - a second **"Decay rng"** line shows the bracket from **solar maximum (shortest)** to **solar minimum (longest)**, with the assumed level (`mean/min/max/auto`, set in *Settings → Station / display → Decay solar*) in parentheses. In **`auto`** the density scale is derived from the live **F10.7 solar flux**, downloaded with the GP elements and cached (the setting row shows the current value); without a cached flux it falls back to mean. Treat all of this as **order-of-magnitude**: it ignores attitude, lift, and short-term space weather, and B* itself is often a fitted fudge term, so the chart is a "should I worry about this object" cue, not a reentry prediction. The **footprint** is the diameter of the visibility circle on the ground (`2·Re·acos(Re/(Re+h))`) — the widest separation between two stations that can both see the bird at once, i.e. the **longest theoretically possible QSO** through it. It grows with altitude, so for an elliptical orbit the apogee figure is the best case and perigee the worst.
- **Live** — az/el, range, range-rate, Doppler at 145.8/435 MHz, sub-point, mean anomaly/phase, the sunlit/eclipse flag, and the **eclipse depth** — the PREDICT-style angle (degrees) by which the satellite is inside Earth's umbral shadow (positive when eclipsed, negative as a sunlight margin), so you can see not just *whether* it's in shadow but *how deeply*. The depth is the angular gap between the satellite's anti-solar position and the edge of the shadow cylinder: it climbs as the bird moves toward the center of the shadow and crosses zero exactly at the **terminator** (entry/exit), which is why a value near 0° means a sunrise/sunset is imminent — useful for spin-stabilised or solar-powered birds whose behavior changes at the shadow boundary.
- **Next pass** — AOS countdown, duration, max elevation, AOS/LOS azimuths, sunlit fraction, eclipse entry/exit, the **peak eclipse depth** if the bird transits shadow during the pass, the optical-visibility flag, the **slant range at AOS/TCA/LOS** and the **one-way path delay** at closest approach (range ÷ c).
- **Ground track** — the next two orbits projected on an equirectangular map.
- **Doppler** — the Doppler curve across the next pass, computed at a **user-settable beacon frequency** (press `f` to change it; default 145.8 MHz). Below the plot it shows the peak

Doppler shift and the maximum range-rate of the pass.

- **Nodal** — orbit-plane dynamics from the J2 secular model: revolutions per day; **node drift** (RAAN regression, °/day) and **perigee drift** (apsidal precession, °/day); a **sun-synchronous** flag (node drift ≈ +0.986°/day); the **LTAN** (local solar time of the ascending node); an approximate **repeat ground-track** cycle (revs/days, ≤30 days); and the **longest possible pass** (an overhead pass at apogee). The repeat cycle looks for a small whole number of days in which the satellite completes a whole number of revolutions, so its track over the ground closes back on itself — when it exists, the same passes recur on that period (a 3-day repeat means today's geometry returns in three days). These are first-order estimates — handy for understanding why pass times march earlier each day, whether a bird is sun-synchronous, and how long the very best passes can run — not a precision propagator.
- **Sun / Beta** — the orbit's lighting geometry. The **solar beta angle** (the angle between the Sun and the orbit plane) now, the analytic **β*** threshold beyond which the orbit is in continuous sunlight, and whether the bird is currently in a **full-sun orbit** or **eclipsed every revolution** — the verdict and the **eclipse fraction / minutes per orbit** come from sampling a full orbit with the same geometric shadow test the Illumination screen uses, so the two never disagree. A final line scans ahead to the **next transition** (the date the orbit next crosses into full sun, or back into eclipse, with a day countdown). Useful for solar-powered birds and for anticipating eclipse-season power behavior.
- **Pass outlook** — a planning summary over the next 7 days: the total number of passes above your mask, how many clear 30°, the longest pass, and the mean gap between passes — plus the **best upcoming pass** (its peak elevation, the date/time it occurs, the countdown to it, and its duration). This answers "when this week is worth operating this bird" at a glance.
- **Orbit position** — where the satellite is in its orbit right now: mean anomaly, true anomaly, and **argument of latitude**, the **time to the next perigee and apogee**, plus argument of perigee, RAAN, the current revolution number, and the element-set age. The three angles describe position three ways: **mean anomaly** advances at a constant rate (it's the orbit clock, uniform in time); **true anomaly** is the real geometric angle from perigee (it runs ahead near perigee where the bird moves fast, behind near apogee — the two are equal only for a circular orbit); and **argument of latitude** (ω + true anomaly) is the angle up from the equator, which tells you how far through the north/south part of the orbit it is. For eccentric orbits the perigee/apogee timing shows when the bird is moving slowest and sitting highest — i.e. when the longest-dwell passes occur.
- **Phys** — physical and identity data. The satellite's **instantaneous orbital velocity** (from the vis-viva equation, so it is correctly faster near perigee and slower near apogee), with the apogee/perigee velocity spread; and the **launch year, launch number, and time in orbit**, derived from the COSPAR International Designator (e.g. `1974-089B` → launched 1974). The launch figure is year-granular — the designator carries no launch day — but it's a nice bit of context when you're working a decades-old satellite. The page also shows the **element-set number** (which elset revision you are propagating) and the **launch siblings** — the other cataloged objects that shared the same COSPAR launch, listed by name (AO-7 rode up with companions, for example), wrapped across lines and clipped with "..." if a very large batch would reach the footer.
- **Explore** — a what-if sandbox. It seeds **apogee, perigee, and inclination** from the tracked satellite, then lets you edit them (`;/.` pick a row, type a value, `x` reseeds from the real elements) and watch the derived characteristics recompute live: period, revs/day, eccentricity, apogee/perigee velocity, footprint radius, longest possible pass,

nodal drift, and whether the orbit is sun-synchronous. It never alters the satellite's real elements, and all quantities stay metric (km, km/s, min) — useful for answering "what would this bird's passes look like 200 km higher?" without touching the catalog.

Higher orbits (0.9.59): pass prediction is no longer LEO-only. For any orbit with a period over ~225 minutes (Molniya-class ellipses, GTO, geosynchronous — deep-space SDP4 has always been in the propagator) CardSat scans elevation itself and bisects the horizon crossings to ~1 s, instead of using the LEO-tuned library search that misses them. A bird that stays above your minimum elevation for the whole prediction window — a geostationary transponder in view — honestly reports one continuous horizon-long "pass". Verified against Skyfield in the host audit harness (Molniya crossing accuracy ≤ 0.04°, coverage 194/194 samples).

Next Passes (schedule)

One merged list of the next pass for **every favorite**, soonest first. Each row shows the countdown to AOS (or **NOW** if the bird is already up), the satellite name, maximum elevation, and pass length. A red `!` flags a satellite whose elements are stale (see §14). A yellow `*` marks a **visually observable** pass (satellite sunlit, sky dark, high enough) when visible-pass prediction is enabled in Settings — see §7 → Visible passes.

- `;`/`.` move.
- **ENTER** — jump straight to **Track** for that satellite.
- `m` — open the live **World map** (all footprints; see World map).
- `t` — open **Sky at a glance**, a horizontal timeline of upcoming passes (below).
- `r` — recompute the schedule.
- `z` — **deep-sleep** until ~60 s before the next AOS (see §12).
- `` ` `` — back.

Sky at a glance (`t`). A horizontal timeline of the **next few hours** for all your favorites: time runs left to right across the top (hour ticks, with a green "now" line at the left edge), and each favorite that has a pass in the window gets a row with one bar per pass. Bars are colored by **peak elevation** — **green** for a high pass (≥ 30°) and **yellow** for a lower one — the same convention as the **Overhead now** screen, and a white tick on a bar marks an **optically visible** pass. It's the fastest way to see which birds are coming up and when they overlap. `r` recomputes; `;`/`.` or `` ` `` return to the list.

Passes

Upcoming passes for the selected satellite, in UTC: **AOS time, duration, max elevation, LOS time.** The element-set age is shown top-right, color-graded. A yellow `*` to the right of a pass marks it as **optically visible** — the satellite is sunlit while your sky is dark and the pass clears the visibility elevation gate (see Settings → Visible passes).

- `;`/`.` — select a pass.
- `d` — open the **pass-detail plot** (below).
- `t` or **ENTER** — go to **Track**.
- `n` — add a **manual transponder** for this satellite (see below).
- `r` — recompute the pass list.
- `x` — **mutual window**: enter a remote grid to find co-visibility passes (below).
- `v` — **10-day pass overview** (InstantTrack-style visibility chart; below).

- `V` — **visible-pass list**: every optically-visible pass for this satellite over the next 10 days, as a scrollable list (below).
- `i` — **illumination** (60-day Sun/eclipse raster; below).
- ` — back to where you came from (Satellites, or Home if you opened Passes from the Home menu).

Visible-pass list (V from Passes)

Figure 3: Visible-pass list

A scrollable list of every **optically-visible** pass for the selected satellite over the next 10 days — filtered to passes where the satellite is **sunlit**, your **sky is dark** (Sun below the configured gate), and the **peak elevation** clears the visibility minimum. Each row shows AOS time, duration, peak elevation, and the **rise compass direction** (where on the horizon to look as it comes up). `;`/`.` scroll; **ENTER** opens the **pass-detail plot** for the highlighted pass (and returns here when you back out); `r` recomputes; ` returns to Passes. This complements the **10-day chart** (`v`), which shows *all* passes graphically — the `V` list is just the ones you could actually see. Requires the **UTC clock** and your **location** to be set.

Pass detail

An **elevation-vs-time curve** for the selected pass. The curve is drawn **yellow where the satellite is sunlit** and **blue where it is in eclipse**. A cyan vertical line marks "now" if the pass is in progress. Below the plot: AOS time and azimuth, LOS time and azimuth, maximum elevation, pass duration, and the percentage of the pass spent in sunlight. Press **`p`** for a **polar view of this pass** — its ground track across the sky with **A/L** (AOS/LOS) markers and an arrow showing the direction of travel; `p` toggles back to the curve. ` or ENTER returns to Passes.

Mutual windows (co-visibility)

From **Passes**, press **`x`** and enter a remote station's **Maidenhead grid** (4 or 6 characters, e.g. `IO91` or `JN58td`). CardSat scans the selected satellite's upcoming passes **over the next 10 days** and lists every window where **both you and that station have the satellite above your horizons at the same time** — the periods you could actually make a contact through the bird. Each row shows the **start time (UTC)**, the **duration** (m:ss), and the **maximum elevation at each end** — yours (`me`) and theirs (`dx`) — so you can judge how workable the window is. `;`/`.` scroll; **`d`** opens the **DX Doppler table** for the highlighted window (below); ` returns to Passes.

> Co-visibility is computed to the **0° horizon** for both stations; a window with very low max elevations may be hard to use in practice. The remote grid is assumed at sea level.

8.0.16.1 DX Doppler table (`d` from a mutual window) For a short window, the hard part of a coordinated contact is that **the dial frequencies are different at each end** — your Doppler and the DX station's Doppler aren't the same, because you're moving relative to the satellite by different amounts. This table solves that. It lists, **every 30 seconds across the window**, the predicted **RX (downlink)** and **TX (uplink)** dial frequencies for **both your station and the DX station**, for the selected transponder. Each 30-second step shows two lines — your dials (green, "me") and the DX station's (cyan, "DX") — so the four frequencies never run together. Press **`t`** to cycle which transponder is used (the same selection you can also make with `t` on the Satellites screen).

Three tracking modes, cycled with **`m`**:

- **True rule** — the operating point is held fixed *in the satellite's passband*; every dial Doppler-tracks, each station with its own geometry. Both of you stay on the same transponder channel the whole pass.
- **Fixed downlink** — the **anchor** station holds its **receive dial constant** in real RF (the knob they don't touch). The passband operating point drifts to absorb that station's downlink Doppler, and the other three dials are computed to follow.
- **Fixed uplink** — the anchor station holds its **transmit dial constant**; the other three dials follow.

Press **`a`** to cycle which dial is the **anchor** (my RX, my TX, DX RX, or DX TX). For a **linear** transponder you also choose **where in the passband** to operate: the table **opens at the center of the passband** (and re-centers when you switch transponders with **`t`**). The header shows the operating point **relative to the center of the passband's downlink** — `ctr` at center, or a signed offset like `+7.5k` / `-12.5k` — so you always know how far off-center you are.

How the **`,`**/**`/`** keys step depends on the mode:

- In a **fixed** mode (fixed downlink or fixed uplink), `,`/`/` step the **anchored dial** to the next **round 1 kHz**, grid-aligned to the center of the passband. This is what lets you park your fixed RX or TX on a clean, easy-to-call number: cycle the anchor to the dial you care about (say *my TX*), and each press lands that transmit dial on 145.949, 145.950, 145.951 MHz — never 145.9502. CardSat nudges the passband so the anchored dial sits exactly on the kHz, then recomputes the rest of the table around it.
- In **true rule** mode (nothing anchored), `,`/`/` simply nudge the passband operating point by 1 kHz, as before.

RX values are shown in green, TX in yellow. **`;`**/**`.`** scroll through the 30-second steps; ` returns to the mutual window list.

> Frequencies include your station calibration offsets. The DX station is assumed at sea level with no local calibration. For an SSB contact, treat the numbers as the center of where to listen/transmit and fine-tune by ear.

10-day pass overview (v)

From **Passes**, press `v` for an at-a-glance chart of the selected satellite's passes over the **next 10 days**, modeled on InstantTrack's "Multiple Days for Single Satellite" visibility screen. Each row is one day (today at the top), drawn as a 24-hour timeline (00–24 h UTC, left to right) with faint gridlines at 06/12/18 h. The window is aligned to UTC midnight and spans ten **full** days, so every row is filled edge to edge — the chart is not cut off partway through the last day at the time you opened it. Every pass is a colored bar from AOS to LOS, shaded by peak elevation — **dim green** below 15°, **green** to 40°, **yellow** above — so high passes stand out. A red tick on the top row marks the current time. **`;`/`.` scroll one day at a time** — the oldest day falls off the top and a new day appears at the bottom (you can scroll forward indefinitely, but not before today). `r` recomputes; ` returns to Passes.

Illumination (i)

From **Passes**, press **`i`** for a **60-day solar-illumination raster** in the style of DK3WN's *illum*. The horizontal axis is date (today at the left edge, **+60 d** at the right); the vertical axis is one **orbital period** (its length in minutes is printed at the right). Each cell is shaded **yellow when the satellite is in sunlight** and left **dark when it is in Earth's shadow (eclipse)**, so the eclipse band — and the "sunline" at its edge — stands out, widening and narrowing as the orbit plane precesses relative to the Sun and vanishing entirely during **full-sun seasons** (handy for judging solar-panel charging). Below the raster a live readout shows the **current status** (`SUN`/`SHADOW`), the **eclipse minutes per orbit** and percentage for the current orbit, and the time to the next Sun↔shadow transition. **`,`/`/` jump a full 60-day window** at a time (forward indefinitely, not before today). `r` recomputes; ` returns to Passes.

> Eclipse uses a cylindrical-shadow model (no penumbra) and the raster is sampled, so the band edges and transition times are good to about a minute.

Track

The main operating screen. Top to bottom it shows: **azimuth / elevation** (and GP age), **range / range-rate** (and an orange **ECL** flag in eclipse), the selected **transponder**, the **downlink (DN) and Doppler-corrected receive (RX)** frequencies, the **uplink (UP) and transmit (TX)** frequencies, the **passband** position line, the **calibration** line, and the **radio status**. On an FM bird that needs a subaudible **PL/CTCSS tone**, a **`PL nn.n Hz`** line appears (orange when the radio is on and the rig supports CAT tone, gray otherwise, or `PL nn.n Hz (rig n/a)` if the selected rig can't set tone over CAT).

Controls:

- `m` — switch between **TUNE** and **CAL** modes.
- `d` — cycle the **Doppler tune mode** (linear birds): FULL One True Rule (tune the **downlink** knob) → downlink-only → uplink-only → FULL One True Rule on the **uplink** knob → hold-both. The passband line shows the active mode (`FULL` / `DL` / `UL` / `FULLu` / `TUNE`). The two FULL modes differ only in which VFO you turn: use `FULLu` when the uplink rig is the one with the tuning knob (common in a two-radio setup).
- `l` — **Log QSO** (also on the Polar screen): capture the contact you're working.

- `i` — **"I heard it"**: submit an **AMSAT status report** for the bird you're tracking. Press `i` once to arm (the status line shows what will be sent), and `i` again to send. Your callsign, grid and the satellite are filled in; needs WiFi and your callsign set. On a multi-mode bird with no clear catalog fit, a picker opens instead. Radio control keeps running while you fill in the entry.
- `t` — cycle to the next transponder.
- `n` — **jump to the beacon**: select the satellite's beacon (a downlink-only entry, or one whose description names a beacon) so the radio tunes straight to it with Doppler correction. Useful for finding a bird by its telemetry/CW beacon before working it. Also available on the **Big readout** screen. If the satellite has no beacon listed, a status note says so. On an Icom rig a receive-only transponder like this turns the rig's **satellite mode off** and tunes the downlink on the **MAIN** band (which also reads back more reliably), matching how OscarWatch and the SDR-Control apps handle receive-only birds.
- `c` — set the **CTCSS/PL tone** for this satellite (numeric entry: a tone in Hz, `0` to force it off, or blank to revert to the built-in default).
- `N` — edit a **per-satellite operating note**: a short free-text reminder that travels with the bird (active modes, schedule, "PL 67.0, use high passes," your own observations). It's stored by NORAD and shown on the Track screen — a `*` appears next to the satellite name in the header and the note text on its own line. Leave it blank to clear. Notes persist across reboots and reflashes (in `notes.txt`).
- `k` — on a **linear (SSB/CW) transponder**, toggle **CW mode on both legs** so you can work CW through the bird instead of SSB. CardSat sets CW on the uplink and downlink; on an inverting transponder the sideband flips but CW is CW on both ends, so you just zero-beat your downlink. The choice is **per channel** (it resets when you change satellite or transponder), shows a `CW` tag on the Track and large-font screens, and applies live whether the radio is already engaged or you turn it on afterward. On an FM bird the key is a no-op ("CW: linear birds only"). Available on both the Track and large-font screens.
- `r` — turn radio output **on/off** (sets modes and begins Doppler service).
- `o` — turn **rotator** pointing **on/off** (if a rotator is configured; sends live az/el to the selected backend, parks on stop). See §17.
- `p` — open the **Polar** plot.
- `a` — open the **point-here arrow**: a big compass arrow to the satellite's azimuth with an elevation bar, for aiming a handheld antenna; radio and rotator keep running underneath. On a board with the IMU (the Cardputer ADV), **`g`** turns on a **pointing aid**: aim the device north and press **`n`** to set the reference, and from then on a **cyan needle** on the compass rose shows where you are physically pointing (tracked by integrating the gyroscope) and a **cyan tick** on the elevation bar shows the device's tilt (from the accelerometer). Rotate until the cyan needle overlaps the green target arrow and tilt until the tick meets the target — the bottom line gives live "Turn L/R, Tilt up/dn" guidance and reads **ON!** when you're within a few degrees of both. The heading is gyro-only (no compass), so it **drifts** over a minute or two; press **`n`** again any time to re-zero north. It's an aid for rough hand-aiming, not a calibrated rotator.
- `z` — open the **large-font readout** (see below). A quick way to read RX/TX, az/el and the AOS/LOS countdown at arm's length; radio and rotator keep running.
- `y` — toggle **tilt tuning** on/off on the fly (only if the board has the sensor and the feature is otherwise available; see Settings). Lets you flip accelerometer tuning on for a tricky stretch and back off without opening Settings.
- `v` — start/stop a **voice memo** (recorded to the microSD card via the ADV's built-in

mic); radio and rotator tracking keep running while you record.
- `q` — for ~60 s after LOS a prompt offers **deep sleep until the next favorite pass**; press `q` inside that window to take it (see §12).
- `g` / `w` / `e` — show the **workable grids / US states / DXCC** reachable under the footprint **right now**, refreshing live; radio and rotator control keep running while you look. (The same three lists are also available per-pass from the Passes screen.)
- `f` — open **Manual mode** (frequency calculator; see below). Useful when you have no CAT-controlled radio and are tuning by hand.
- **ENTER** — save the current calibration **for this satellite**.
- `` ` `` — leave the Track screen, **returning to the screen you came from** (Passes, Next Passes, or Home). If the radio and/or rotator are engaged, they **keep tracking in the background** — the loop's Doppler and rotator service runs regardless of which screen you're on, so you can browse passes, log a QSO, check the map, etc. while the rig stays Doppler-corrected and the antenna stays pointed. A green **RAD** / **ROT** / **R+R** tag in the header on those other screens shows tracking is still live (and that the rig may be transmitting). Use **`r`** to stop the radio and **`o`** to stop (and park) the rotator. When neither is engaged, `` ` `` is just a plain exit. Background tracking stays locked to the satellite it was started on: if you **select a different satellite** while it's running (for example by opening another bird from the Satellites list), tracking **stops and the rotator parks** rather than silently retargeting the rig — re-engage with `r`/`o` once you've opened the new satellite's Track screen.

TUNE-mode keys: `,`/`/` tune down/up the passband, `s` cycle step (100/1000/5000 Hz), `x` re-center. CAL-mode keys: `,`/`/` trim downlink, `;`/`.` trim uplink, `s` cycle step (10/100/1000 Hz), `x` zero. See §9 and §10. When **Tilt tuning** is enabled (see Settings), in TUNE mode on a linear bird you can also roll the Cardputer left/right to move through the passband — a small **TLT** marker on the passband line shows it's armed.

Large-font readout (z from Track)

A stripped-down, glanceable view for operating a pass without squinting: the **RX** and **TX** frequencies in the largest digits that fit, with **Az/El** and the active **Doppler tune mode** below them. Small badges show **RAD**/**ROT** (radio and rotator on/off), **TILT** if tilt tuning is armed, and the transponder index. The bottom line echoes the tune mode (**FULL / DL / UL / TUNE / CAL**) and, on a linear bird, the passband position — so the big view follows whatever Doppler tuning option you selected on the Track screen.

The radio, rotator and Doppler tracking keep running exactly as on Track — this is just an alternate view of the same live session. All the in-place Track controls work here too: `,`/`/` tune through the passband, `s`/`x` step/recenter, `m` TUNE/CAL, `d` cycle tune mode, `t` next transponder, `r` radio, `o` rotator, `y` tilt, and `l` to log a QSO. Press `z` or `` ` `` to return to Track.

(The earlier AOS/LOS countdown was removed from this view to give the frequencies more room; the countdowns remain on the regular Track and Passes screens.)

Manual mode (f from Track)

Manual mode is for operating without a CAT-controlled radio: it shows the same live data as Track but **never commands a radio or rotator**. Instead, you fix one leg — the frequency you'll hold on your own rig — and it shows the **Doppler-corrected frequency to tune the other leg to**, updated live, with your saved calibration applied.

The two frequency rows are marked **HOLD** (the leg you park on your radio, shown at its nominal value without Doppler) and **TUNE>** (the leg you must follow, shown Doppler-corrected). Press `u` to toggle which leg is fixed:

- **Linear birds** — fixing the downlink shows the uplink to transmit (and vice versa). `,`/`/` move the fixed frequency through the passband, `s` cycles the step, `x` recenters. The HOLD leg's parked value stays put; the TUNE> leg follows with Doppler. Because the goal is to keep hearing *yourself* on the fixed leg, the TUNE> leg is corrected for the **round-trip** Doppler — it cancels both the uplink and downlink shift, in **either** direction (hold the downlink and tune the uplink, or hold the uplink and tune the downlink), since where your own signal lands depends on where the bird heard your transmission. (Fixing a single satellite-passband point instead — the convention used when CardSat drives a real radio on the Track screen — lets the downlink drift on the ground; here the goal is the opposite, a stationary fixed leg you can park on.)
- **FM birds** — pick which leg is fixed with `u` (typically the **VHF** leg, which has little Doppler and is the one you park). The other (UHF) leg shows the Doppler-corrected frequency to chase. FM legs are independent channels, so no round-trip correction applies. A hint line notes which leg is fixed and whether it's VHF.
- **Downlink-only birds** — just the computed downlink to tune your receiver to.

`m` toggles **CAL** (trim the same per-satellite calibration as Track), `t` cycles transponder, `l` logs a QSO, `p` opens the polar plot, `g` opens live workable grids — and **the log, polar, and grid screens all return here** rather than to Track. **ENTER** saves calibration for this satellite. `` ` `` or `f` returns to Track.

Press `z` for a **large-font version of the Manual calculator** — the HOLD and TUNE legs in big digits, with the fixed/derived leg labeled, for reading at arm's length in the field. The in-place keys (`u` swap leg, `m` CAL, `,`/`/` tune, `s`/`x`, `t`) work the same there; `z` or `` ` `` returns to the normal Manual view. If the board has the motion sensor and **Tilt tuning** is on, you can roll the device to move through the passband in Manual mode too, exactly as on Track.

Polar

A sky plot centered on the zenith: N/E/S/W cardinal labels, elevation rings at 30°/60°, the horizon as the outer ring. A green dot marks the satellite; a yellow glyph marks the **Sun** (when it's above the horizon). The satellite's **path across the sky for the current pass** is drawn as a cyan arc with **A** (AOS) and **L** (LOS) markers and a white arrowhead showing the **direction of travel**. When the bird is below the horizon the arc shows the **next** pass instead, and the readout gives that pass's AOS time. The right-hand readout also shows az/el, range, range-rate, **Sun az/el**, and whether the satellite is **SUNLIT** or in **ECLIPSE**. `l` logs a QSO and `v` starts/stops a **voice memo** (microSD, ADV) without leaving the plot. `p`, ENTER, or `` ` `` return to Track.

OSCARLOCATOR (k from Satellites)

A live **azimuthal-equidistant** plotting board — the classic OSCARLOCATOR view — reached with `k` on the **Satellites** screen. Unlike the **Polar** screen (which plots a single pass in az/el over your sky), this plots the satellite's **sub-point on the Earth** in real time, the way a paper OSCARLOCATOR dial does. It complements the tabular **EQX table** (`e`): the EQX table lists the equator-crossing times and longitudes, and this view shows the resulting geometry.

Two modes, toggled with `m` (the view **opens in polar mode** by default):

- **Polar** (default) — a pole sits at the center and the satellite is plotted by latitude (distance from the pole) and longitude (angle). CardSat **automatically chooses the North or South polar sheet** from the satellite's current hemisphere and **flips between them live** as the bird crosses the equator, so the satellite always stays on the visible chart.
- **QTH-centered** — your station sits at the center, with range/elevation rings around it. The satellite is drawn at its true bearing and great-circle distance from you, so you can see at a glance where it is relative to your horizon and how far away it is.

Both modes draw a coarse coastline, a lat/lon graticule, the satellite marker (yellow when sunlit, cyan in eclipse) and its **ground footprint** circle. They also draw a **QTH range ring** as a dashed amber circle — the satellite's footprint radius at the bird's mean altitude, centered on your station, so anything the satellite's sub-point reaches inside that ring is a workable pass (dashed so it stays distinct from the same-size instantaneous footprint) — and the **ground-track arc** in blue: the satellite's full sub-point track across the disc — a complete orbit's worth of ground track centered on the current time, so it sweeps the whole projection like a real OSCARLOCATOR (portions outside the plotted radius are clipped). Green and orange markers show where the current pass's **AOS** and **LOS** fall along that track, and a white arrowhead at the satellite shows the direction of travel. The right-hand readout shows the sub-satellite lat/lon, az/el, slant range, altitude, and sunlit/eclipse state; an **(off chart)** note appears when the satellite is outside the plotted area (e.g. below the horizon in QTH mode, or in the other hemisphere in polar mode just before the view flips). `m` toggles the mode; ` returns to Satellites.

3D Globe (3 from Satellites)

A three-dimensional wireframe Earth, reached with `3` on the **Satellites** screen. Where the world map is a flat projection and the OSCARLOCATOR is a flat azimuthal disc, this is an **orthographic globe** — the Earth as a sphere you're looking at from space. Only the near hemisphere is drawn; anything on the far side of the Earth is hidden behind the curve, just as it would be in real life.

The globe **auto-follows the selected satellite**: it rotates continuously so that satellite's sub-point stays at the center of the disc. Drawn on it:

- a **graticule** (meridians and parallels every 30°) and a coarse **coastline**, both clipped at the limb so only the visible hemisphere shows;
- a **day/night terminator** in yellow — the line dividing the sunlit and dark halves of the Earth, computed from the Sun's current sub-point;
- your **QTH** as a white cross;
- **all your favorites** as small dim-green dots (those on the near hemisphere), with the **selected satellite** drawn larger and centered (yellow when sunlit, cyan in eclipse);
- the selected satellite's **ground footprint** as a green circle, which wraps around the limb of the globe naturally as the bird nears the horizon;
- the selected satellite's **ground-track trail** in blue — a full orbit's worth of sub-points projected onto the sphere, so you can see where the bird came from and where it's heading;
- optionally a **second (DX) location**: press `g` and enter a Maidenhead grid square (e.g. `IO91`). The DX station is drawn as an orange box with its own footprint ring at the selected satellite's altitude; where that ring overlaps the satellite's own footprint is the **live mutual-visibility region** for that bird — the area from which both you and

the DX station could work it at once. Press `G` (shift-g) to clear the DX location.

The right-hand readout shows the selected satellite's sub-lat/lon, az/el, altitude, sun-lit/eclipse state, the view mode (**follow** or **free**), and either the DX grid (when set) or the favorite count. **Arrow keys** turn the globe by hand — this drops into **free-look** so you can inspect any part of the Earth; **ENTER** re-snaps to auto-follow. ` returns to Satellites.

Sat-to-sat visibility (2 from Satellites)

Press `2` to find the windows when the selected satellite **and a second satellite** are **both above your horizon at the same time** over the next five days — for cross-satellite relay experiments, or simply to plan back-to-back working of two birds on one outing. The second satellite is taken from your **favorites** list.

You first land on a **pick screen** that shows the currently-selected second satellite and its position in your favorites (e.g. "3 of 7"). Press `n` / `p` to step forward/back through favorites — this is **instant**, because no search runs while you are choosing. When you have the satellite you want, press **ENTER** (or `r`) to run the window search. After results appear, `n`/`p` return you to the pick screen to choose a different second satellite, so you never sit through a calculation just to scroll past a bird you didn't want.

Each row is one overlap window, showing its **start time** (**UTC**), **duration**, and the **peak elevation of each satellite** during the window (the first in green, the second in cyan) so you can judge how usable it is. `;`/`.` scroll; ` returns to Satellites.

The search runs in the background with a **progress bar** while it scans, so the screen stays responsive (the back key works throughout) and always finishes.

Both satellites are evaluated to your **0° horizon**. A window with very low peak elevations on either bird may be hard to use. The search looks three days ahead and lists up to sixteen windows.

Voice memo (v, SD card required)

Figure 4: Voice memo recording

Press `v` on the **Track**, **Manual**, **large-font readout** (Track or Manual), or **Polar** screen to record a short spoken note — for example "worked W1ABC, good signal" during a pass — without leaving the screen. A red **REC** badge with a countdown appears top-right while recording, and **radio, rotator, and web control keep running** throughout (the memo is captured one small block at a time between the normal tracking updates). Press `v` again to stop, or it stops automatically at the 30-second cap.

Starting a memo from an operating screen also appends a **log stub** — time, your call and grid, the satellite, mode, and the live frequencies, with the callsign left blank and a note naming the memo file — so the contact can be completed after LOS instead of typed mid-pass. Unwanted stubs delete like any entry (`x` twice).

Memos are saved as 16 kHz mono WAV files under **`/CardSat/audio/`** on the SD card, named by UTC timestamp and (when a satellite is selected) the bird being tracked, e.g. `memo_20260617_203145_AO-91.wav`. **An SD card is required** — with no card, `v` reports "Memo: SD card required" and does nothing.

Voice Memos browser (Log → Voice Memos). Memos can be reviewed on-device from the **Log** menu's **Voice Memos** entry. The browser lists every memo on the card **newest first**, showing each one's date, time, the **satellite** it was recorded on (or `-` for a standalone memo), and its length. Keys:

- **ENTER** — play the selected memo through the speaker. Playback streams from the SD card a block at a time; **any key stops** it early.
- **`n`** — record a **new standalone memo** right from the browser, not attached to any satellite. A `REC` timer shows while it records; **any key stops and saves**.
- **`d`** — delete the selected memo (with a confirm prompt).
- **`r`** — refresh the list; ` returns to the Log menu.

You can still copy the WAV files off the card on a computer as well.

> **Cardputer ADV — voice memo requires building against ESP-IDF 5.4.x.** The ADV captures through an ES8311 codec (the original Cardputer used a PDM mic), and the codec's record clock is only driven correctly when CardSat is built against the **ESP-IDF 5.4.x** toolchain. In practice that means the **Espressif "esp32" Arduino core 3.2.x** (which bundles IDF 5.4.2). On the newer 3.3.x core / IDF 5.5.x the mic path produces a constant (silent) signal — an upstream regression, not a CardSat bug (see espressif/esp-idf#18621). Everything else in CardSat builds and runs on either toolchain; only voice memo is affected. To confirm your build is correct, the boot log should report an IDF 5.4.x SDK version. Recording uses M5Unified's `M5.Mic`, so an up-to-date M5Unified is recommended.

Workable grids (g)

The 4-character Maidenhead grid squares currently inside the satellite's footprint — every grid from which a station could work the bird at the same time you can. Reached two ways:

- `g` from **Passes** — the **union** of grids covered across the selected pass (sampled about once a minute from AOS to LOS), computed once on entry.
- `g` from **Track** — the grids under the footprint **right now**, refreshed about every 3 s. Radio and rotator tracking keep running while you view it.

Grids are listed six per row in alphabetical order. The **workable count** is shown on its own cyan line at the top of the list (the header had no room for it), and when the list spills past one page that line also shows the visible window, e.g. `1370 workable  (1-48)`. `;`/`.` scroll a row at a time and `{`/`}` page. Coverage is computed with a per-grid bitset, so there is **no cap** on the number of grids — it works for any amateur satellite, including high orbits (a ~2500 km bird floods roughly 4500 grids). ` returns to whichever screen opened it.

Press **`f`** to set a **prefix filter** that narrows the list to grids beginning with what you type: `EM` shows every workable EM grid, `EM2` narrows to EM2x, and `EM21` shows just EM21 if it's workable. The filter is entered in upper case (the same capitalization rule as all grid entry — lower case is accepted and converted), and only well-formed Maidenhead characters are kept (two field letters A–R, then two digits 0–9). With a filter active the count line shows it, e.g. `EM2: 9 of 1370`, and **c** clears it. The filter persists as you move between passes until you clear it.

Workable US states (`w`)

The US states (and DC) currently inside the satellite's footprint — the state equivalent of the workable-grids screen, reached the same ways and in the same modes:

- `w` from **Passes** — the **union** of states covered across the selected pass, computed once on entry.
- `w` from **Track** (or **Manual**) — the states under the footprint **right now**, refreshed about every 3 s, with radio and rotator tracking still running.

States are listed by their two-letter USPS code, six per row, alphabetically, with the same cyan workable-count line and `;/.` · `{/}` scrolling as the grids screen. Membership is decided by a point-in-polygon test against bundled **simplified** state boundaries (about 0.1°/11 km resolution), so a footprint grazing a state line may briefly claim both neighbors — fine at footprint scale, where both are in fact workable. AK, HI and DC are included. ` returns to whichever screen opened it.

Workable DXCC (`e`)

The DXCC entities currently inside the footprint — the same idea again, for **DXCC chasing**, reached the same ways (`e` from Passes for the per-pass union, or from Track / Manual for live now). Entities are listed by common prefix (e.g. `DL`, `JA`, `VK`, `9V`), five per row, with the same workable-count line and scrolling.

Coverage and accuracy. All **340 current DXCC entities** are included, via a hybrid model: the major countries use simplified boundary polygons (so the right country is picked from the footprint geometry), while the long tail of islands and micro-entities is represented by each entity's reference coordinate from `cty.dat` and counted as workable when that point falls within the footprint (plus a small claim radius). Country borders are coarse, so a footprint near a border may list a neighbor too, and a single-point entity is claimed as a unit rather than by exact shape. Treat this as **chasing guidance** — which entities are reachable on the pass — and confirm the actual entity worked from your own log. ` returns to whichever screen opened it.

Location

- `e` / `o` / `a` — edit latitude / longitude / altitude.
- `g` — set position from a **Maidenhead grid** square.
- `p` — enable/disable **GPS**.
- `s` — cycle the **GPS source** (Grove 9600, Grove 115200, Cap LoRa868, Cap LoRa1262).
- `c` — set the **UTC clock** manually (`YYYY-MM-DD HH:MM:SS`).
- `v` — open the **live GPS position** screen (below).
- **ENTER** — open the **GPS sky plot** (GNSS satellites in view).
- ` — back (applies the position to the predictor).

8.0.30.1 Live GPS position (`v`) A full-precision readout of where you are, for rovers, portable ops, and grid-line activations. It shows your **latitude and longitude in degrees-minutes-seconds** to 0.001″ (about 3 cm — finer than the fix itself), the same coordinates in **decimal degrees** to six places, your **altitude**, your **Maidenhead grid**, and — from a live fix — your **ground speed** (km/h and knots) and **course over ground** (degrees true with a cardinal label). The top line shows the fix status, the number of satellites used, and the HDOP. The grid recomputes live as you move, so you can watch it flip as you cross a grid boundary. ` returns to Location.

LoRa Messages (Home → Messages)

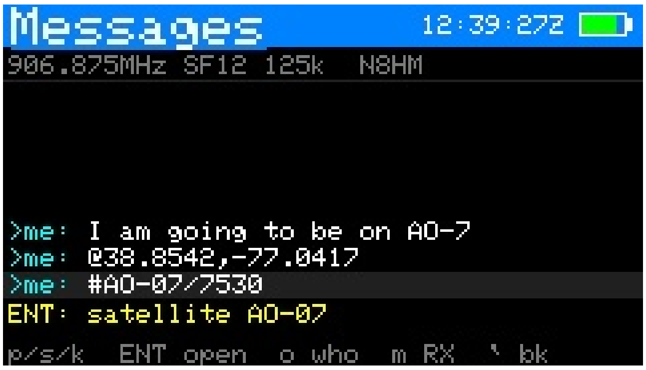


Figure 5: Messages

CardSat-to-CardSat **text messaging** over LoRa, using the M5Stack **Cap LoRa** (**SX1262**) module. It is a **broadcast group chat**: every CardSat tuned to the same frequency, spreading factor and bandwidth sees every message — there is no addressing or routing, which keeps it simple and reliable for a group on an outing, a club net, or a SOTA/portable activation.

> **Hardware-verified.** Two-way LoRa text messaging is confirmed working on the M5Stack Cap LoRa (SX1262), tested against a LilyGo T-LoRa unit running the companion CardSat Pager firmware. It's built into the standard binaries (RadioLib is a required build dependency). **Know your band's rules** — pick the **LoRa region** preset that matches where you operate (see Settings below):
>
> - **US (default)** — the **33 cm amateur band (902–928 MHz)**. The US 70 cm band is held to ~100 kHz occupied bandwidth, which is tight for 125 kHz LoRa, so 33 cm is the home for amateur LoRa here. Default **906.875 MHz**, clear of the busy 915 MHz ISM center.
> - **EU** — the **70 cm amateur band (430–440 MHz)**. Default **433.775 MHz**, the LoRa-APRS standard, at 125 kHz.
> - **Japan** — the **430 MHz amateur band (430–440 MHz)**. Default **431.000 MHz**. (Japan's 920 MHz band is ISM, with certification rules — not amateur — so amateur LoRa belongs on 430 MHz.)
>
> These are starting points within each band, not the only legal frequencies; you can still set any carrier 150–960 MHz by hand, and you remain responsible for operating within your license and local regulations.

Using it. Set your callsign (Settings → Station, or the QRZ screen), enable **LoRa msg** in Settings → Network / data, pick your **LoRa region**, then fine-tune the frequency/SF/bandwidth to match the other stations. Open **Messages** from the Home menu. Press `n` to write a message on the full keyboard and **ENTER** to send; `;`/`.` scroll back through history. Your own messages show in cyan as `>me`; received messages show in green by the sender's callsign. A message too long to fit on one line wraps onto a second, indented line so nothing is cut off. The top line shows the current frequency, SF, bandwidth and your callsign. History holds the most recent 24 messages (a fixed ring — no SD card needed, lost on reboot).

Selecting a message. Scroll with `;`/`.` to move the selection through the list. The **newest message always sits on the bottom line** (chat style — older messages scroll up off the top), and the selected message is highlighted. When a message is selected, **ENTER acts on it** (see the next section); if it contains no actionable content, ENTER simply reports that.

Actionable messages (position, satellite, sked). A message is still plain text on the air — the wire format is unchanged, so this interoperates with any other CardSat — but CardSat recognizes three markers anywhere in a message's text and turns a selected message into an action with **ENTER**:

- **`@lat,lon`** (e.g. `@38.8542,-77.0417`) — a **position**. ENTER opens a **bearing compass**: a north-up dial with an arrow to the sender, plus the great-circle **distance and bearing** and the message's age. The Cardputer ADV has no magnetometer, so this is a *computed bearing* (degrees from true north) for you to orient by, not a live magnetic compass.
- **`#SAT`** (e.g. `#SO-50/27607`) — a **satellite**. ENTER opens a **satellite detail** screen with the NORAD ID and the **next pass** for your location. CardSat appends the satellite's **NORAD catalog number** after the name (`name/norad`) so the far end can resolve the right bird even if its list uses a different display name — a satellite that one station calls `RS95S` and another calls `QMR-KWT2` shares the same NORAD number, so the reference still lands. A message with just a name (no `/norad`) still works by name.
- **`!SAT YYYY-MM-DD HH:MM`** (e.g. `!AO-91/43017 2026-07-04 18:30`) — a **schedule proposal**. ENTER opens the **sked editor pre-filled** with the satellite (resolved by NORAD, so it maps to your name for it), date and start time (and the sender's callsign), so you can review and save it.

When the selected message carries one of these, a **yellow hint** on the line above the footer shows exactly what ENTER will open.

Sending an actionable message. Three keys on the Messages screen compose and send these markers for the **currently selected satellite** (from the Satellites screen) and **your own location**:

- **`p`** — send your **position** as `@lat,lon` (from your set location or GPS).
- **`s`** — send the current **satellite** as `#name`.
- **`k`** — propose a **sked**: you're prompted for a date, then a time, and CardSat sends `!SAT date time`. The date field is pre-filled with today (UTC) as a starting point.

These compose from your own state, so they work even on an empty message list. The other station simply receives the text; if they're running CardSat, their copy decodes it the same way yours does.

Station roster (`o` key). Press **`o`** on the Messages screen to open the **roster** — a list of every station heard reporting a position, newest first. Each entry shows the **callsign**, its **Maidenhead grid**, the **distance and bearing** from your location (when you have a fix), the last **signal** (RSSI), and how long ago it was heard. Scroll with `;`/`.`, press **ENTER** to open the **bearing compass** to the selected station, or press **`p`** to send your own position (a presence ping) so others can add you to theirs. The roster is built from ordinary `@lat,lon` messages — no special packet — so pressing `p` on any CardSat both announces you and populates everyone else's roster; the grid is computed locally from the shared lat/lon, so the on-air format is unchanged. (The roster is held in memory and clears on reboot.)

Sharing satellite elements over LoRa (`L` key). From the **Satellites** screen, press `L` to broadcast the selected satellite's **GP orbital elements** to any nearby CardSat over LoRa — handy in the field for pushing a freshly-fitted set (e.g. a pre-launch state vector from the *State vector → GP* tool, which also offers `L` on its result screen) to a rove group without WiFi. The element set is split into a few small chunks with a checksum and sent one frame at a time; the on-air status shows the progress. On the receiving unit, CardSat reassembles the object, verifies the checksum, and shows an **"Import satellite?"** prompt with the sender, name, NORAD, and key elements — press `y` to add or update it in your GP data, or `n` to decline. Nothing is imported without your confirmation, and a corrupted transfer is rejected rather than accepted (ask the sender to resend). This rides on the **untested** LoRa path — treat it as experimental, and confirm any imported orbit against a known pass before relying on it. Requires LoRa enabled on both units on the same frequency/spreading-factor.

Automatic position reply. With **Auto position reply** on (Settings → Network / data, **off by default**), CardSat answers a received position report with its own `@lat,lon` automatically — handy for a net where everyone wants to see who's where without each operator pressing `p`. Because it broadcasts your location, it is opt-in. Two safeguards keep it well-behaved: a short random delay before replying (so simultaneous listeners don't all transmit at once), and loop guards — CardSat won't auto-reply to the same station more than once every few minutes, and never more than once every 30 seconds overall — so two auto-replying units can't ping-pong endlessly.

Settings. In Settings → Network / data: **LoRa msg** on/off (turning it on brings the radio up); **LoRa region** (US 33cm / EU 70cm / JP 430 — selecting one seeds a legal default frequency + 125 kHz bandwidth for that band); **LoRa freq** (arrow-adjust in 100 kHz steps, or ENTER to type an exact MHz value, 150–960 MHz); **LoRa SF** (7–12; 12 = maximum range and sensitivity, the default; lower = faster but shorter range); **LoRa BW** (62.5 / 125 / 250 kHz); **LoRa TX pwr** (0–22 dBm); **Msg notify** (off / banner / banner+beep, described below); and **Auto position reply** (off / on, described above). Both ends must use the same frequency, SF and BW to hear each other.

Notifications. CardSat listens for messages continuously in the background (the receiver is left on whenever LoRa is enabled), so a message can arrive while you're on any screen. When one does, an **envelope badge with an unread count** appears in the header (top, just left of the battery) on every screen, and — depending on the **Msg notify** setting (Settings → Network / data) — a brief banner shows the sender's callsign, optionally with a short beep. Opening **Messages** clears the badge. **Msg notify** options: *off* (silent badge only), *banner* (the default, a cross-screen banner on arrival), or *banner+beep* (adds an audible chirp — handy when the screen is parked, but off by default so it can't surprise you mid-pass). In the Charge / Sleep screen the badge updates silently and no banner or beep fires, keeping that mode minimal.

The build. LoRa is a **standard, always-built feature** — official binaries ship with it compiled in (`CARDSAT_HAS_LORA` defaults to `1`), so **RadioLib is a required build dependency**: install it via the Arduino Library Manager (search "RadioLib" by Jan Gromes) before compiling. The `#if CARDSAT_HAS_LORA` guards remain in the source only so the tree still compiles if you deliberately override the define to `0` in your own build flags; you should not need to.

> **If the single-file `CardSat.ino` fails to link** showing `dangerous relocation: l32r: literal target out of range`, that is an Xtensa toolchain limit: the very large single translation unit pushes RadioLib's code out of the reach of the `l32r` instruction's literal pool. CardSat mitigates this by constructing the SX1262 on

the heap (a pointer, not a static global) so its destructor/vtable stay out of the sketch's static-init code. If the single-file build still fails to link, **build from the modular `src/` folder instead** (use the PlatformIO build): there `lora.cpp` is its own small translation unit and the literal-range problem does not arise.

LoRa RX / hex monitor (Messages → `m`)

Figure 6: LoRa RX monitor

A general-purpose tool to receive and inspect **any** LoRa signal on the Cap LoRa (SX1262) — not just satellites, and with no network or satellite data involved. Useful for checking whether a LoRa transmitter is on the air, what a beacon's bytes look like, and for peaking reception by ear. Press `m` on the Messages screen to open it (LoRa must be enabled in Settings; it works even with no messages in the log).

It takes over the shared radio while open, so CardSat messaging is paused until you leave — the two share one SX1262 and one channel.

Config screen. Set the full SX1262 receive parameter set, then press **ENTER** to start receiving:

- **Freq** — carrier. Press **ENTER** on this row to type the frequency directly in **MHz** (e.g. `433.775`) on a numeric entry screen. For fine tuning, `,//nudge it by the current step and **s`'** cycles the step size (1 kHz → 10 kHz → 100 kHz → 1 MHz). (ENTER on any other row starts receiving.)
- **SF** — spreading factor 7–12.
- **Bandwidth** — the full SX1262 LoRa ladder: 7.8, 10.4, 15.6, 20.8, 31.25, 41.7, 62.5, 125, 250, 500 kHz.
- **Coding** — coding rate 4/5–4/8.
- **Sync** — sync word (0x00–0xFF; **0x12** is the LoRa "private" default, **0x34** the public/LoRaWAN value).
- **Preamble** — preamble length in symbols (4–64).
- **CRC** — whether a payload CRC is expected (on/off).

All of these are **saved and restored** across reboots, and are kept separate from the LoRa Messages parameters, so tuning the monitor never changes your messaging setup.

Monitor screen. A classic hexdump of received frames that **updates automatically as frames arrive** (it is not keypress-driven). Under the standard title bar, a status line shows the live frequency, SF, BW, CR, a **PAUSE / RX / `--`** state indicator, and a packet counter. Each frame is shown **16 bytes per row: a line of hex with the ASCII characters directly beneath it** (non-printable bytes shown as `.`), with a byte offset at the left. The newest frame is shown; `;/.` scroll back and forth through the last dozen frames.

On a busy channel, press `p` to **pause** the display so you can read a frame in detail: the radio keeps receiving into the buffer while paused (a `+N new` indicator counts what arrived), and the frozen frame stays put until you press `p` again to resume. You can also **tune live** without going back: ,//nudge the frequency by the current step, **f** cycles the step, and **s** / **b** / **c** cycle SF / BW / CR — each change re-applies to the radio immediately, so you can sweep parameters until frames appear when you don't know a signal's exact settings. **x** clears the buffer. **`` "** (back-tick) or **ESC** returns to the config screen (the radio keeps running); pressing it again on the config screen leaves the mode and restores messaging.

> **Receive-only.** This is a monitor — it never transmits. As with all RF features, use an antenna appropriate to the band you tune (e.g. a 70 cm antenna for 433 MHz); the Cap LoRa's SX1262 has no band-pass filter, so any frequency the chip supports works with the right antenna. Frames are stored up to 64 bytes for display; a longer on-air frame is shown truncated (its true length is still reported). LoRa modulation only — FSK/GFSK signals are not received.

Update

- `k` or **ENTER** — update GP data from your configured source and sync the clock (NTP). The same action also refreshes the AMSAT OSCAR **activity marks** shown on the Satellites list, the **space-weather** data (solar flux + Kp), **and** the terrestrial **weather** for your site, so one press brings everything current. The Update screen notes this so it's clear `k` does more than GP.
- `f` — **fast update**: refresh the orbital elements (GP), the AMSAT activity marks (a single bulk fetch, so it's included), and the transponder data for your **favorites only** — skipping the space-weather and terrestrial-weather fetches that `k` also pulls. This is the quick way to bring your regularly-worked birds current without the longer full refresh — handy in the field. (If you haven't marked any favorites, it refreshes the currently active satellite instead.)
- `a` — fetch and cache **all** transponders for offline use. This runs **in a single session** — no reboots. CardSat fetches every satellite's transponder data in one pass, showing a running count on the Update screen (e.g. "TX 24/90: AO-91") and finishing on "Cached all N transponders". The whole pass takes a few minutes on a reasonable WiFi link; a per-satellite retry absorbs the occasional transient miss. (Earlier firmware split this across automatic reboots to work around a TLS memory limit; the BearSSL migration removed that limit, so it now completes in one go.) Satellites with no transmitters in the SatNOGS database are cached as an empty list, which is expected.
- `w` — connect WiFi only (no download).
- `` ` `` — back. Diagnostics print to the serial monitor at 115200.

Settings

Settings are grouped into six submenus — **Radio / CAT**, **Rotator**, **Passes / alerts**, **Display / sound**, **Station / logging**, and **Network / data** (each shown with its item count). `;`/`.` move; ENTER opens a submenu (or, inside one, edits a text field or runs an action); `,`/`/` change an adjustable row; `` ` `` backs out to the submenu list, then home. Press `h` anywhere for the on-screen key reference. The notable rows:

Row	Action
Radio	,// select model (auto-sets address + baud)
CI-V addr	ENTER → edit (hex); Icom only
CI-V wiring	,// cycle **TX/RX (G2/G1)** / **1-pin G2** / **1-pin G1** — Icom wired CI-V only; single-pin is **confirmed on an IC-821** but still needs correct 5 V/3.3 V level interfacing, and TX/RX is the simplest path (see docs/interfaces/CIV_SINGLE_PIN.md)
CAT baud	,// cycle 1200…115200 (incl. 57600) — applies to all radio protocols
Min pass el	,// 0–30°
Decay solar	,// cycle assumed solar activity **mean → min → max → auto** for the orbital-analysis decay estimate (changes the headline number and the bracket). **auto** uses the live F10.7 flux fetched with GP data
Weather temp	,// toggle the **Weather** temperature unit **°C ↔ °F**. Under *Display / sound*.
Weather wind	,// cycle the **Weather** wind unit **km/h → mph → m/s**. Independent of the temp unit.
Weather pressure	,// toggle the **Weather** pressure unit **hPa ↔ inHg**. Independent of the others.
WiFi SSID	ENTER → edit · **s** → scan for networks and pick one
WiFi pass	ENTER → edit
WiFi 2 SSID	ENTER → edit an **optional second network** tried if the first fails (field use: a second router, or a phone hotspot). Leave blank to disable
WiFi 2 pass	ENTER → edit
Save & test WiFi	ENTER → connect and report OK/FAIL (tries the primary network, then the second if set)
AOS alarm	,// or ENTER toggle on/off
IR pass beacon	,// or ENTER toggle on/off — also flash the built-in IR LED on each pass alert, with a distinct flash count per event (see §12). SD not required; off by default

Row	Action
Rotator (+ type / host / port / baud / ranges / offsets / deadband / park / pre-point)	`,`/`/` adjust, ENTER edits host/port. **Rotator** type cycles **None → GS-232 → rotctl (net) → PstRotator → Yaesu (direct) → Easycomm I → Easycomm II → Easycomm III → SPID Rot2Prog** — select **None** if you have no rotator (this replaces the old separate on/off row); see §17
GP source	ENTER → **source picker**: AMSAT (default), any CelesTrak JSON-PP category (Amateur Radio listed first), or **Custom URL...** — see §14
VFO Type	`,`/`/` or ENTER toggle *Main Up/Sub Dn* ↔ *Main Dn/Sub Up*
Sat mode	`,`/`/` or ENTER toggle the rig's satellite mode on/off
CAT rate	`,`/`/` adjust the CAT update period in 10 ms steps (default 500 ms; soft-floored to what the CAT baud can service)
Downlink LO	ENTER → edit the **downlink transverter** local-oscillator frequency in MHz (`0` = no transverter). CardSat displays the real downlink but sends the rig **real − this** as its IF. See §10a
Uplink LO	ENTER → edit the **uplink transverter** local-oscillator frequency in MHz (`0` = no transverter). CardSat displays the real uplink but sends the rig **real − this** as its IF. See §10a
CAT delay	`,`/`/` adjust the pause after each command, 0–200 ms in 2 ms steps (default 70 ms; CI-V/Icom only)
Dopp FM band	`,`/`/` the FM-leg write deadband, 0–2000 Hz in 25 Hz steps (default 300 Hz). CardSat only re-sends an FM frequency once Doppler has moved it more than this — FM's wide passband absorbs the rest, so a loose value avoids needless CI-V chatter
Dopp linear band	`,`/`/` the SSB/CW-leg write deadband, 0–1000 Hz in 10 Hz steps (default 50 Hz). Tighter than FM because linear modes need close tracking; near closest approach CardSat tightens this automatically

Row	Action
Dopp lead	`,`/`/` the predictive-lead cap, 0–100 ms in 5 ms steps (default 50 ms; `0` = off). On fast overhead passes CardSat can compute Doppler slightly ahead to mask CAT latency, tapering the lead to zero near closest approach. Raise it if your rig's CI-V is slow; set `0` to disable
Screen sleep	`,`/`/` cycle off / 30 s / 1 min / 2 min / 5 min — blanks the backlight after that idle time
Brightness	`,`/`/` adjust the active screen brightness in ~6% steps; previews live. Under *Display / sound*
Volume	`,`/`/` adjust the speaker volume, 0–100% in ~9% steps; plays a short blip at the new level as you adjust so you can hear it. Applies to the AOS alarm, game sounds, and voice-memo playback, and audio plays uninterrupted while CardSat is on the network (a fetch or upload no longer pauses the speaker). Saved and restored across reboots. Under *Display / sound*
Tilt tuning	`,`/`/` or ENTER toggle **accelerometer passband tuning** on/off. Shown as **n/a (no IMU)** on boards without one (only the Cardputer **ADV** has the sensor). When on, roll the device left/right in TUNE mode on a linear bird to move through the passband. Under *Display / sound*
My callsign	ENTER → enter your station callsign (stored uppercase); used in the log and ADIF `STATION_CALLSIGN`

Row	Action
LoTW DXCC / CQ zone / ITU zone / primary / secondary / IOTA	A chained set of pickers for the **LoTW upload** station location. **DXCC** opens a full entity list (subdivision-bearing entities grouped at the top, marked `>`; type to filter). **CQ/ITU zone** are typed. **Primary** (labeled state/province/oblast/prefecture/kunta per your DXCC) opens a gated picker — required for US/AK/HI, **(n/a)** for entities without one. **Secondary** is county (US) or city/gun/ku (Japan), gated by the primary; **(n/a)** elsewhere. **IOTA** (e.g. `NA-005`) is optional for any entity. Inside a picker: `;/.` move, ENTER selects, typing filters, ` clears/back. Under *Station / logging*. See §8 → LoTW upload.
Cloudlog URL / key / station ID	ENTER → enter your self-hosted **Cloudlog** (or Wavelog) base URL (`https://…` or `http://…`), a **read-write API key**, and the numeric **station profile id** to file QSOs under, for **Log → Upload to Cloudlog**. Under *Station / logging*. See §8 → Cloudlog upload.
Console to file	`,//` or ENTER toggle mirroring the **entire serial console** to `/CardSat/Logs/console.log` (default **off**; capped ~32 KB on internal flash, ~512 KB on SD). Exists because engaging USB takes the console away — see §16. Under *Station / logging*.
QRZ user / QRZ pass	ENTER → enter your QRZ.com username / password for the **QRZ Lookup** screen (requires a QRZ XML-data subscription). Password shown masked. Under *Network / data*.
Backup config+favs → SD	ENTER → copy config + favorites to `config.bak` / `favs.bak`
Restore config+favs	ENTER → restore them from the backup files
Reset all data	ENTER → type **ERASE** to wipe everything (red row)

Row	Action
CAT type (+ host / port / user / pass)	,// cycle **Wired CI-V → Icom LAN → rigctl (net) → rigctl (Grove) → USB serial** (USB serial absent only in builds with USB CAT compiled out; see §3). Icom LAN drives the **IC-9700** over RS-BA1 (see §3); **rigctl** drives a radio attached to a remote Hamlib **rigctld** server over TCP (set host/port to your rigctld; Hamlib's conventional port is 4532). Under *Radio / CAT*.
Rot wire	,// cycle the serial-rotator transport: **I2C bridge** (default) → **Grove G1/G2 → USB adapter**. Applies to GS-232/Easycomm/SPID; independent of **Rot type**. Grove needs CAT on USB/LAN and GPS off Grove — CardSat enforces the one-Grove-port rule both ways. See §17. Under *Rotator*.
Radio USB / Rot USB	,// pick which USB adapter each port uses when more than one is plugged in (**Auto** = the only adapter that is not the other port's). Each menu skips the other port's pick — but only while that other port is *actually* a USB device (a radio set to CI-V/LAN/None, or a rotator set to a non-USB type or None, no longer reserves an adapter). Selections persist by adapter serial number where one is reported, else VID:PID + address. Under *Radio / CAT* and *Rotator*.

Radio and rotator over USB at the same time (v0.9.64). With two USB-serial adapters plugged in, you can assign the radio to one and the rotator to the other, and run both at once over CardSat's single USB host — set each port's adapter in its **Radio USB** / **Rot USB** row. The engage order doesn't matter: whichever you turn on second waits for its own adapter to enumerate before reporting a problem — reliably so when each adapter has a unique serial number; serial-less adapters (many CH340 clones, identical pairs) are keyed by USB address, so re-check the selections after a reboot, hub change, or replug. With only one adapter present the two ports can't share it, and the picker says which port owns it. Turning either port off (or leaving its screen, or **stop-all**, or entering charge mode) releases that device and frees the memory it held; each re-engages on demand when you turn it back on. | Scan USB adapters | ENTER brings the USB host up and enumerates adapters (shows "n seen" after). A deliberate keypress: it **closes the serial console for the session**. Present in both *Radio / CAT* and *Rotator*. | | Rigctld server (+ port) | ,// enable a **rigctld server** so a PC (Gpredict, WSJT-X, a logger) drives the radio through CardSat over TCP (default 4532); VFOA=downlink, VFOB=uplink. Under *Radio / CAT*. | | Rotctld server (+ port) | ,// enable a **rotctld**

server so a PC (Gpredict, ...) drives **whichever rotator CardSat is configured for** through CardSat over TCP (default 4533). Under *Rotator*. | | Web control (+ port) | `,`/`/` or ENTER enable the **mobile web control page** served over WiFi (default port 80). When on and connected, the row shows the device's IP address to browse to (e.g. `192.168.1.42` — open it as `http://192.168.1.42`). Under *Network / data*. Plain HTTP on the LAN, **no authentication** — see §18. | | Rotator: manual control | ENTER opens a screen to jog az/el by hand with live read-back. For a **Yaesu (direct)** rotator this is also where you **calibrate the ADC**: it shows live ADC counts and you capture the axis endpoints with `1`/`2`/`3`/`4` (az 0 / az full / el 0 / el full) — see §17 and ROTOR_INTERFACE.md. Under *Rotator*. |

WiFi scan

Reached from **Settings** by pressing `s` on the **WiFi SSID** row. CardSat scans for nearby networks and lists them strongest-signal first, with each network's RSSI (dBm) and a `*` for secured networks.

- `;`/`.` — select a network.
- **ENTER** — use it: the SSID is saved and you're taken to password entry (open networks skip the password and return to Settings).
- `r` — rescan.
- `` ` `` — back to Settings.

Edit

A simple text entry box (for the fields above and manual GP/transponder/time entry). Type to append, **DEL** to backspace, **ENTER** to confirm, `` ` `` to cancel.

Adding a manual satellite (from Satellites → `n`): you'll be prompted in order for the **name**, **NORAD ID**, **epoch** (`YYYY-MM-DD HH:MM:SS`, UTC), then the GP mean elements — **inclination**, **RAAN**, **eccentricity** (e.g. `0.0006190`), **argument of perigee**, **mean anomaly**, **mean motion** (rev/day), and **BSTAR** (drag; `0` is fine if unknown). The satellite is stored with the downloaded ones and persists across GP refreshes.

> **Only have a TLE? Convert it with `tools/tle2gp.py`.** Manual entry asks for GP (OMM) mean elements, but some objects are still published only as a classic **two-line element set (TLE)**. The repo includes a small, dependency-free Python helper that decodes a TLE into exactly the fields above — handling the conversions that are easy to get wrong by hand: the TLE epoch (`YYDDD.dddddddd` → ISO date/time), the implied-decimal eccentricity, BSTAR's exponent notation, and the derivative scaling (TLE stores n-dot/2 and n-ddot/6; GP reports the full values). Run it on a file of one or more TLEs, or paste a 2–3 line set on standard input:
>
> ```
> python3 tools/tle2gp.py mysat.txt        # file with one or more TLEs
> python3 tools/tle2gp.py                   # then paste 2–3 lines, Ctrl-D
> python3 tools/tle2gp.py --json mysat.txt  # AMSAT-style GP/OMM JSON instead
> ```
>
> The default output lists each element with its label and units, ready to type straight into the `n` prompts in order (epoch, inclination, RAAN, eccentricity, argument of perigee, mean anomaly, mean motion, BSTAR). The `--json` form emits an AMSAT-style GP/OMM record, handy if you'd rather host the set at a

Custom URL GP source (see §7) than type it in. Either way the numbers are identical to what CardSat would have downloaded — the script only *re-packages* the same SGP4 mean elements a TLE already contains, so accuracy still decays with the element set's age and you'll want a fresh TLE periodically.

Adding a manual transponder (from Passes → `n`): you'll be asked, in order, for **Downlink low (Hz)**, **Uplink low (Hz, 0 = none/beacon)**, **Downlink high (Hz, 0 = single channel/FM)**. If you gave a downlink high above the low *and* an uplink, it's treated as **linear** and you'll also be asked **Uplink high (Hz, 0 = same bandwidth)**, **Inverting? (y/n)**, and finally the **Mode**. Single- channel entries skip straight to Mode. Manual transponders are stored separately from the SatNOGS cache so a GP/transponder refresh won't erase them. Up to **64 transponders** are held per satellite (SatNOGS plus your manual ones) — enough for even the busiest birds, such as the ISS (~50 transmitters on SatNOGS).

09

Doppler tuning and the One True Rule

A satellite's motion shifts both the frequency you **receive** (downlink) and the frequency the satellite **receives from you** (uplink). CardSat corrects both, continuously.

CardSat follows the AMSAT **"One True Rule"** (Paul Williamson, KB5MU): *tune both the transmitter and the receiver to achieve a constant frequency at the satellite.* In practice the firmware holds your chosen spot in the transponder fixed **as the satellite sees it**, and applies the Doppler correction to **both** the receive and transmit frequencies every cycle. The result: you tune somebody in, let go, and nobody drifts through the passband.

The corrections are:

```
RX = downlink × (1 − β) + calDl      (tune your receiver here)
TX = uplink   ÷ (1 − β) + calUl      (transmit here)
β  = range-rate ÷ c                  (c = speed of light)
```

where `downlink`/`uplink` are the satellite-side frequencies of your chosen passband spot, and `calDl`/`calUl` are your calibration offsets (§10).

Choosing your spot in the passband

On a **linear transponder** you can move through the passband two ways:

- **TUNE mode** (device keys): press `m` until the passband line shows `<TUNE>`. `,`/`/` move your operating point down/up; `s` cycles the step; `x` recenters. The `PB` line shows your offset from band center, the half-width, and `INV` for inverting birds.
- **Radio-knob mode** (One True Rule, the natural way): press `d` to cycle the tune mode. In **FULL** (passband line shows `<FULL>` in orange) just **turn the radio's tuning knob** to move around the passband — CardSat reads your downlink, works out where you are, and keeps both legs Doppler-corrected around that fixed satellite frequency. Let go and you stay put. Pressing `d` cycles on through **downlink-only** (`<DL>` — One True Rule on the downlink, uplink left alone), **uplink-only** (`<UL>` — only the transmit leg is corrected; handy when an SDR or second receiver handles the downlink, and it needs no frequency read-back so it works even on set-only rigs), **FULL on the uplink knob** (`<FULLu>` — the mirror of FULL: turn the **uplink** rig's knob and CardSat reads *that*, works out your passband spot from it, and keeps both legs corrected around it), and back to **hold-both** (`<TUNE>`, device-key tuning). The two FULL modes exist because in a two-radio station the knob you'd naturally reach for isn't always on the downlink rig — if your uplink radio is the one with the good VFO (and the downlink is a receiver you'd rather leave parked), pick `<FULLu>` and tune that. FULL, downlink- only, and uplink-FULL need a rig that reports frequency; the cycle skips them otherwise.

For an **inverting** transponder the uplink moves opposite to the downlink (tune the downlink up, the uplink goes down); for a non-inverting one they track together. CardSat handles the mapping automatically.

Tuning with the radio's knob

When you turn the radio's knob in FULL or downlink-only mode, CardSat distinguishes a deliberate dial move from the rig's own tuning-step rounding and read-back jitter using a **mode-aware threshold** (about 30 Hz on SSB/CW, 250 Hz on FM, never below the rig's tuning step). The moment it detects a real move it adopts your new spot and then **holds off its own Doppler writes to the downlink for a short grace window (~400 ms)** so it never tugs against the knob while you're turning — it resumes downlink correction once you let go. The uplink isn't connected to your knob, so it **keeps following** your new passband point immediately and tracks the move without lag. (In `<FULLu>` the roles are mirrored: the **uplink** is the knob it reads and holds off, and the **downlink** follows immediately — everything else, including the edge handling below, works the same way.) If tracking ever feels slightly sticky or slightly loose for your operating style, the `KNOB_MOVE_SSB_HZ`, `KNOB_MOVE_FM_HZ`, and `TUNE_GRACE_MS` constants in the firmware are what to adjust.

If you turn the knob **past either edge of the passband**, CardSat holds you at the edge (it can't operate you outside the transponder) and a flashing red **"OUT OF PASSBAND"** banner appears at the bottom of the screen, naming the edge you ran into (low or high), while it pulls the downlink back to that edge. Tune back inside and the banner clears. It's a transient nudge, not a persistent alarm — just enough to tell you why the dial stopped moving you through the band.

Sidebands

For linear transponders CardSat sets the rig's modes for you: **USB on the downlink, LSB on the uplink** (because an inverting linear transponder flips the spectrum, so an LSB uplink is heard as a normal USB downlink). The exception is any bird with an uplink or downlink **below 30 MHz (HF)**, which uses **USB up and USB down**. FM and single-channel birds use the transponder's own mode on both legs.

Working CW through a linear bird. Press `k` on Track to toggle **CW mode on both legs** of a linear transponder, so you can work CW through the passband instead of SSB. CardSat sets CW on the uplink and downlink; on an inverting transponder the sideband still flips but CW is CW on both ends, so you simply zero-beat your downlink. The choice is **per channel** — it resets when you change satellite or transponder — and a `CW` tag appears on the Track and large-font readouts while it's active. On an FM bird the key does nothing ("CW: linear birds only").

> **Frequency representation and display.** CardSat stores every frequency as a 32-bit count of hertz, so the highest frequency it can represent is about **4294 MHz** — comfortably above the amateur-satellite bands in use (2 m, 70 cm, 23 cm, 13 cm). Frequencies above that can't be entered or tracked. On screen, readouts **shed decimal places as the integer part grows** so they always fit the panel: sub-GHz birds keep five decimals (e.g. `145.99000`), while higher bands show fewer (`1296.500`, and so on). The large-font views trim one more decimal than the normal views to keep the big digits inside the screen. This only affects the *displayed* precision — the underlying tuning is always full-resolution.

Tilt tuning (accelerometer, opt-in, ADV only)

A third way to move through a linear transponder's passband, alongside the device keys and the radio knob. The Cardputer **ADV** has a motion sensor (the original Cardputer does not). When **Tilt tuning** is switched on under *Settings → Station / logging*, you can roll the device left and right to move through the passband instead of (or alongside) the `,`/`/` keys. It's deliberately a **rate** control, not an absolute one: a gentle tilt nudges slowly for fine work, a firmer tilt slews faster, and holding the device level holds the frequency. There's a dead-zone of a few degrees around level so a hand-held device doesn't drift, and the rate saturates past roughly 35°.

It only acts on the **Track**, **large-font**, and **Manual** screens, in **TUNE** mode, on a **linear** bird — everywhere else it does nothing. A **TLT** (Track / Manual) or **TILT** (large-font) marker shows when it's armed. On a board without the sensor the setting reads **n/a (no IMU)** and can't be turned on. Tilt tuning is off by default; many operators will prefer the keys, since tilting the device also moves your antenna and your eyes — it's offered as an option, not the default. Once the board has the sensor you can flip it on and off mid-pass with `y` on the Track or large-font screen, without opening Settings; the change is saved either way.

10

Calibration

Real radios and satellites have small oscillator offsets. **CAL mode** lets you null them out per satellite.

1. On **Track**, press `m` until the cal line shows `<CAL>`.
2. Find a known signal (your own downlink, or the satellite's beacon).
3. Trim the **downlink** with `,//` and the **uplink** with `;/..` `s` cycles the step (10/100/1000 Hz); `x` zeroes both.
4. Press **ENTER** to save the calibration **for this satellite** — it's stored and reloaded automatically next time you track that bird.

The passband position is *not* persisted (it's per-pass); calibration *is*.

Editing calibrations on the SD card

Per-satellite calibrations are stored as a plain-text file, so you can author or bulk-edit them on a computer instead of nudging each bird by hand. With the microSD removed and put in a computer, open **`/CardSat/calib.txt`**. Each line is one satellite:

```
# norad  downlink_Hz  uplink_Hz      (lines starting with # or ; are comments)
43017   -250   300
25544    120     0
```

- The first field is the **NORAD catalog number**; the next two are the **downlink** and **uplink** offsets in **Hz** (signed — negative lowers the frequency). These are the same `calDl`/`calUl` values you trim in **CAL** mode.
- Whitespace-separated, one satellite per line. Blank lines and lines beginning with `#` or `;` are ignored, so you can annotate the file freely.
- The file is read each time you select or track a satellite, so edits take effect the next time you open that bird — **no reflash needed**. Saving a calibration on the device (CAL → **ENTER**) rewrites this file but preserves your comment lines.

CTCSS tone overrides work the same way in **`/CardSat/tones.txt`**, one line per satellite as `norad tone_tenths` (tenths of a Hz, so `670` = 67.0 Hz; `0` forces the tone off):

```
# norad  tone_tenths
25544   670
```

Both files live on the microSD card if one is present, otherwise in the device's internal flash. If a satellite has no line in `calib.txt`, the global calibration from **Settings** is used.

10a. Transverter LO offsets

The supported radios tune up to the 1.2 GHz (23 cm) band. To work a satellite whose uplink or downlink is on a **microwave** band — 2.4 GHz (13 cm), 5.6 GHz (C), 10 GHz (X/3 cm), and up — you use a **transverter**: an external converter that shifts a lower **IF** band your rig *can* tune up to the real microwave band on transmit, and back down on receive. The rig stays on, say, 2 m or 70 cm; the transverter does the rest.

CardSat drives this with two settings under **Settings → Radio**:

- **Downlink LO** — the local-oscillator frequency of your **receive** transverter, in MHz.
- **Uplink LO** — the local-oscillator frequency of your **transmit** transverter, in MHz.

Enter each as the LO in MHz; enter `0` (or clear the field) for a leg that has no transverter. The relationship a transverter enforces is:

```
real on-air frequency  =  rig IF frequency  +  LO
rig IF frequency       =  real on-air frequency  −  LO
```

So if your 10 GHz downlink converter has a **10368 MHz LO** and the satellite's real downlink is **10489.550 MHz**, the rig tunes an IF of **121.550 MHz** (2 m). You tell CardSat the *real* frequencies (from the transponder data — nothing to convert by hand), set **Downlink LO = 10368**, and CardSat:

- **displays** the real 10489.550 MHz downlink on the Track screen,
- applies **Doppler** to that real frequency, and
- sends the rig **10489.550 − 10368 = 121.550 MHz** (plus Doppler) as its dial.

The order matters and CardSat gets it right: **Doppler is applied to the real frequency first, and the LO is subtracted last.** Doppler scales with frequency, so the shift on a 10 GHz downlink is roughly 70× larger than it would be on a 144 MHz IF — correcting the IF directly would under-correct by that ratio and walk you out of the passband. CardSat always computes the real-frequency Doppler, then converts to the IF.

The two legs are independent, which is exactly what split-band microwave satellites need. For a QO-100-style bird (2.4 GHz up, 10 GHz down) you set the uplink LO for the 13 cm up-converter and the downlink LO for the 3 cm down-converter; the rig works a comfortable 2 m/70 cm IF on both legs while CardSat shows the microwave dial.

On the **Track** screen, the **DN** and **UP** labels show a trailing `x` on any leg that currently has a transverter LO set (**DNx**, **UPx**). It's a reminder that the number shown is the *real* on-air frequency and the rig itself is sitting on the IF. The web page and `/api/status` report the rig's **IF** read-back for the commanded frequencies (that's what the radio actually returns); add the LO to recover the real frequency.

> Any residual LO error (a transverter whose LO is slightly off its nominal value) shows up as a fixed frequency offset and can be trimmed with the ordinary per-satellite or global calibration — the calibration offset is applied in the same path.
>
> **Not yet hardware-verified end to end.** The LO transform and the microwave-band math are exercised by the repository's orbit-audit harness, but neither bench radio used to develop CardSat operates above 1.2 GHz, so the full path through a real transverter has not been confirmed on hardware. If you run this with a transverter, reports are very welcome.

11

Working a pass, step by step

An FM bird (e.g. a repeater satellite):

1. **Update → k** to refresh GP data (and the clock) if you have WiFi.
2. **Satellites** → highlight the bird → `f` to favorite it → **ENTER** to open Passes.
3. Pick the pass you want; press `d` to preview its elevation curve if you like.
4. At AOS, **ENTER/t** to Track. Press `r` to start sending to the radio.
5. CardSat sets FM on both legs and Dopplers the single channel. Talk.

A linear (SSB/CW) bird:

1–4 as above. On Track you'll start in **TUNE** (press `d` to cycle to FULL radio-knob mode). 5. Press `r` to enable radio output. CardSat sets **USB down / LSB up** (or USB/USB on HF) and corrects both legs. 6. Tune to a clear spot — with the device `,//` keys, or by turning the rig's knob in `<FULL>` mode. Your downlink stays put; the uplink tracks (inverted if the bird inverts). 7. If your own signal isn't centered, switch to **CAL** (`m`), trim, and **ENTER** to save.

Watch the **ECL** flag and the polar Sun glyph — some birds change behavior in eclipse.

12

AOS alarm and deep sleep

AOS alarm (toggle in Settings → *AOS alarm*): when enabled and you have favorites and a set clock, CardSat tracks the soonest upcoming favorite AOS in the background. It **beeps** at T-60 s, T-30 s, and T-10 s, then sounds a longer double-beep and shows a blinking **"AOS!"** banner at acquisition. Once a pass is underway it also chirps at **TCA** (closest approach / peak elevation — a double mid-tone) and at **LOS** (a descending two-tone), so you can follow a pass by ear without watching the screen. All of these sounds are governed by this one setting: turning the AOS alarm off silences the AOS, TCA and LOS cues together. A small orange countdown banner appears on any screen within the last minute.

AOS lead alert (Settings → *AOS lead alert*, **off** by default; **2 / 5 / 10 / 15 min**): an earlier "get ready" cue, a distinct rising three-tone chirp and a screen flash, sounded that many minutes *before* AOS — enough lead time to get to the radio and point an antenna. It's separate from and in addition to the final 60/30/10/0-second countdown above, and (like the countdown) it's gated by the AOS alarm toggle.

Next favorite pass on Home. The home screen shows a persistent line for the soonest upcoming favorite pass — the satellite and a live AOS countdown (or **IN PASS** while one is up) — so "what's next and when" is answered without opening Passes. It reflects the same pass the alarm is tracking.

IR pass beacon (toggle in Settings → Station → *IR pass beacon*, off by default): when enabled, every pass-alert event *also* emits a burst of flashes from the Cardputer's **built-in IR LED** (GPIO 44), with a **distinct flash count per event** so external hardware can tell them apart. The flashes accompany the existing beeps — they don't replace them — and are gated by the same AOS-alarm machinery, so they only fire when the AOS alarm is on. Each flash is a ~60 ms burst of standard **38 kHz IR carrier** (with ~140 ms gaps), the kind any common IR receiver/ demodulator detects:

Event	Flashes
T-60 s to AOS	1
T-30 s to AOS	2
T-10 s to AOS	3
AOS (pass start)	4
TCA (peak elevation)	5
LOS (pass end)	6

CardSat only **transmits** these counts — what you do with them is up to you. Point a 38 kHz IR receiver module (e.g. a TSOP38238 or a Vishay TSOP4838) at the Cardputer, count the pulses in each burst, and trigger whatever you like: key a relay to power up a rotator or preamp at T-10, flash a shack light at AOS, start an SDR recording, drive a bigger external alert, or log events on a second microcontroller. The flashing is fully non-blocking — it runs one burst at a time between the normal tracking updates, so radio, rotator, and web control keep running throughout. Build whatever receiver and logic you want around the counts above.

> **The IR pass beacon is host-verified only.** The carrier timing and flash-count logic were checked off-device, but the actual IR output and a receiver decoding it have not been confirmed on hardware. The 38 kHz carrier, duty cycle, and burst/gap timing may need tuning for your particular receiver; treat the counts as the stable contract and the exact waveform as adjustable.

Deep sleep (Next Passes → `z`): CardSat computes the next favorite AOS and puts the ESP32 into deep sleep until **~60 s before** it, dramatically extending battery life between passes. The screen shows how long it will sleep and for which satellite. On wake, the unit reboots straight to **Next Passes** with the pass imminent and the alarm ready. (Press the reset button to wake early.) The clock is preserved across deep sleep, but make sure it was set — via NTP or GPS — before sleeping.

At get-ready time the battery is checked too: at or below **30%** the flash carries a yellow **BATT n%** note (with a low warning tone), so a pass is not lost to a dying cell that a quick top-up before AOS would have saved.

13

Sun, Moon, weather, and reference tools

CardSat computes the Sun's position and whether the satellite is illuminated:

- **Polar screen** — a yellow Sun glyph at its az/el (when above the horizon), the Sun's azimuth/elevation in the readout, and **sat SUNLIT** (green) or **sat ECLIPSE** (orange).
- **Track screen** — an orange **ECL** flag when the satellite is in Earth's shadow.
- **Pass detail** — the elevation curve is colored yellow (sunlit) / blue (eclipse), and a sunlit-percentage is given for the pass.

This uses a low-precision Sun ephemeris and a cylindrical Earth-shadow model — the sunlit↔eclipse transition is a hard edge rather than the few-second penumbral fade, which is plenty for knowing whether a bird has power or is optically visible.

Sun / Moon antenna tracking

Sun / Moon on the main menu shows the live position of both bodies from your location. It opens in a **graphical sky view**: a polar dome (zenith at center, North up, horizon at the rim) with the Sun drawn as a rayed yellow disc and the Moon as a cyan crescent, so you can see at a glance where each one is. A compact panel on the right lists azimuth, elevation and above/below-horizon for both. A body below the horizon is shown faintly just outside the rim so its bearing is still readable. Press `g` to toggle between the graphic and a plain azimuth/elevation data list. `;`/`.` selects which body the rotator follows (marked with a green ring, not a highlight bar). Press `o` to drive the rotator at the selected body — useful for antenna gain checks against Sun noise, EME pointing, or rotator calibration against a visible target. `x` stops; ` parks and disengages.

Behavior notes:

- The rotator has **one master at a time**: engaging Sun/Moon tracking disengages satellite rotator tracking and vice versa, and opening **Rotator manual** control disengages both.
- While the selected body is **below the horizon** the rotator parks and waits; tracking resumes automatically when it rises.
- Tracking keeps running if you navigate to other screens — an orange **SUN** / **MOON** tag in the header shows it's active. Going back from the Sun/Moon screen parks and disengages.
- The Moon ephemeris is a low-precision series (~arc-minutes after topocentric parallax correction) — far finer than any amateur antenna beamwidth. The Sun is good to ~0.01°.

13.0.1.1 Sky sources (`s`) Press `s` on the Sun/Moon screen for a secondary **sky-sources** plot: the classical planets and the strongest cosmic radio sources, on the same sky dome (zenith center, North up, elevation = radius). It's an antenna-pointing aid and a reference for the brightest RF sources crossing the sky — handy for a Sun-noise-style gain check against a strong source, or simply to know what's overhead.

Radio sources are drawn as orange crosses and **planets** as cyan dots; an object below the horizon sits just outside the rim in gray so its bearing is still readable. `;`/`.` step through the objects (a green ring marks the selection); the right-hand panel shows the selected object's azimuth, elevation, above/below-horizon status, and whether it's a radio source or a planet. ` returns to Sun/Moon.

The catalog covers the planets **Mercury, Venus, Mars, Jupiter, Saturn** (computed live) and the fixed radio sources **Cassiopeia A** (the brightest galactic source), **Cygnus A**, the **galactic center (Sgr A*)**, the **Crab nebula (Tau A)**, and **Virgo A (M87)**, plus a few bright stars (Polaris, Vega, Antares) for orientation. Positions use the same low-precision ephemeris as the Moon — far finer than any amateur beamwidth.

13.0.1.2 Sun / Moon transits (`t`)

Press `t` on the Sun/Moon screen to open the **transit finder** for the active satellite. It scans the next **48 hours** for times the satellite passes in front of (or close to) the **Sun or Moon** as seen from your location — the dramatic astrophotography event where the ISS is silhouetted on the solar or lunar disc. The scan runs incrementally with a progress bar (it never blocks the device), then lists each close approach: the body (Sun/Moon), a countdown to the event, the minimum angular separation, the body's elevation, and whether it's a true **TRANSIT** (the satellite crosses the disc) or a near **conjunction**. Press `r` to rescan, `;`/`.` to move the selection.

Because the ground path of a transit is only a few kilometers wide, this is a **point prediction for your exact location** — moving a few km changes it, and fresh elements matter for the center-line. **Never observe a solar transit without proper solar filtering on your eyes and optics.**

13.0.1.3 EME / moonbounce (`e`)

> **Analysis & planning (0.9.61):** the live screen adds the **spatial polarization offset** (Pol) — the Moon's parallactic angle at your station, the main reason 144 MHz echoes fade with the Moon up — and a coarse 144 MHz **Faraday** figure when solar flux is known. Press `p` for the full report: polarization (with a sked note to subtract the DX station's), a per-band table of Doppler / Faraday / sky-temperature-K / libration spread from 50 MHz to 10 GHz, an absolute round-trip **path-loss** budget, and a **ground-gain** window flag. The planning table (`p` from the EME screen) now spans **90 days** and stars the good days (high northern declination, low degradation). Every EME view prints.

Press `e` on the Sun/Moon screen for the **EME (Earth-Moon-Earth) screen** — the numbers a moonbounce operator needs to find their own echo and judge whether the Moon is workable. It reads live from your location and clock:

- **Self-echo Doppler** for **50 / 144 / 432 / 1296 / 10368 MHz** — the total round-trip frequency shift on your own signal returning off the Moon. This is dominated not by the Moon's orbital motion but by **your own station's rotation** as the Earth turns, so CardSat computes it **topocentrically** (your position *and* velocity projected onto the Earth–Moon line). The figure is small at 50 MHz but swings to a few kHz at 1296 and tens of kHz at 10 GHz across a Moon pass — which is exactly the offset you tune out to hear your echo. A geocentric-only number would be wrong by thousands of Hz at microwave.
- **Range and range-rate** — the current Earth–Moon distance and how fast it is changing (the quantity the Doppler is derived from).
- **Path degradation** — the extra path loss relative to perigee (roughly +2 dB at apogee, from the inverse-fourth-power round-trip law), with a **near perigee** / **near apogee** note. Perigee is the easiest Moon; apogee the hardest.

- **Sky-noise flag** — a coarse **cold sky / warm / HOT** indicator from the Moon's galactic latitude. Near the galactic plane the 144 MHz sky-background temperature is high, degrading weak-signal reception; well away from it the sky is cold and quiet.
- **Mutual-Moon window** — press `m` and enter a DX station's grid to scan the next two weeks for the spans when the Moon is **above the horizon for both stations at once** — the common EME window in which a QSO is geometrically possible. Press `g` in that view to change the grid.
- **30-day planner** — press `p` for one row per day: the Moon's **declination** at 12:00 UTC and the **path degradation**. High declination means long, high Moon tracks (for the northern hemisphere — southern operators want it negative, and the raw numbers are shown so either reading works); low degradation means a near-perigee Moon. A green star marks days with both. This is how EME operating weekends are actually picked.
- **Sun-proximity warning** — whenever the Moon is up and within about **10°** of the Sun, a red **SUN** flag appears: with the Sun in the beam, solar noise buries weak echoes regardless of everything else.
- **Rotator** — press `o` to aim a connected rotator at the Moon and track it (the same one-master-at-a-time rule as Sun/Moon tracking); `x` stops. `` ` `` returns to the Sun/Moon screen.

The Moon position uses the same low-precision series as the Sun/Moon screen (arc-minute class, far finer than any amateur beamwidth); the Doppler figures are intended as an operating aid — "where is my echo, roughly, and which way is it moving" — rather than a sub-Hz prediction. Cross-check against a dedicated EME calculator (MoonSked, the ARRL EME tool) for your grid when precision matters.

Space weather

Space Wx on the main menu summarises the indices that matter most for propagation: the **solar 10.7 cm radio flux** (F10.7, a proxy for solar activity and ionospheric ionisation), the **planetary Kp index** (geomagnetic disturbance, 0–9), and the **running A index** (the daily-equivalent geomagnetic amplitude, shown when the feed provides it). Each value is labeled in plain terms — flux low/moderate/good/very-high, Kp quiet/unsettled/minor-storm/moderate-storm/major-storm, and A quiet/unsettled/active/storm — and color-coded, with a short **operating outlook** line translating the numbers into what to expect on HF and satellite paths, plus a note of how old the data is.

Opening **Space Wx** shows the last cached values immediately, then — if WiFi is up — fetches an update in the background, showing an **"Updating Space Wx"** banner on the bottom status bar and then a brief result (**Space Wx updated/unchanged**, **update failed**, or **WiFi failed**) before the banner clears. The indices are also refreshed whenever you run **Update**, and `r` forces a refresh on demand. The Kp and A fetch is independent of the flux fetch, so a hiccup in one never suppresses the other.

This is a planning cue, not a forecast: the flux and Kp are observed values, and the outlook text is a simple heuristic reading of them, not a calibrated propagation prediction. A high Kp (storm) is the main thing to watch — it warns of auroral flutter on VHF and disturbed high-latitude HF.

13.0.2.1 HF / 6m propagation guide (`p`) Press `p` on the Space Wx screen for a **propagation guide** that turns the same two indices — the 10.7 cm solar flux and the Kp index — into band-by-band operating guidance, for the operator who works HF and 6 m as well as the birds:

- **MUF estimate** from the solar flux and Kp: an approximate maximum usable frequency for day and night. Higher flux raises the MUF, so the high bands open as the flux climbs; the band-by-band day/night outlook lives on the `b` outlook page (below).
- **Geomagnetic effect** from Kp: quiet / unsettled / storm, and what that means for HF (a storm degrades paths and can black out polar routes).
- **Auroral VHF** likelihood: a high Kp is the trigger for **6 m / 2 m auroral** propagation, so the screen flags when it's possible and reminds you to beam toward the pole.
- **D-layer absorption** on the low bands (worse when the field is disturbed).

When a previous sample exists (six or more hours older), each index carries a **delta** — "142 sfu +5" is a rising flux, and direction is half of what these numbers tell an HF operator. The previous sample survives reboots via the cache.

Press `o` for the **outlook page**: the day/night band-by-band outlook (40/20/15/10 m open / fair / weak / shut), the MUF estimate, the current flare state, solar wind Bz + speed, the 3-day Kp forecast, and the VHF flag — everything the printed guide shows, on screen. The flare / solar-wind / Kp-forecast rows live here rather than on the core page. Two of the rows draw on data that was already being fetched but not previously shown: the **flare line** reports today's **C/M/X flare counts** and the number of **new sunspot regions** (from the daily solar-indices file the sunspot number comes from) — the daily counts convey activity *level*, which matters more to HF than a single latest-flare snapshot — and a **radio-blackout line** gives today's NOAA **R1–R2** (and R3) radio-blackout probability plus the **S1** solar- radiation-storm chance (from the 3-day forecast's B/C sections). A radio-blackout forecast is a direct HF-degradation predictor. Press `r` to refetch the indices without leaving the screen; `` ` `` returns (outlook page → core page → Space Wx). As on the Space Wx screen itself, these are **climatological rules of thumb, not a real-time model** — the screen says as much — and **6 m sporadic-E**, the dominant summer opening, is *seasonal* and is not predicted by these indices.

Weather

Weather on the main menu (just below Space Wx) shows current conditions and a short forecast for your operating site — handy for portable and field operation. It displays the current temperature and sky condition, wind speed/direction and humidity, then a row for each of the next few days with the day's condition, high/ low, and chance of precipitation.

The data comes from **Open-Meteo** (open-meteo.com), a free, no-key weather service. The location is taken from the same site coordinates the prediction engine uses (your GPS fix or manually set lat/lon), so set your location first. Opening **Weather** shows the cached forecast immediately, then — if WiFi is up — fetches an update in the background with an **"Updating Weather"** banner on the bottom status bar, followed by a brief result (**Weather updated**, **update failed**, **WiFi failed**, or **Set a location first**) before the banner clears. It also refreshes when you run **Update**, and `r` forces a refresh. Like Space Wx, the last result is cached to flash, so it remains viewable offline with a note of its age.

Units are now three **independent** settings under *Settings* → *Display*: **Weather temp** (°C / °F), **Weather wind** (km/h / mph / m/s), and **Weather pressure** (hPa / inHg). Any combination is allowed — e.g. °F with m/s, or °C with inHg — and changing any of them re-labels the cached values immediately without a re-fetch (the fetch stores values in canonical units and converts at display time). Press `p` on the Weather screen to **print** the current conditions, the field figures, and the forecast to the configured print sinks; it is also in the *About* → *Print* menu as "Weather report".

Press `f` for a second **field conditions** page aimed at operating away from home. It adds what the summary omits: the **feels-like** (apparent) temperature next to the air temperature and a day/night marker; **wind gusts** alongside the steady wind, flagged **MAST** when gusts reach a level worth thinking twice about for a portable mast; the **barometric pressure** with its **3-hour trend** (rising / steady / FALLING — a falling barometer is the classic sign of deteriorating weather moving in); the day's **UV index** with a risk word (low → extreme), for judging sun exposure over a long operating session; and today's **sunrise / sunset** times, useful both for planning setup and teardown in daylight and for working **grayline** propagation. Below those is a per-day table of UV, peak gust, and sun times for the forecast days. All of it comes from the same cached fetch, so the page works offline too. Press `f` again (or `` ` ``) to return to the summary.

Weather data by Open-Meteo.com, licensed under CC BY 4.0.

The same fetch now also pulls **hourly cloud cover for the next 48 hours**. It is not shown on the Weather screen itself; it feeds the **visible-pass list** and the **Sun/Moon transit finder**, which append a color-coded cloud percentage to each row — the go/no-go a visible pass or a transit photo actually hinges on. Cached with the rest of the forecast, so the readouts work offline until the window ages out.

Transponder database

From the **Satellites** list, press `t` to browse every transponder and beacon entry the on-device catalog holds for the selected satellite (sourced from SatNOGS with the transponder cache). Each entry is shown as a short block: its description, the **downlink** (a range for linear transponders) with mode and — when SatNOGS lists it — the **baud rate** (data rate; a CW transmitter's rate is shown in WPM), and the **uplink** with any CTCSS tone and inverting/linear flags. The baud tells you what decoder or terminal settings a telemetry beacon needs. `;`/`.` scrolls through the list. Entries are **ordered by usefulness**: two-way transponders first, then amateur-band before non-amateur (so any out-of-band TT&C or telemetry downlinks fall to the end), active before inactive. Transmitters SatNOGS marks **inactive** are **dimmed and tagged "(off)"** so you can tell at a glance which are no longer believed operational. It's a quick offline reference for a bird's frequencies and modes — handy for checking what a satellite carries without a radio connected. If nothing shows, the transponders haven't been cached yet; run **Update** with WiFi on.

QRZ callsign lookup

QRZ Lookup on the main menu looks up a callsign in the **QRZ.com** database and shows the operator's name, mailing address, country, grid square and license class. It uses QRZ's XML data service, which **requires a QRZ XML-data subscription**.

To use it, enter your QRZ **username** and **password** in *Settings → Network / data* (rows **QRZ user** / **QRZ pass**). Then open **QRZ Lookup**, press **ENTER**, type a callsign, and press ENTER again. CardSat logs in to QRZ, caches the session key, and displays the result; the key is reused for subsequent lookups until it expires (then it re-logs in automatically).

The screen handles the obvious cases plainly:

- **No WiFi** — it simply says WiFi isn't connected; connect via Update or Settings and try again.
- **No credentials** — it explains that a QRZ XML subscription is required and that you need to enter your username and password in Settings.

- **Not found / login error** — the QRZ error message (e.g. "Not found", "password incorrect") is shown in the status line.

Your QRZ password is stored on the device the same way the WiFi and radio-LAN passwords are; it is shown masked in Settings. CardSat talks to QRZ over HTTPS.

Grid distance & bearing (Grid dist/bearing)

Figure 7: Grid distance

Grid dist/bearing on the main menu (just before QRZ Lookup) is a great-circle calculator for terrestrial VHF/UHF work — tropo, contests, or just aiming a beam at a known grid. Press `g` and enter a **Maidenhead grid**; CardSat shows the **distance** (km and miles) and the **beam heading** from your station, both **short path and long path**.

- Press `o` to **point a connected rotator** at the computed bearing. This is a terrestrial heading, so the elevation is set to 0.
- The heading is shown both as **T** (true) and **M** (magnetic), with the local **declination** in parentheses. If you are hand-tracking with a magnetic compass, or your rotator is referenced to magnetic north, use the **M** figure. (Whether CardSat *sends* true or magnetic to the rotator is a separate setting — see below.)
- Press `q` to **look up a callsign's grid**: this opens a small **QRZ → grid** screen (separate from the main QRZ Lookup, which is untouched). Enter a callsign; on **ENTER** it seeds the calculator with that operator's grid so you get distance and bearing to them directly.

13.0.6.1 True vs magnetic rotator bearings By default CardSat sends **true** bearings to the rotator, because most rotators are either aligned to true north or already apply their own declination correction. If your controller expects **magnetic** bearings and does *not* correct for declination, set **Rot bearing: magnetic** in Settings (rotator section). CardSat then subtracts the local magnetic declination from every commanded azimuth — tracking, park, pre-position, Sun/Moon, and manual alike. The declination comes from a built-in IGRF model; it is **approximate** (a few degrees, worst in the South Atlantic anomaly) — fine for pointing, but not a survey instrument. The value is also readable in BASIC as `MAGDECL`.

The magnetic heading is also **displayed** wherever you would use a compass to point by hand: the manual rotator screen shows the magnetic target heading and the local declination, and the tracking side panel, OSCAR display, and globe view append the magnetic azimuth (**M###**) to the true azimuth.

Your own grid comes from your set location, so set that first on the **Location** screen. ` returns to the main menu.

Upcoming activations (Activations)

Activations on the main menu shows the **upcoming satellite activations** that operators have scheduled on **hams.at** — roves, grid activations, and special operations announced ahead of time. It's a quick way to see who's planning to be on which bird, from where, and when, so you can be listening for a rare grid or a wanted station.

Opening **Activations** shows the last-known list immediately from the on-device cache — even with no WiFi — then, if WiFi is up, fetches an update in the background with an **"Updating Activations"** banner on the bottom status bar, followed by a brief result (**Activations updated/unchanged**, **update failed**, or **WiFi failed**) before the banner clears. Each row shows the **date, callsign, satellite and grid**. Move with `;`/`.`, press **ENTER** on a row for the full detail — **start and end times (UTC), mode, frequency, and the activator's comment** — and ` to go back to the list. Press `r` at any time to refresh.

Do I actually have a footprint? (0.9.43) When you open an activation's detail, CardSat checks whether the listed satellite is really above the horizon at the same time as *both* you and the activator — a genuine mutual window. Because hams.at keplerian data can lag, the check searches **±30 minutes around the listed start** and reports the co-visibility window it finds, so a pass that has drifted since the alert was posted is still caught. The detail screen shows a **footprint note**: the mutual window's start–end and duration in green if one exists, "No footprint near listed time" in yellow if not, or a short reason ("set clock first", "no activator grid", "sat not in your list") when it can't be computed. The activator's comment is often longer than the screen; scroll it in place with `;`/`.`. Press `a` to set a SKED reminder (T-60/30/10 beeps and a flash), and — when a footprint exists — `w` to open the **mutual-window screen**.

The **mutual-window screen** draws a small **polar plot** of the pass: the satellite's track as seen from **you** (cyan, "M") and from the **DX** (orange, "D") across the window, with the **date, AOS, LOS, duration and peak elevation for each station** beside it. From there, `d` opens a **DX Doppler table tailored to this activation**: CardSat reads the activation's frequency (from the freq field, then the comment) and, if it matches one of the satellite's real **two-way** transponders, pre-selects that transponder and locks the DX dial to the listed frequency as a **fixed downlink or uplink** (whichever leg the number falls in). On a bird with **more than one transponder** (AO-7, for example), the listed frequency is **remembered and re-applied as you cycle transponders** with `t` — it stays locked onto whichever transponder it belongs to and is not lost by stepping through the others. If no usable frequency is found, it opens the normal DX Doppler table instead. The existing DX Doppler (reached from the **Mutual** schedule) is unchanged; the activation path is a separate, pre-seeded view.

The feed is fetched over HTTPS from `https://hams.at/feeds/upcoming_alerts`, parsed (up to 30 activations), and **cached to flash** so the last-known roster survives a reboot and shows instantly offline. The **Update** screen's `k` also refreshes it alongside the GP update. Times are shown exactly as the feed provides them, in UTC. Some activations list "(none)" for frequency or elevation — those simply weren't specified by the operator, and the activator's comment often fills in the detail.

14

GP age and accuracy

Theory — what GPs and SGP4 actually are

Why a satellite needs "elements" at all. You cannot just store a satellite's position and velocity and expect it to stay useful — it sweeps through space at ~7.5 km/s and the orbit itself slowly changes shape. Instead, the satellite is described by a compact set of numbers that, fed to a matching math model, *regenerate* the position at any time. Those numbers are the **orbital elements**; the model is **SGP4**. The two are a matched pair: the elements only mean what they mean *because* SGP4 is the model that will consume them.

TLE, OMM and "GP". The historical packaging is the **Two-Line Element set** (**TLE**) — two 69-character lines, a fixed-column format dating to punch cards, holding the elements plus epoch, drag term, and catalog bookkeeping. **OMM** (Orbit Mean-elements Message) is the modern, self-describing replacement carrying the *same* numbers as JSON, XML or KVN instead of fixed columns. CelesTrak's umbrella term for "current mean elements in whatever format" is **GP** (General Perturbations). CardSat downloads **GP data as JSON/OMM** (it never has to parse the brittle column format off the wire), but internally it renders each set back into a TLE line-pair — epoch encoded as `YYDDD.dddddddd`, B* in TLE exponent notation, with the line checksums — and hands that to the SGP4 library, which still ingests elements the classic way. The element values are identical; only the wrapper changes. This matters more than it once did: the 18th Space Defense Squadron has announced plans to retire the legacy TLE format in favor of OMM — the punch-card wrapper is finally being sunset after half a century — so CardSat's GP-native pipeline needs no change when the old format disappears. (Because they're the same numbers, a TLE can be mechanically decoded into the GP fields CardSat's manual entry asks for — that's what the bundled **`tools/tle2gp.py`** helper does; see §8.)

Oversized sources. CardSat holds up to 150 satellites in memory; several CelesTrak groups carry more (some far more). Since v0.9.54 an oversized source is handled honestly: satellites on your **favorites** list are loaded first — guaranteed, even if they sit past the 150th object in the file — the remaining slots fill in file order, and the truncation is *visible*: the status line reports "Loaded X of Y" and the boot log prints `[gp] parsed X of Y satellites (truncated; favorites kept)`. Downloads are also preflighted against free storage — a group too large for the active filesystem is refused with "file too big for storage" *before* writing, so it can't fill an internal-flash unit mid-write or disturb the previous good catalog. If you need a bird from deep in a big group, favorite it (Satellites → `f`) and refresh.

What SGP4 is. SGP4 (Simplified General Perturbations 4) is the analytic propagator that NORAD/USSPACECOM elements are *fit to*. "Analytic" means it isn't numerically integrating forces step by step; it applies closed-form expressions for the perturbations that matter for a near-Earth satellite over days-to-weeks:

- **Earth's oblateness** — the equatorial bulge (the **J2** zonal harmonic, with smaller J3/J4 terms) that precesses the orbit plane and rotates perigee. This is the dominant non-Keplerian effect and the reason pass times march earlier each day.
- **Atmospheric drag** — modeled through the **B*** ("B-star") term, a fitted ballistic-and-

density coefficient that slowly shrinks the orbit. (A companion model, **SDP4**, extends the same scheme to deep-space orbits with period > 225 min by adding lunar/solar gravity and resonance terms — the pair is often written "SGP4/SDP4." In practice CardSat's birds are all near-Earth, so the SGP4 branch is what runs.)

Crucially, the **"mean" elements are not osculating elements.** They are deliberately *detuned* values chosen so that, after SGP4 adds its periodic terms back in, the output matches reality. You cannot mix them with a different model, average them, or read the mean motion as an instantaneous rev-rate and expect physical truth — they are model-specific fitting constants. This is also why CardSat propagates with the **WGS72** gravity constant set: that is the set the elements were fit against, and using WGS84 would introduce a small systematic error.

From element to az/el — the frames. Propagating gives position and velocity in the **TEME** frame (True Equator, Mean Equinox — the quirky inertial-ish frame SGP4 outputs). To point an antenna, CardSat rotates the observer's geodetic latitude/longitude/altitude into the same TEME frame using **GMST** (Greenwich Mean Sidereal Time — Earth's rotation angle for the instant), differences the two position vectors to get the slant vector, and resolves that into topocentric **azimuth/elevation**. Range-rate (for Doppler) is taken directly from the projection of the relative *velocity* onto the slant direction, evaluated at the exact fractional second rather than by differencing whole-second range samples — cleaner right at closest approach where range-rate changes fastest.

Where the error comes from. SGP4 near a fresh epoch is good to roughly a kilometer. Error grows with **element age** because the small unmodelled accelerations (drag fluctuations from space weather, higher-order gravity, solar radiation pressure) integrate over time, and drag is the worst offender — it depends on upper-atmosphere density, which swings with solar activity and isn't known in advance. So a low, draggy orbit (ISS, cubesats) can develop noticeable along-track timing error (the satellite arriving early or late along an otherwise-correct path) within a week or two, while a high, stable orbit stays usable far longer. Along-track error shows up to you as **pass-time slip**; cross-track error shows up as **pointing/Doppler error**. The practical defense is simply to refresh elements often enough for the orbit you care about.

GP age indicator

SGP4 predictions degrade as the GP mean elements age. CardSat shows the **age of the element set** (days since the GP epoch) on the Passes and Track screens, color coded:

- **Green** — under 14 days (fresh).
- **Yellow** — 14–28 days (getting old).
- **Red** — over 28 days (stale; expect timing/pointing error). In the Next Passes schedule, stale sats are flagged with a red `!`.

Refresh with **Update → k** whenever you have WiFi. For low orbits, weekly (or fresher) elements are best.

Choosing the element source

By default CardSat pulls the **AMSAT** daily GP bulletin (all amateur satellites, JSON, including the friendly `AMSAT_NAME`). To change source, open **Settings → GP source** and press ENTER for a picker:

- **AMSAT (amateur)** — the default bulletin.

- **CelesTrak categories** — any CelesTrak GP group in JSON-PP format. Amateur Radio is listed first, followed by SatNOGS and the Special-Interest, Weather & Earth Resources, Communications, Navigation, Scientific and Miscellaneous groups. CelesTrak omits `AMSAT_NAME`, so CardSat falls back to `OBJECT_NAME` and the data parses correctly.
- **Custom URL...** — type any URL that returns OMM JSON / JSON-PP.

Move with `;`/`.` (or `{`/`}` to page) and press ENTER to select. The choice is saved immediately and used by the next **Update → k**.

15

Working offline

CardSat is designed to operate with no network in the field:

1. With WiFi, go to **Update** and press `k` (GP + clock) and then `a` (cache **all** transponders). Both are written to flash. The transponder cache runs in a single session and finishes on "Cached all N transponders" — let it run to completion before going offline.
2. After that, everything — pass prediction, transponders, Doppler, the schedule — works with WiFi off. The clock keeps running (and survives deep sleep); use GPS or manual `c` entry to set it where there's no NTP.

Cached data persists across power cycles until you refresh it or perform a reset.

Hand-curating your own GP data (SD card, no online update)

You don't have to use the online update at all. CardSat reads its orbital elements from a plain JSON file on the storage card — **`/CardSat/gp.json`** — and you can write that file yourself: hand-pick exactly the satellites you want, drop the file on the microSD card, and run CardSat with no network. This is handy for field use with no WiFi, for a small fixed set of birds, or for pinning known-good elements so an automatic refresh can't replace them mid-operation.

The file is a **JSON array of OMM objects** — the same JSON format CelesTrak and Space-Track export when you request "JSON", so the easiest path is to download real OMM JSON for the satellites you want, delete the objects you don't, save it as `gp.json`, and copy it to `/CardSat/gp.json` on the card. Each object needs at minimum `NORAD_CAT_ID`, `EPOCH` (ISO 8601 **with the time of day** included), and `MEAN_MOTION`, plus the full Keplerian set (eccentricity, inclination, RAAN, argument of perigee, mean anomaly) and `BSTAR` for correct predictions. If you only have a classic TLE, the bundled **`tools/tle2gp.py`** helper converts it to the JSON object CardSat expects.

Two things to remember: the clock must be set (GPS or manual entry) since you're offline, and **running the online Update (`k`) with WiFi overwrites `/CardSat/gp.json`** — so to stay fully offline, just don't run it, or keep a backup of your curated file.

> See **docs/guides/OFFLINE_GP_DATA.md** for the complete field reference, a full worked example, and a step-by-step curation checklist.

What persists and works without WiFi. Every network data source is cached to flash and reloaded at boot, so after one online refresh the unit is fully usable offline in the field:

- **GP / orbital elements** — passes, Doppler, and all tracking run entirely from the cached elements.
- **Weather** — the current conditions, the multi-day forecast, and the 48-hour cloud cover (used by the visible-pass and transit screens) all render from cache.
- **Space weather** — the F10.7 flux and Kp, their trend deltas, and the propagation guidance derived from them.

- **AMSAT status** — the per-satellite activity marks (active / not-heard / telemetry) and the catalog name map, so the AMSAT status screen is populated offline.

Opening any of these screens shows the cached data immediately, then attempts a refresh only if WiFi is reachable; if it is not, the cached values stay on screen with a brief "WiFi failed" note rather than a blank screen. A failed or interrupted refresh never discards good cached data — downloads are staged and only swapped in on success. The one thing that inherently needs a live connection is *submitting* an AMSAT report and fetching the per-report who-heard-it detail; those say so and leave everything else working.

16

Radio-specific notes

CardSat speaks three CAT dialects, selected by the **radio model** in Settings. Defaults are editable; the **CI-V address** field applies to **Icom only**.

Radio	Family / protocol	Default baud	Interface	Read-back
IC-820	Icom CI-V	9600	CI-V 5 V single-wire	yes
IC-821	Icom CI-V	9600	CI-V 5 V single-wire	yes
IC-910	Icom CI-V	19200	CI-V 5 V single-wire	yes
IC-970	Icom CI-V	9600	CI-V 5 V single-wire	yes
IC-9100	Icom CI-V	19200	CI-V 5 V single-wire	yes
IC-9700	Icom CI-V	19200	CI-V 5 V single-wire	yes
FT-847	Yaesu 5-byte	57600	serial (TTL/RS-232)	yes[1]
FT-736R	Yaesu 5-byte	4800	serial (TTL/RS-232)	no
TS-790	Kenwood ASCII	4800	RS-232 (MAX3232)	yes
TS-2000	Kenwood ASCII	57600	RS-232 (MAX3232)	yes

Protocol command sets follow the Hamlib backends (`icom`, `yaesu/ft847`, `yaesu/ft736`, `kenwood/ts2000`, `kenwood/ts790`). Serial framing is set automatically: **Icom 8N1, Yaesu 8N2, Kenwood 8N1 — except 8N2 at 4800 baud.** The TS-790 generation (IF-232C interface: TS-450/690/790/850/950) requires **two stop bits at 4800 baud** (one stop bit at higher rates such as the TS-2000's 57600); CardSat selects this by baud automatically. Note the TS-790's CAT is via the **optional IF-232C** adapter — its operating manual documents the interface but not the command set, so the TS-790 mapping leans on the shared Kenwood protocol and is the least bench-verified of the rigs here.

The CAT *interface circuit* differs by family. **Icom CI-V** uses a 3.3 V-safe single-wire interface (**CIV_INTERFACE.md**); you can wire it as **separate TX/RX** (G2/G1, the default) or **single-pin** (one shared open-drain wire, confirmed on an IC-821) — set this with **CI-V wiring** in Settings and see §3 → Icom CI-V and **CIV_SINGLE_PIN.md**. **Yaesu** and **Kenwood** use RS-232-level serial — build a **MAX3232** level shifter as in **RS232_INTERFACE.md** (**!** untested; build at your own risk).

Automatic PL / CTCSS tone for FM satellites

Several FM birds require a subaudible **PL (CTCSS) tone** on the uplink. CardSat applies it automatically: when the active transponder is **FM with an uplink** and the satellite is in its built-in tone table, it enables the rig's **TX CTCSS encoder** at the right frequency the moment radio output is on, turns it **off** again when you switch to a transponder that doesn't need one, and disables it when you turn radio output off (so your rig isn't left transmitting a tone). The tone in use is shown on the Track screen as a `PL nn.n Hz` line, and a status flash confirms it (`PL 67.0 Hz on uplink`).

Built-in tones (operating tone, by NORAD id): **ISS** 67.0, **SO-50** 67.0, **AO-91** 67.0, **AO-92** 67.0, **PO-101** 141.3. SatNOGS carries no structured tone field, so this list lives in

`SatDb::knownCtcssHz()` — extend it there as new FM satellites appear. **SO-50 note:** the 67.0 Hz figure is the *operating* tone; arming its 10-minute timer with a 74.4 Hz burst is a separate manual step on your radio.

Setting tones yourself. Press `c` on the Track screen to set a tone for the current satellite. Enter a frequency in Hz (e.g. `67.0`, `141.3`) and it is snapped to the nearest standard CTCSS tone; enter `0` to force the tone **off** for that bird; leave it **blank** to clear your override and fall back to the built-in default. The override is stored per satellite (by NORAD, in `tones.txt`) and wins over the table, so it survives reboots and a GP/transponder refresh, and it's how you add a tone for any FM bird not already in the built-in list.

Tone-over-CAT is supported on **IC-910/9100/9700, FT-847, and TS-2000** (the `hasTone` flag in `radio_profiles.h`); on other models the `PL` line reads `(rig n/a)` and no tone command is sent. Encoders by family: **Icom** CI-V `1B 00` (tone freq, BCD) + `16 42` (encoder on/off); **FT-847** sat-TX opcodes `2B` (tone, via the CAT code table) + `4A…2A/8A…2A` (encoder on/off); **TS-2000** `TN` (tone number) + `TO` (encoder on/off). All are taken from the Hamlib backends; like the rest of CAT they're host-verified but not yet exercised on a real radio — watch the serial trace and confirm the rig shows the tone.

Icom. CardSat drives MAIN/SUB directly and forces the rig's built-in satellite mode **off** at startup on the rigs that expose it over CAT (IC-910, IC-9100, IC-9700). Downlink on SUB, uplink on MAIN. Read-back uses `0x03`; it is confirmed working on hardware for the **IC-821** and supported in firmware for the IC-820/910/970/9100/9700. Wrong-VFO fixes live in `radio_profiles.h`.

> **IC-820/821/970 satellite mode and band pairing are MANUAL.** Unlike the IC-910/9100/9700, these three rigs have **no working CI-V command** to toggle satellite mode (the IC-821's sat-mode command is a no-op on real hardware — confirmed on an IC-821) or to assign which band sits on MAIN vs SUB. Their `0x07 D0`/`D1` bytes are band *access* (which band a read/write targets), not a MAIN/SUB assignment. So on these radios you must **engage satellite/full-duplex mode, set up the band pair, and set any uplink PL tone on the radio's front panel**; CardSat then only Doppler-tunes within that layout. See **RADIO_SETTINGS.md** for the full per-radio CAT-vs-manual breakdown. This matches how Hamlib, OscarWatch, SatPC32, and Gpredict treat the same radios.
>
> By contrast, the **IC-910, IC-9100, and IC-9700** ***can*** **set up MAIN/SUB over CAT**, and **CardSat now orients MAIN/SUB automatically** on all three when you turn radio control on for a two-way transponder (once at engage, never per tick), so you don't pre-arrange the bands: - **IC-9100/9700** issue `0x07 D2 00/01 <bandcode>` to set MAIN/SUB band selection directly. - **IC-910** reads MAIN (`0x07 D1` then `0x03`) and, if it's the wrong band, issues one swap (`0x07 D0`) — the 910's MAIN/SUB can't share a band, so one check fixes both. This also **fixes a latent bug**: the 910's MAIN-select byte was previously `0x07 D0` (the *swap* command); it is corrected to `0x07 D1`.
>
> **! Host-verified only — untested on real 910/9100/9700** (the author runs an IC-821, which has none of these commands); a leg outside 2 m/70 cm/23 cm is skipped and set manually. The 820/821/970 remain fully manual for band setup.
>
> **IC-910 satellite-mode & tone commands differ from the IC-9100/9700.** The IC-9100/9700 toggle satellite mode with CI-V `0x16 0x5A`, but the **IC-910 uses a different command group entirely: `0x1A 0x07`** (verified from the IC-910 CONTROL

COMMAND table — on the IC-910, command `0x16` has no satellite-mode sub-command). Likewise the CTCSS tone encoder is `0x16 0x42` (Repeater tone) on the IC-9100/9700 but **`0x16 0x43` (Subaudible tone) on the IC-910** — its `0x42` is the auto-notch filter. CardSat sends the right command for each rig. (Earlier firmware sent `0x16 0x07`/`0x16 0x42` to the IC-910, so sat mode never engaged and the tone key toggled the notch; both are fixed.)

IC-820H vs IC-821H MAIN/SUB select. These two rigs use the *same* CI-V command (`07`) for band select but with the **two sub-commands swapped** — a quirk confirmed from each radio's own manual (CI-V command table):

Radio	Address	Main band access	Sub band access
IC-821H	`4C`	`0x07 D0`	`0x07 D1`
IC-820H	`42`	`0x07 D1`	`0x07 D0`

CardSat ships the correct (reversed) mapping for each, so both tune the right VFO out of the box. If you ever swap one rig's CI-V address onto the other's profile, remember the band-select bytes differ too. The frame is otherwise identical: preamble `FE FE`, controller `E0`, command `07`, end `FD`.

Icom over the network. The **IC-9700** can be driven over WiFi/Ethernet instead of the CI-V bus — set **CAT type → Icom LAN** in Settings (with the IC-9700 selected as the radio model). CardSat carries the *same* CI-V frames shown below inside the RS-BA1 UDP protocol, so MAIN/SUB, read-back, sat mode and CTCSS behave identically; only the transport changes. Other networked Icoms (IC-705, IC-7610, IC-785x) use the same RS-BA1 protocol so the link would establish, but they are single-receiver radios without the MAIN/SUB satellite architecture and are not supported. Setup is in §3; the wire protocol is documented in `docs/interfaces/ICOM_LAN_PROTOCOL.md`.

Yaesu (FT-847, FT-736R). 5-byte CAT (four data bytes + opcode), big-endian BCD at 10 Hz. CardSat enables CAT at startup and sets the satellite RX (downlink, opcode `0x11`) and TX (uplink, `0x21`) VFOs directly.

[1] **FT-847 read-back** uses "read freq & mode" (`0x03`, patched to `0x13` for the SAT-RX/downlink VFO), which returns 4 big-endian BCD bytes + a mode byte. This works only on **firmware-updated** FT-847s — early units have no read capability and stay silent (CardSat times out and holds the passband). In satellite mode the radio can occasionally return the uplink VFO instead (Hamlib #1286); CardSat rejects any read that jumps more than 1 MHz from the commanded downlink, so a stray reply holds steady rather than jerking the passband. So **radio-knob (One True Rule) tuning works on a firmware-updated FT-847**; on older units, use the device **TUNE** keys.

The **FT-736R** can't report frequency over CAT at all (TUNE keys only), and its native opcodes differ from the FT-847 — the proven path is an **FT-847-emulating** interface (KA6BFB / HS-736USB), in which case select **FT-847**. Put either radio in **VFO** (and its satellite mode in VFO) before enabling CAT.

Kenwood (TS-790, TS-2000). ASCII commands terminated by `;`: downlink on **VFO A** (`FA<11-digit Hz>;`), uplink on **VFO B** (`FB…;`), mode via `MD<n>;`, read via `FA;`. Connects over **RS-232** (DB-9) — use a **MAX3232** level shifter, not the CI-V circuit. On the TS-2000, bridge **CTS/RTS** (or use "RTS +12 V") for its handshake quirk and verify the VFO-A/B

↔ main/sub-band behavior for your firmware. The TS-790 supports a subset of the same commands.

> **All Yaesu/Kenwood sat rigs:** CAT **cannot switch the band pair.** You select the uplink/downlink bands and engage the rig's own satellite / full-duplex mode **on the radio**; CardSat Doppler-tunes within that. (Same as SatPC32.)

Watching the CAT trace

CardSat prints **every CAT frame it sends** to the serial monitor at **115200 baud**, decoded, with the radio's reply where the protocol provides one. Connect over USB, open a serial monitor at 115200, and you'll see traffic like:

> **If you drive the radio over USB, the console drops while engaged.** Engaging USB CAT (or a USB rotator) claims the S3's one USB PHY, so there is no serial monitor to watch while it's engaged (since v0.9.64 the console returns when you disengage). Two ways round it: run the radio on **Grove** and keep the console, or turn on **Settings → Station / logging → `Console to file`** and read the trace from `/CardSat/Logs/console.log` afterwards (web files portal, or pull the card). See *Logs on disk* below.

Icom (CI-V):

```
[CI-V TX] FE FE A2 E0 07 D1 FD  sel-band SUB
[CI-V TX] FE FE A2 E0 05 00 60 58 45 14 FD  set-freq 145456000 Hz
[CI-V RX] radio ACK (FB)
[CI-V TX] FE FE A2 E0 03 FD  read-freq req
[CI-V] SUB freq read: 145456000 Hz
```

Yaesu / Kenwood:

```
[CAT TX] 14 59 00 00 11  set-freq RX 145900000 Hz
[CAT TX] 43 59 00 00 21  set-freq TX 435900000 Hz
[CAT TX] FA00145900000;
[CAT TX] MD2;
[CAT RX] FA00145900000;
[CAT] VFO-A (downlink) read: 145900000 Hz
```

How to read it:

- **`[CI-V TX]` / `[CAT TX]`** — a frame sent to the radio (raw bytes, or the ASCII string for Kenwood) plus a decode: which VFO, set-freq `<Hz>`, set-mode, read request, or CAT-on.
- **`[CI-V RX] radio ACK (FB)`** — Icom accepted the command; **`NAK (FA)`** means it rejected it. (Yaesu set commands aren't acknowledged.)
- **`… freq read: <Hz>`** — the frequency read back (Icom/Kenwood) used by radio-knob One True Rule tuning; `no valid reply` if it failed.

This is the fastest way to confirm your **wiring, model, address, and baud**, and to watch the **Doppler corrections** stream during a pass. The **Icom LAN** backend adds its own `[NET]` traces for the connect/auth handshake and keepalives (the CI-V frames it carries are the same ones shown above). Silence a backend by setting `CIV_DEBUG` / `YAESU_DEBUG` / `KW_DEBUG` / `ICOMNET_DEBUG 0` at the top of `civ.cpp` / `yaesu.cpp` / `kenwood.cpp` / `icomnet.cpp` and rebuild.

Logs on disk (`/CardSat/Logs`)

Everything below is retrievable through the **web files portal** (or by pulling the card), and works identically whether CardSat is on an SD card or bare internal flash.

`/CardSat/Logs/usb.log` — the USB story for *both* ports, in one file, in order. Written automatically; nothing to turn on. CAT lines are prefixed `cat:` and rotator lines `rot:`, because once a radio and a rotator are both in play an unprefixed line is ambiguous:

```
20268 cat: allocating USB host heap=85720 largest=31732
20437 rot: adapter[0] addr=1 FTDI FT232R 0403:6001 key=0403:6001/A50285BI
20514 cat: bind: done heap=56648 largest=31732
```

Every engage, bind, failure and teardown lands here with the free heap and largest contiguous block at that moment — so the same file that names a failure also profiles what it cost. A refusal always writes its reason (`No USB serial adapter detected`, `Only adapter is the radio's`, `That adapter is the rotator's`).

`/CardSat/Logs/console.log` — **Settings → Station / logging → `Console to file`**, default **off**. Mirrors the *entire* serial console to disk. This exists because engaging USB takes the console away for the session, which is exactly when you most want a trace.

Size is capped and cannot run away. Each log is limited and rotates once to `.1`: roughly **32 KB total on flash**, **512 KB on SD**. An uncapped log on a flash-only Cardputer would fill the same filesystem the GP cache and your config live on. Delete the `/CardSat/Logs` folder to reclaim all of it.

What console capture costs. Writes are buffered, so with capture on it is about **0.5% of the main loop** while Doppler tracking, and nothing measurable when off. The honest caveat: a **hard crash loses whatever is still in the buffer** — usually the last line or two. The buffer is flushed on every reboot CardSat performs itself and before a USB engage, and the USB trace above is written *unbuffered*, so the one trace that has to survive a freeze always does.

> **Leave it off unless you are diagnosing something.** It is a diagnostic, not a feature — the cost is small but it is not zero, and the log will happily record thousands of lines of Doppler chatter you will never read.

CAT self-test

Settings → Radio / CAT → Run CAT self-test exercises the whole CAT link in one shot, on the device, with no satellite pass needed. For a wired (CI-V/LAN) radio it runs straight away; for a **USB CAT** radio that isn't already engaged, it brings USB CAT up itself and asks you to re-run once the transport is live (a moment later). Leaving the test turns USB CAT back off again if the test was what engaged it. The test sets and reads back the **downlink** and **uplink** frequencies, sets the **downlink/uplink modes**, sends a **band select**, toggles **sat mode**, and probes **PTT read** and **CTCSS** — reporting each step as `[PASS]`, `[FAIL]`, or `[INFO]` (the latter for things a given radio doesn't support over CAT, like sat-mode commands on the older rigs or PTT read where the protocol has none). `;`/`.` scroll the results, `` ` `` exits. It's the quickest way to confirm a new radio, address, baud, and wiring are all correct before a pass — a `[FAIL]` on the set+read lines points straight at a CI-V address / baud / wiring problem, while `[INFO] … not supported` lines are normal and just reflect that radio's capabilities.

> **It's a bench test, not a fully state-preserving check.** The self-test sets both VFOs to test frequencies, changes the downlink/uplink modes, toggles satellite mode

(left off), and briefly enables CTCSS (left off). It restores the frequencies only if CardSat had cached non-zero dial values, and it does **not** restore the prior modes, sat-mode state, tone, or selected VFO. After running it, re-engage tracking (`r` on Track) or reselect the transponder to put the rig back into a known operating state. It never keys the transmitter, so it's safe to run off-air.

17

Antenna rotator (GS-232, Easycomm, SPID, rotctl, PstRotator, Yaesu direct, rotctld server)

CardSat can both **drive** a rotator and **act as a rotator server** for other software. The network *client* backend is labeled **rotctl** in Settings — it connects to a rotctld server, it is not itself the daemon. Separately, CardSat can run its **own rotctld server** (Settings → Rotator → *Rotctld server*, default port **4533**) so a networked PC (Gpredict, ...) drives whichever rotator CardSat is configured for; a `P` from the PC disengages CardSat's own tracking. A **manual control** screen (Settings → Rotator → *Rotator: manual control*) jogs az/el by hand with live read-back. With the azimuth **el range** set to 180°, the *flip* decision is made once per pass and held to LOS. The `rotctld` and `rigctld` servers have **no authentication** — run them on a trusted LAN only.

CardSat can drive an az/el antenna rotator through one of several interchangeable backends, chosen in Settings with the **Rotator** type selector (set it to **None** if you have no rotator — this is the default, and replaces the separate on/off row used in earlier versions):

- **GS-232** — a directly-attached controller speaking the **Yaesu GS-232A/B** protocol (a Yaesu G-5500 with a GS-232B, a SPID controller in GS-232 mode, or any GS-232 emulator: K3NG / RadioArtisan / ERC).
- **Easycomm I / II / III** — the open, plain-ASCII tracking protocol used by **SatNOGS, K3NG, ERC** and most homebrew rotator controllers (the same protocol Hamlib's `easycomm` backends speak). Choose the version your controller expects: **II** is the common decimal-degree form (`AZ123.4 EL045.0`), **I** is the older integer form, and **III** shares II's positioning grammar. Runs over any **`Rot wire`** transport (below).
- **SPID Rot2Prog** — the binary protocol of **SPID MD-01 / MD-02** (Alfa/ RFHamDesign) controllers, as documented in Hamlib's `spid` (`rot2prog`) backend. Whole-degree positioning over any **`Rot wire`** transport (below). (If your SPID box is set to GS-232 emulation, you can use **GS-232** instead.)
- **rotctl (net)** — connects to a **Hamlib `rotctld` server** anywhere on the same WiFi network. CardSat is the TCP client (the same role Gpredict plays), so any rotator Hamlib supports can be driven over the LAN with no extra CardSat wiring.
- **PstRotator (net)** — a **PstRotator** (YO3DMU) instance on the LAN, driven over **UDP** (the same datagrams Gpredict/SatPC32 send it). CardSat sends <PST>...</PST> control messages to PstRotator's UDP Control port; PstRotator then drives whatever controller it is configured for.
- **Yaesu (direct)** — a Yaesu **G-5500**-class az/el controller wired **straight to CardSat**, with no GS-232 box: an I2C **ADS1115** reads the position-feedback voltages and an I2C **PCF8574** drives the four direction lines, both on the same I2C bus the GS-232 bridge would use. CardSat closes the pointing loop itself. This is a hardware build with its own calibration step — see **ROTOR_INTERFACE.md** (**!** untested; build at your own risk). Calibrate from **Settings → Rotator → Rotator: manual control** (capture keys **1/2/3/4** at the axis endpoints).

Which wire a serial rotator runs on is a separate setting: `Rot wire`. The protocol (`Rot type`) and the transport are independent, so any of the three serial protocols — GS-232, Easycomm, or SPID — works over any of these three wires (the network backends need no wire at all, and Yaesu (direct) uses its own I²C hardware):

- **I2C bridge** (default) — an **SC16IS750/752 I2C→UART bridge** on a second I2C bus. This is what every pre-0.9.58 configuration meant, and it is still the only option that coexists with *both* a Grove radio and a Grove GPS.
- **Grove G1/G2** — the Cardputer's Grove port, driven directly. **The Cardputer has one Grove port**, and wired CI-V CAT and the Grove GPS use the same two pins, so this needs **CAT on USB or LAN** and **GPS not on Grove**. CardSat enforces that both ways: it refuses a Grove rotator while Grove is taken, and if you later put CAT or GPS on Grove the rotator **yields** to the I2C bridge and tells you.
- **USB adapter** — a USB↔serial adapter on the USB-C port, sharing the resident USB host with USB CAT. Engaging USB claims the S3's one USB PHY, so the **serial console closes for the rest of the session** (see §16). Rotator-only works; radio *and* rotator together is supported but **experimental** — see *Two USB adapters* below.

  The Easycomm and SPID backends are host-verified only. The ASCII formatting/parsing and the SPID binary frame encode/decode were checked off-device, but neither has been exercised against a physical controller yet. Treat them as untested; verify carefully before trusting them with real hardware.

Only one rotator is active at a time. The pointing logic — alignment offsets, deadband, flip mode, park-on-LOS — is identical for all of them; **Rot type** only changes how the final azimuth/elevation reaches the hardware.

17.0.0.1 Two USB adapters: radio and rotator at once (experimental) With **one** adapter there is nothing to configure: whichever port engages takes it, and the other is refused rather than both binding the same wire.

With **two** — say an IC-9700 on a hub plus a rotator adapter — assign them explicitly:

1. **Settings → Radio →** `Scan USB adapters` (the same row exists under Rotator). This brings the USB host up and enumerates. It is a deliberate keypress because it closes the serial console for the session; that should be your choice, not a side effect of opening a menu.
2. `Radio USB:` — cycle to the adapter driving the radio.
3. `Rot USB:` — cycle to the other one. The radio's adapter is **skipped**: visible in the list but not selectable, so you can see it is taken. The same applies in reverse.

Selections persist by a stable key — the adapter's **serial number** where it reports one (FTDI and CP210x usually do; CH340 usually does not), else VID:PID plus address. Two adapters of the *same model with no serial numbers* can only be told apart by address, so **re-check the selection after replugging those**.

`Auto` (the default) means *"the only adapter that is not the other port's"*. With two adapters and neither nominated, engage **refuses** rather than guessing — enumeration order is not a decision.

> **Untested.** This path has never had a radio and a rotator plugged in together. The hazard it guards against is real: left at the library's default, two ports both bind the *first* enumerated adapter and the radio's Doppler writes land on the rotator. CardSat binds explicit addresses to prevent that, but treat the combination as experimental.

IC-9100 / IC-9700 owners: those radios present an internal hub carrying both a serial port and a **USB Audio** device. CardSat does not use USB audio, and audio can never be mistaken for the CAT port — a serial candidate needs a *bulk* endpoint, and audio streams are *isochronous*. What audio costs is a device slot's worth of enumeration: a 9700 is a composite device (one USB address, two interfaces), so a 9700 plus a rotator adapter is **3 devices** of the 4-slot budget. If you ever see a slot-exhaustion error, that is the signal — it is reported explicitly rather than as a vague "no device".

Wiring

Chain: **Cardputer I2C (Wire1) → SC16IS750 → MAX3232 → DB-9 → GS-232 controller.**

- The bridge runs on the Cardputer-ADV expansion I2C bus, **G8 = SDA / G9 = SCL** (`ROT_I2C_SDA` / `ROT_I2C_SCL` in `config.h`), on a second I2C controller (Wire1) so it never touches the keyboard/IMU bus. These are confirmed from the M5Stack Cap LoRa-1262 pinmap and don't collide with CAT (G1/G2), the GPS UART (G13/G15), the LoRa SPI (G3/G4/G5/G6/G14/G39/G40), or the SD card (CS G12).
- GS-232 uses RS-232 voltage levels, so a **MAX3232** sits between the bridge's TTL TXD/RXD and the controller's DB-9 (three wires: TXD, RXD, GND).
- The SC16IS750 needs a 14.7456 MHz crystal (the common breakouts have one); set `ROT_XTAL_HZ` if yours differs, and its I2C address (A0/A1 strap) in `ROT_I2C_ADDR` (default 0x4D).
- On the Cap LoRa-1262, G8/G9 is exactly the I2C bus broken out to the cap's **HY2.0-4P Grove Port.A**, so a Grove SC16IS750 bridge plugs straight in. That bus also carries the cap's PI4IOE5V6408 IO expander (~0x43/0x44, used only for the LoRa RF switch, which CardSat never drives), so keep `ROT_I2C_ADDR` (default `0x4D`) clear of those — or add a TCA9548A mux if anything clashes.

Settings

Enable and tune the rotator in **Settings** (scroll past the radio rows):

Setting	Meaning
Rotator	off / on (builds the selected backend)
Rot type	**GS-232** (I^{2}C bridge), **Easycomm I/II/III** (I^{2}C bridge), **SPID Rot2Prog** (I^{2}C bridge), **rotctl (net)** (TCP), **PstRotator (net)** (UDP), or **Yaesu (direct)** (I^{2}C ADC + output expander, no GS-232 box)
Net host	server IP / hostname (rotctl or PstRotator backend)
Net port	server port — rotctld TCP **4533**, PstRotator UDP **12000**
Rot baud	GS-232 serial speed (1200 / 4800 / 9600); GS-232 backend only
Rot deadband	degrees; suppress moves smaller than this (anti-chatter)
Rot park az	azimuth the rotator parks at on LOS / when disabled
Rot pre-point	lead before AOS to pre-aim at the rise bearing (off / 30 s / 1-5 min)

Setting	Meaning
Rot Az offset	added to commanded azimuth (mount alignment)
Rot El offset	added to commanded elevation
Rot az range	azimuth travel: **0..360** (default), **-180..+180** (centered on N), or **0..450** (90 deg overlap)
Rot az lookahead	,// 0–10 s (default 3 s; 0 = off). On a **0..450** rotator only: CardSat predicts the bearing this many seconds ahead and, when a pass is about to cross north, commits early to the 361–450° overlap band so it makes a short move instead of unwinding ~360°. Has no effect on 0..360 or flipped passes. Tune to your rotator's slew speed
Rot el range	**90 deg**, or **180 deg** = flip over the top for high passes

Rot type and Net port adjust in place with ,//; Net host and Net port also open a text editor with ENTER.

Using it

On the **Track** screen press **o** to start/stop pointing. The status line shows **Rot ON / off / n/c** (*n/c* = no link: the I2C bridge wasn't found, or the rotctld server isn't reachable). Pressing o re-attempts the link on the spot, so you can engage as soon as the controller or server comes up. While on, CardSat sends the satellite's azimuth and elevation about once a second — the GS-232 W aaa eee command, rotctld's P <az> <el>, or PstRotator's <PST><AZIMUTH>..</AZIMUTH><ELEVATION>..</ELEVATION></PST> datagram — but only when the position has moved past the deadband, since rotators are slow and mechanical. When the satellite sets, or you press o again or leave the screen, the rotator **parks** at the configured azimuth/elevation.

Pre-positioning before AOS. A slow rotator can take most of a minute to slew across the sky, so by the time it reaches the rise bearing the pass has already begun. With **Rot pre-point** set to a lead time (default **2 min**; off disables it), CardSat predicts the next pass for the tracked satellite and, once AOS is within that window, aims the rotator at the **rise azimuth on the horizon** instead of the idle park position — so it is already pointed when the satellite appears and live tracking takes over smoothly. Set the lead a little longer than your rotator's worst-case slew time. Outside the window it still rests at **Rot park az**.

Azimuth range (0-360 / ±180 / 0-450). By default CardSat assumes the azimuth axis runs **0 to 360 deg**, 0 deg = North, 180 deg = South. **Rot az range** offers two alternatives for rotators built differently:

- **-180..+180** — the axis is centered on North and swings ±180 deg. CardSat re-expresses each bearing into that range (e.g. 270° → −90°), the same option Gpredict offers.
- **0..450** — the rotator has **90 deg of overlap** past North (a common Yaesu/SatPC32 arrangement). A bearing of 0-90 deg is reachable either directly or as 360-450 deg, so CardSat commands whichever is **nearer its current position**: a pass crossing North

continues up into the 360-450 region instead of unwinding a full turn — no dead-zone spin at culmination.

Both affect what is sent on the **rotctld** path (e.g. `P -90.0 12.0` or `P 372.0 8.0`). GS-232 controllers natively accept 0-450 and re-wrap negatives, so the GS-232 backend follows the overlap but the $\pm$180 framing has no visible effect there.

For overhead passes, setting **Rot el range** to **180 deg** enables a **flip**: near culmination CardSat commands elevation past 90 deg (over the top) together with a 180 deg azimuth flip, so an antenna on a 0-180 deg elevation rotator rides through zenith without the fast 180 deg azimuth swing a 0-90 deg mount would need. Leave it at **90 deg** for a conventional elevation rotator. CardSat points **open-loop** from its own SGP4 prediction (it does not poll the controller's heading); the GS-232 `C2` read-back exists in the backend for diagnostics but is not used in the tracking loop.

Network rotators (rotctld and PstRotator)

To use the network backend, run `rotctld` on any machine that can reach your rotator (or use a controller that speaks rotctld natively), set **Rot type** to **rotctl (net)**, and enter the server's address in **Net host** (an IP is simplest) and **Net port** (Hamlib default **4533**). A dummy server for a bench test:

```
rotctld -m 1 -t 4533 -vvvvv
```

`-m 1` is Hamlib's dummy rotator model and `-vvvvv` makes rotctld log every command it receives, so you can watch CardSat's `P`/`S` commands arrive without moving real motors (without the verbose flag the server is silent, which can look like nothing is being sent); swap in your rotator's model and serial port for the real thing. CardSat opens the socket when you enable the rotator and reconnects on its own (throttled to a few seconds) if it drops, so a server that is briefly down or rebooted recovers without intervention. Pointing is open-loop from CardSat's SGP4 prediction — it does not read the heading back.

Hardware that speaks rotctld natively works the same way, with no PC in between. For example, the **MuseLab AntRunner** (BG5DIW) portable az/el rotator (360 deg azimuth / 180 deg elevation, controlled over WiFi by its onboard ESP32) and the **AntRunner-Pro** (fixed-install, 360 deg / 90 deg, controlled over Ethernet) both present a Hamlib `rotctld`-compatible network service. Set **Rot type** to **rotctl (net)** and point **Net host** / **Net port** at the AntRunner's rotctld service (or at a `rotctld` instance bridging to it over USB).

> rotctld has **no authentication** — keep it on a trusted LAN and never expose the port to the internet. Several clients may connect to one rotctld and can contend for the rotator.

PstRotator (UDP). If you already run **PstRotator** on a shack PC, CardSat can drive it directly instead of a rotctld server. In PstRotator, open **Communication > UDP Control Port**, set the port (default **12000**), and tick **UDP Control** in Setup. On CardSat set **Rot type** to **PstRotator (net)**, **Net host** to the PC's IP, and **Net port** to that UDP port — *remember to change it from rotctld's 4533 to 12000.* CardSat then sends <PST><AZIMUTH>az</AZIMUTH><ELEVATION>el</ELEVATION></PST> to point and <PST><STOP>1</STOP></PST> to stop. PstRotator does its own azimuth range, offsets and flip, so CardSat sends a plain 0-360 bearing here; leave CardSat's **Rot az range** / **Rot el range** at default and let PstRotator manage overlap and flip (or use CardSat's and keep PstRotator neutral — just don't double up).

This backend also drives the **WA4MCM PSR-100** (Don Friend, WA4MCM) — a popular portable, lightweight az/el satellite-rotor kit for field/portable work. The PSR-100 accepts the same PstRotator-style `<PST>` UDP az/el datagrams over its WiFi interface, so CardSat points it directly: set **Rot type** to **PstRotator (net)** and aim **Net host** / **Net port** at the PSR-100's UDP interface — no PC running PstRotator in between.

> The PstRotator UDP control is **unauthenticated and connectionless** — keep it on a trusted LAN. Because UDP has no connection to lose, CardSat reports **Rot ON** whenever WiFi is up and a host is set; it cannot tell whether PstRotator is actually listening, so confirm the antenna follows on the first pass.
>
> The rotator only points while you are on the Track or Polar screen. It only moves the antenna — it does not change the radio's bands, and CAT on the Yaesu/Kenwood sat rigs can't switch the band pair either, so set those by hand.

The GS-232 backend is bench-reasoned against the GS-232A/B manuals and Hamlib's gs232a/gs232b backends, and host-tested for the bridge baud math and command formatting — it has not been run against a physical SC16IS750 or a real rotator. The I2C pins (G8/G9) are confirmed from the Cap LoRa-1262 pinmap; still confirm the SC16IS750's strapped address and the controller's baud before keying real motors. The rotctld backend follows the published Hamlib `rotctld` protocol and is the one rotator path you can fully bench-test without trusting hardware encoders — point it at `rotctld -m 1` and watch the commands — but it has not driven a physical rotator either. The **PstRotator** UDP backend is host-verified for message formatting against the PstRotator manual (Rev. 7.5); it has not been run against a live PstRotator instance.

18

Mobile web control

CardSat can serve a small **mobile-friendly web page over your WiFi network**, so a phone, tablet, or laptop on the same network can drive it without touching the keypad. It's **off by default**.

Enable it: *Settings → Network / data → Web control* (`,`/`/` or ENTER to turn on; **Web port** sets the port, default 80). Once it's on and CardSat is connected to WiFi, the **Web control** row shows the address to open — for example `http://192.168.1.42`. Type that into any browser on the same network.

What the page does:

- **Satellite selection** — pick one of your favorites from the drop-down and tap **Track** to make it active. (If you haven't marked any favorites, the list falls back to the catalog.) A **filter box** narrows a long list as you type, and the **star** button beside the selector adds or removes the chosen satellite from your favorites, just like marking one on the device.
- **In-pass indicator & AOS countdown** — the header shows a live **"IN PASS — LOS in m:ss"** while the active satellite is up (with the peak elevation and time under the sky plot), and a **"Next AOS in m:ss"** countdown otherwise, so you know at a glance when to be ready. A small green **RAD** / **ROT** / **R+R** badge in the header lights up whenever the device's radio and/or rotator are tracking — mirroring the on-device header tag — so you can tell from your phone that the rig may be transmitting even after you've left the device's Track screen.
- **Live sky plot** — a polar plot (N up, elevation as distance from the rim) that **always draws the pass arc with a direction-of-travel arrowhead** and an AOS marker: while the satellite is up it shows the **current** pass path with the live position dot overlaid; between passes it shows the **next** pass path so you can see where it will rise and set. The green dot appears the moment the bird comes up.
- **Adaptive refresh** — the page polls **faster during a pass** (sub-second, so the fast-moving Doppler frequencies stay current) and **slows down between passes** to save battery and network traffic, backing off automatically if the connection drops and showing **"reconnecting..."** until it recovers.
- **Live readout** — downlink (RX) and uplink (TX) frequencies, azimuth/elevation, and the current tune mode. Beside the RX label a small **±N.NN kHz** figure shows how much **Doppler shift** is being applied right now. **Tap either frequency to copy it** to the clipboard (handy for setting a second radio by hand). Below the mode, an **AMSAT activity line** shows the current bird's live status when it has been reported — e.g. **"AMSAT: Heard 3h ago · 5 rpt"**, colored green (heard), yellow (telemetry), or red (not heard).
- **Transponder selection** — a drop-down lists the active satellite's transponders by name and mode; choosing one selects it on the device (the same selection the `t` key cycles), so you can pick the FM channel or the linear passband from your phone.
- **Radio & rotator control** — the **tuning cluster** nudges the passband down/up with a **visible, tappable step** (100 Hz / 1 kHz / 5 kHz — the same step the `s` key cycles), plus **Center** and **TX→**. Buttons also switch **radio** and **rotator** output on/off. These do exactly what the corresponding keys do on the Track screen — the web page drives the

same controls, it doesn't second-guess them. Rapid taps are debounced so a burst can't queue up stale commands.

- **Fast calibration pad** — the centerpiece for correcting drift **during a pass**: **RX cal** and **TX cal** each get big one-tap **−/+** buttons with a **tappable step** (10 / 100 / 1000 Hz) and a live readout, so you can net onto a signal without a keyboard. **Zero cal** clears both in one tap. An **Exact...** toggle reveals the type-an-exact-value-in-Hz path when you want a precise number. Calibration is saved per-satellite, the same as on the device; for a linear bird the current passband position and bandwidth are shown. The RX/TX dials are always computed live from the transponder plus Doppler plus this calibration, so calibration is the operator-settable correction rather than a free-typed absolute frequency.
- **Manual (no-radio) tuning** — tap **Manual** to switch the control card to the hand-tuning calculator. It shows the **HOLD** leg (the frequency to park your own radio on) and the **TUNE** leg (the Doppler-corrected frequency to follow), exactly as the on-device Manual screen does, with the same round-trip Doppler correction on linear birds. **Swap leg** chooses whether you hold the downlink or the uplink; Tune ±, next transponder, CAL, and recenter work as on the device.
- **Orbital analysis** — tap **Orbit** for a read-only summary of the same numbers the on-device orbital-analysis pages compute: altitude and footprint, period, apogee/perigee, inclination/eccentricity, decay estimate, live range-rate and sub-point, sunlit/eclipse state, beta angle, J2 node and perigee drift (with a sun-synchronous flag), mean/true anomaly, time to perigee/apogee, the instantaneous **orbital velocity**, the **launch year and time in orbit** (from the COSPAR designator), the element-set number and launch-sibling count, and the multi-day pass outlook. It's view-only — nothing on the device changes.
- **Upcoming passes** — a table of the next passes (AOS, peak time, max elevation, duration, rise→set azimuth) with a **star** marking each **optically-visible** pass (sunlit satellite, dark sky). Above it, a compact **timeline bar** shows the same passes laid out in time — visible ones in orange, a white tick for "now" — so you can eyeball spacing at a glance. **Tap a pass** (row or bar) to arm a **browser AOS notification**: your phone alerts you when that pass begins (grant the notification permission when prompted).

The page is **responsive** — a single column on a phone, and a two-column layout on a tablet or computer. It coexists with the rigctld/rotctld servers and the normal on-device UI; the device keeps tracking and you can use its keypad at the same time.

Files (download). A **Files** link in the page header opens a simple file browser at `/files` for pulling files off the device without removing the SD card or rebooting into a Launcher. It lists the `/CardSat` tree — rove plans, screenshots, logs, notes, and the like — with sizes and modified times; click a folder to descend, click a file to download it. To grab several at once, tick the checkbox beside each file and press **Download selected** — the browser saves them one after another. (The device still streams a single file per request, so multi-select adds no memory cost on the unit; a browser may ask permission the first time a page hands it several downloads.)

Access is confined to `/CardSat` (a path guard rejects anything outside that tree), and the page is **download-only** — there is no upload path, so it cannot write to or modify the device's filesystem. Like the rest of web control it's unauthenticated on the LAN, so the same trust caveat applies.

> **Developers:** the page is backed by a small **HTTP + JSON API** (`/api/status`, `/api/sats`, `/api/passes`, `/api/orbit`, `/api/tx`, and a few `POST` controls)

that a third-party client can call directly. It's fully documented — every endpoint, parameter, and response field, plus the access model — in `docs/interfaces/WEB_API.md`.

Security. This is plain **HTTP on the local network with no password** — anyone who can reach your WiFi can open the page and operate the radio and rotator. Use it only on networks you trust, and don't forward the port to the internet. Turning **Web control** off stops the server immediately. In the field, pairing this with a **phone hotspot** (see the second-WiFi network in §7) gives you a private network for phone-to-CardSat control with no other infrastructure.

Network control roles: client vs server

Besides the web page, CardSat speaks the **Hamlib network protocols** — and it can sit on *either* end of them. It helps to keep the two roles straight, because the names differ by a single letter:

- **CardSat as the client** — *it* drives a remote device. **`rigctl (net)`** (a CAT type, §3) makes CardSat connect to someone else's **`rigctld`** server to control a radio; **`rotctl (net)`** (a rotator type, §17) makes it connect to a **`rotctld`** server to point an antenna. In both cases CardSat is the one issuing commands, and the daemon name ends in **-d**.
- **CardSat as the server** — *other software* drives **CardSat**. The **Rigctld server** (Settings → Radio / CAT) and **Rotctld server** (Settings → Rotator) make CardSat itself a **`rigctld/rotctld`** daemon, so a PC running Gpredict, WSJT-X, a logger, or plain `rigctl/rotctl` can drive whatever radio and rotator CardSat is connected to — CardSat does the Doppler and pointing, the PC just sends frequencies or bearings.

The rule of thumb: the **client** option in CardSat's menus is named for the *client tool* (`rigctl` / `rotctl`), while the **server** option is named for the *daemon* it becomes (`rigctld` / `rotctld`). Default ports follow Hamlib convention — **4532** for rig, **4533** for rotator — and **none of these network services use authentication**, so keep them on a trusted LAN.

19

Printing to a receipt printer

CardSat can print text reports to a **network thermal receipt printer** — the pocket battery kind — over WiFi. It speaks **ESC/POS over raw TCP port 9100**, which virtually every receipt printer supports; the reference target is an **Epson TM-P20II (WiFi model)**. (Bluetooth printers are *not* supported: the ESP32-S3 has no Bluetooth Classic, and most cheap BT receipt printers are Classic-only. See docs/design/PRINTING_SCOPE.md.)

Setup. Put the printer on the same WiFi, note its IP, and enter it in **Settings → Network / data → Printer IP** (port defaults to 9100). **Printer format** sets the page language to match your hardware: **ESC/POS (receipt)** (default — thermal receipt printers like the TM-P20II or GZM8022), **Plain text** (no control codes; the universal fallback most raw-9100 printers accept), **PCL (HP)** (HP LaserJet / OfficeJet and other office printers — a PJL-wrapped job with a fixed-pitch font and page-eject), **PostScript** (PostScript office printers — a Courier document with automatic page breaks), **ESC/P2 (Epson)** (Epson page / inkjet / dot-matrix printers), **Star Line** (networked Star thermal printers, TSP-series), **ZPL (Zebra label)** (network label printers — each report line becomes a positioned field on a label). Serial and file outputs are always plain text.

> The ESC/POS, plain-text, PCL and PostScript paths are the primary targets; **ESC/P2, Star Line and ZPL are provided but untested against real hardware** — verify on your device before relying on them, and use the /CardSat/Reports file as a fallback.

Raster printing for driverless printers (AirPrint / IPP Everywhere / Mopria). Most network printers sold today are "driverless" — they accept raster page images over IPP rather than a page-description language, and PWG Raster is the format the great majority of them support (it is *required* by the IPP Everywhere standard, and many AirPrint printers accept it too). CardSat can generate this on the device: select **Printer format → PWG raster (AirPrint)**, or **URF raster (AirPrint)** for the Apple-Raster variant that some AirPrint printers require instead. CardSat renders the report to a 300-DPI grayscale page one scanline at a time (~4 KB working memory, no full-page buffer) and streams it over IPP (port 631). Both encoders were validated byte-identical to the reference `ppm2pwg`/`ppm2pwg -f urf` tools and confirmed printing on a real AirPrint printer.

Which raster format? Try **PWG raster** first — it reaches the most printers. If the printer accepts only Apple Raster, use **URF raster**. To find out what your printer supports, use **Test printer (probe formats)** in the printer settings: it asks the printer over IPP and shows which document formats it accepts (e.g. "PWG URF PDF JPEG"). If PWG or URF appears, the matching raster format will work. Raster is heavier than the text formats — prefer a text format (PCL/PostScript/ESC-POS) where the printer supports one; use raster when it is the only thing the printer accepts, which is the common case for home printers.

Suggested settings by printer type. Set these under Settings → Network / data. When unsure on a network printer, run **Test printer** first to see what it accepts.

Printer type	Example	Printer format	Paper width	Transport
Pocket WiFi receipt (58 mm)	Epson TM-P20II	ESC/POS (receipt)	58 mm (32 col)	Raw : 9100
Desktop WiFi receipt (80 mm)	GZM8022	ESC/POS (receipt)	80 mm (48 col)	Raw : 9100
HP office laser/inkjet	LaserJet, OfficeJet	PCL (HP)	64 col	Raw : 9100 (or IPP)
PostScript office printer	workgroup printers	PostScript	64 col	Raw : 9100 (or IPP)
AirPrint / IPP Everywhere home printer	most modern home printers	PWG raster	64 col	IPP : 631 (auto)
AirPrint printer, URF-only	some HP LaserJets	URF raster	64 col	IPP : 631 (auto)
Epson page/inkjet (raw)	ESC/P2 models	ESC/P2 (Epson)	64 col	Raw : 9100
Star thermal (network)	TSP-series	Star Line	80 mm (48 col)	Raw : 9100
Zebra label printer	ZPL models	ZPL (Zebra label)	32–48 col	Raw : 9100
No printer	—	any	any	Serial / file only

The raster formats (PWG, URF) switch to IPP on port 631 automatically — you do not need to change the transport. Only set **Printer transport → IPP** by hand for a PCL/PostScript office printer that exposes IPP but not raw 9100.

Printer transport chooses how the network job reaches the printer: **Raw (:9100)** — the JetDirect raw-socket path used by all the formats above (the default), or **IPP (:631)** — the Internet Printing Protocol, an HTTP job for office printers that expose IPP but not raw 9100. IPP carries your selected page language (use **PCL** or **PostScript** with it) as the document. **Important limitation:** IPP on CardSat sends PCL or PostScript; it does **not** rasterize. Many modern home/AirPrint printers are **raster-only** — they advertise only `image/urf` or `image/pwg-raster` and have no PCL/PostScript interpreter. Those printers will **accept an IPP job and then silently discard it** (your computer prints to them only because the computer rasterizes first). CardSat cannot drive a raster-only printer — there isn't enough RAM on the ESP32-S3 to render a full-page bitmap. For such a printer, use **Save to /CardSat/Reports** and print the .txt from a computer. To check what a printer supports, the `tools/ipp_probe.py` script queries it and sends test pages. **Printer paper** sets the width: **58 mm (32 col)**, **80 mm (42 col)**, **80 mm (48 col)**, or **Font B (64 col)** — every report reflows to match, and the wider paper adds columns where it helps (the day-sheet gains an LOS time and full-width names; the log puts each QSO on one line).

Three ways to receive a report — you don't need a printer. Any report can go to any combination of three places, set in Settings → Network / data: - the **network printer** (Printer IP above), - the **USB serial console** (*Print to serial: on*) — the report prints in your serial monitor, ready to copy and paste, so an operator with no printer still benefits, - a **text file** (*Save to /CardSat/Reports: on*) — an 80-column `.txt` under `/CardSat/Reports` on the SD card, named per report (e.g. `passes-123456.txt`), to pull off later.

If none of the three is enabled a print action says so; if the printer is unreachable but serial or file is on, the report still goes there and the status notes the printer failed.

Printing is contextual — print from the screen that shows the data. Press `p` on the relevant screen: the **Passes** screen prints the day-sheet; **Mutual windows**, **DX Doppler**, **EQX**, and **Target search** each print their table; the **Notes** browser prints the highlighted note; the rove-plan viewer prints the plan you're reading. On the **schedule**, `P` prints every favorite's upcoming passes. On a pass's **polar (sky-track) screen**, `P` prints an ASCII sky map of that pass — the satellite's arc across the sky in printer-safe characters (`+` zenith, `.` horizon, `*` track). In the note editor, `Fn+p` prints the note. On the **About** screen, `p` opens a **Print submenu** listing **every** report — the one place to reach the ones without a natural home screen (ticket, satellite card, Keplerian elements, QSO log, workable horizon) as well as all the others; ENTER prints the highlighted one and the list stays open. `a` and `c` on About remain direct shortcuts for the *Support AMSAT* page and your *operator contact card*. Over the USB serial console (115200), the `print` command does any report from a keyboard — `print passes|ticket|card|keps|log|rove|` `horizon|amsat|contact|mutual|dxdopp|eqx|allpass|target|note|orbit|illum|tenday|timeline|` The console also has read-only diagnostics: `heap` (free heap + largest block), `mem` (a full memory baseline — live heap, min-ever high-water, and the static byte cost of the catalog and the large per-screen arrays), and `memtrace` (toggles a one-line heap log on every screen change, for profiling which screens cost the most). The About screen's heap line likewise shows live / largest-block / minimum-ever free heap. These exist to establish a memory baseline ahead of any future RAM optimization.

The reports. *Passes* — favorites' AOS/elevation/azimuth/duration for 24 h, one line each. *Ticket* — an outreach slip for the active satellite (rise time, direction, and the **selected transponder's** downlink/uplink — whichever you have chosen on Track — plus a friendly line) for demos and classrooms. *Card* — the active satellite's transponders (for a **linear** transponder the whole passband is shown, low–high, not just the bottom edge) plus its next three passes. *Keps* — the active satellite's Keplerian elements. *Log* — your recent QSOs as a paper backup. *Rove* — the plan being viewed (viewer `p`), or a fresh export of the current survey from the menu/console. *Horizon* — the workable-horizon union export. *Orbital analysis* — a **permanent record** of the active satellite's orbital characteristics at the current element set: Keplerian elements, epoch, period, SMA, apogee/perigee and their footprint diameters, orbital speeds at apogee/perigee, J2 nodal and apsidal drift, sun-synchronous flag, **LTAN**, repeat-track resonance, longest-possible pass, beta angle and full-sun/eclipse geometry, decay estimate, and launch identity. It deliberately **omits live values** (current range, Doppler, az/el, sub-point, next pass) so the printout is a lasting snapshot of the orbit rather than a momentary look. The one time-dependent entry is **LTAN** (local solar time of the ascending node), which is printed with the timestamp it was computed at and its drift rate in minutes per day: on a sun-synchronous orbit that rate is ~0 and the LTAN is effectively a fixed property, but on other orbits it sweeps quickly (the ISS moves about −24 min/day), so the rate tells you how fast the printed figure goes stale. Printed with `p` from the Orbital-analysis screen.

Illumination — the active satellite's sunlit fraction of each orbit over the next 60 days, plus an **ASCII raster** (one line per day, `#` sunlit / `.` eclipse across the orbit) mirroring the on-screen shading, from the Illumination screen. *10-day passes* — every pass of the active satellite over the coming ten days as a dated table (AOS/LOS/peak elevation) **and an ASCII day-timeline** (a 24 h UTC track per day, passes marked by elevation tier), from the 10-day screen. *6-hour timeline* — all favorites' passes in the next six hours as a sorted list **and an ASCII timeline** (one row per favorite across the six-hour window), from the Sky-at-a-glance timeline.

EME / moonbounce — the Moon's az/el, topocentric range and range-rate, path degradation vs perigee, the sky-noise flag, and **self-echo Doppler for all five bands** (50/144/432/1296/10368 MHz), from the EME screen with **`Fn+p`** (a bare `p` there opens the 30-day plan). *EME 30-day plan* — declination and degradation for each of the next 30 days, from that plan screen with `p`. *EME mutual Moon* — the common-Moon-window list against a DX grid, printed with **`Fn+p`** while that sub-view is showing. *QRZ lookup result* — the callsign, name, address, country, grid and class of the last lookup (`p`). *Station readiness* — the pre-operating checklist exactly as the screen shows it (`p`). *Awards* — QSO/grid/state/DXCC totals plus the per-satellite tally, from the Awards screen (`p`). *Workable US states* and *Workable DXCC* — the **entity lists** themselves, not just the counts (`p` on either screen). *Visible passes* — the optically-visible passes over the next ten days with their rise directions (`p`).

All times are **UTC**. Reports stream a line at a time, so printing costs the device almost no memory.

20

Managing data and factory reset

CardSat keeps all of its data in a **`/CardSat`** folder:

- cached GP data and per-satellite transponder caches,
- your manual GP satellites and manual transponders (kept separate, so a refresh won't erase them),
- favorites, per-satellite calibration, and all settings.

CardSat stores this on the **microSD card** by default (in the `/CardSat` folder), so your configuration travels with the card and is easy to back up. If no card is present at boot it **falls back to the internal LittleFS** flash (same `/CardSat` folder) so the unit still works standalone. The serial monitor reports which it mounted (`[fs] using microSD card for storage (/CardSat)` or `[fs] no SD card - falling back to internal LittleFS`). Insert the card **before powering on** — the filesystem is chosen once at startup. For the LittleFS fallback you must flash CardSat with a partition scheme that includes a data region, e.g. **Huge APP (3MB No OTA/1MB SPIFFS)**.

Screenshots

Press **`b`** on any screen (except the text editors, the Tools list, and the DXCC / character lookups, where letters are input) to save a screenshot of exactly what's displayed. Images are written to **`/CardSat/Screenshots/`** on the SD card as 24-bit BMP files named `shot_0001.bmp`, `shot_0002.bmp`, ... (never overwritten). A short high beep confirms each capture; a low beep means there's no SD card to write to — screenshots require the card, as the LittleFS fallback is too small for images. The key works on every screen **except** the text-entry screen (where `b` types normally), and it is intentionally hidden — no footer hint — so it never appears in the captured image. BMPs open anywhere and convert easily to PNG for documentation.

Factory reset: Settings → **Reset all data** → type **ERASE** to confirm. On the SD card this removes CardSat's own files in `/CardSat` (it never formats your card); on internal LittleFS it formats the data partition. Either way it reboots to a clean first-run state. Use it to start over or before handing the unit to someone else.

---

21 Troubleshooting

No passes / "Clock not set." Set the clock first: Update → `k` (NTP), enable GPS, or Location → `c` to enter UTC manually. Also confirm your location is set.

No transponders on Track. The satellite's transponders weren't cached. With WiFi, open the bird from Satellites (it fetches from SatNOGS), or use Update → `a` to cache everything. You can also add one manually (Passes → `n`).

"file too big for storage" on Update. The chosen GP source is larger than the free space on the active filesystem (most likely a big CelesTrak group on an internal-flash unit, whose LittleFS partition is small). Nothing was written — the previous catalog is intact. Use a microSD card, or pick a smaller group.

Printer unreachable. If a print action reports the printer can't be reached, check that the printer is powered, on the same WiFi, and that its IP in Settings → Network is current (DHCP can move it). "No print output on" means no sink is enabled.

Connects but nothing prints (office printers). An HP or other office printer on port 9100 speaks **PCL or PostScript**, not the ESC/POS receipt codes CardSat sends by default — so it receives the job and silently discards it. Set **Settings → Network → Printer format** to **PCL (HP)** or **PostScript** to match your printer (try PCL first for HP LaserJet / OfficeJet; PostScript for PostScript-capable models). **Plain text** is a safe universal fallback. If unsure, turn on **Save to /CardSat/Reports** and print the .txt from a computer instead.

If the printer exposes **IPP but not raw 9100**, set **Printer transport → IPP (:631)** with format **PCL** or **PostScript**. If the printer is **raster-only** (AirPrint-style, advertising only `image/urf`/`image/pwg-raster`), select **Printer format → PWG raster** (or **URF raster**) instead — CardSat renders the page on-device and prints to it over IPP. Use **Test printer (probe formats)** to see which formats a printer accepts and pick the matching one.

Bench debugging over USB. Connect a serial monitor at **115200 baud** and type `help`: a read-only console reports the firmware version (`ver`), free heap and largest block (`heap`), catalog counts including truncation (`sats`), favorites and the active bird (`fav`), the next pass (`next`), and WiFi state (`net`) — plus `time`, `gps` (fix, coordinates, grid), `bat`, `fs` (storage backend and free space), `up`time, and `pass <sat>` for the next pass of *any* catalog bird by name fragment or NORAD number. It never changes *device* state — the `print` commands only transmit a report to your configured printer (§19) — so it is safe to leave connected.

Radio doesn't respond / wrong VFO moves. Check the **CAT baud** (and, for Icom, the **CI-V address**) in Settings match the radio. Open the serial monitor at 115200 and watch the **CAT trace** (§16): each command is shown decoded, and the radio should reply **ACK (FB)** rather than **NAK (FA)**. If the wrong VFO tunes, the MAIN/SUB select bytes for that model may need adjusting in `radio_profiles.h`.

USB CAT won't engage, or the adapter isn't found. The status line names the refusal: `No USB device detected` means nothing enumerated (cable, hub, power); `Not a known serial adapter: <n> <vid:pid>` means the device is present but its VID:PID isn't in EspUsbHost's allow-list (common with clone bridges — try an FTDI or CP210x). `Only` `adapter is the radio's` / `That adapter is the rotator's` mean the one adapter present is already assigned to the other port — check **Radio USB** / **Rot USB** in Settings. With two identical no-serial-number adapters, re-check the assignment after any replug. Every engage, bind, and refusal is also written to `/CardSat/Logs/usb.log` with the reason. If *USB serial* doesn't appear in the CAT type row at all, the firmware was built without USB CAT (see §3). Remember the serial console is gone once USB engages — use `usb.log` and **Console to file** instead (§16).

Radio-knob tuning feels jumpy or unresponsive. Read-back happens about twice a second, so there's up to ~0.5 s of latency, and a 20 Hz threshold ignores tiny readback jitter. On a slower CAT link (9600 baud) it's less snappy. If your rig quantizes frequency coarsely, that threshold may need tuning.

Predictions seem off. Check the **GP age** indicator — refresh GP data if it's yellow/red. Confirm your location and that the clock is correct (UTC).

"Network problem" reboot prompt. If downloads keep failing with refused connections, CardSat first retries and hard-resets WiFi automatically (this clears a wedged socket pool). If that still doesn't recover, it shows a **Network problem** screen offering a reboot — press **ENTER**/`y` to reboot (the reliable cure once the socket stack is stuck) or `` ` ``/`n` to keep running and try again later. CardSat never reboots on its own; the choice is always yours.

Fewer satellites than expected after Update. The GP file streams straight to flash and is parsed incrementally, so it isn't limited by RAM; the catalog holds up to 220 sats. If the count looks short, open the serial monitor — the line `[net] streamed N bytes ... (declared M)` should show **N == M** (a complete download). All-null placeholder entries in the AMSAT feed (sats with no current elements) are skipped on purpose.

No alarm beeps. Confirm **AOS alarm** is on in Settings, you have **favorites**, and the **clock is set** — the alarm tracks only the soonest upcoming favorite AOS, so with no favorites or an unset clock there's nothing to count down to.

GPS not fixing. Make sure the right **GPS source** is selected (Location → `s`) and that GPS is enabled (`p`). The Grove source shares pins with CAT — don't use both at once.

"fs open" / "No filesystem" when downloading GP (often under a launcher). CardSat needs somewhere to store the GP file. Run directly from flash it uses the internal SPIFFS/LittleFS partition; under a launcher that didn't attach a SPIFFS region it falls back to the **microSD card**. If you see `fs open` / `no filesystem`, either insert a microSD card (the launcher usually boots from one anyway), have the launcher allocate a SPIFFS partition for CardSat, or flash CardSat directly with the **Huge APP (3MB No OTA/1MB SPIFFS)** partition scheme. The serial monitor's `[fs]` … line at boot tells you what mounted.

Rotator shows "n/c" or won't move. First check **Rot wire** — the fix depends on the transport. **I2C bridge:** *n/c* means the SC16IS750 didn't answer; check `ROT_I2C_SDA`/`ROT_I2C_SCL` in `config.h` match your wiring and the bridge address (`ROT_I2C_ADDR`) doesn't clash with another I2C device. If it reaches the bridge but the rotator is silent, confirm the MAX3232 level shifter is in line and the controller's baud matches **Rot baud**. **Grove G1/G2:** the rotator refuses (or yields back to the bridge) while wired CI-V CAT or the Grove GPS hold those pins — move CAT to USB/LAN and the GPS off Grove. **USB adapter:** see the USB CAT entry above; the same status-line reasons and `usb.log` apply, and the rotator's lines are prefixed `rot:`.

22

Screen-by-screen reference

This section catalogs **every screen** in the firmware in a uniform format — what it is for, how you get to it, what it shows, and what each key does. It complements the task-oriented walk-throughs in §8 (which follow the natural operating flow); use this section when you want the complete picture of one specific screen.

Throughout, the navigation keys are the printed arrow legends — `;` up, `.` down, `,` left, `/` right — with **ENTER** to select and ` (or **DEL**) to go back. `b` saves a screenshot on any screen. Only the *screen-specific* keys are listed below.

Home menu

Figure 8: Home menu

- **Purpose** — the top-level launcher.
- **Reached from** — power-on lands here; ` from most top-level screens returns here.
- **Shows** — a scrolling list of the twenty destinations: Satellites, Next Passes (all favs), Passes (sel), Track (sel), World Map, Overhead now, Sun / Moon, Space Wx, Activations, AMSAT status, Weather, Grid dist/bearing, QRZ Lookup, Location, Update, Settings, Log, Messages, About, and Charge / Sleep. The header carries the clock and battery gauge.
- **Keys** — `t` opens the **Tools** hub (below); after LOS, `q` (60 s window) deep-sleeps until the next favorite pass; `;/.` move the highlight; `{/}` page; **ENTER** opens the selected item; any **letter** jumps to the next item starting with it (repeat to cycle — `s` steps through Satellites, Sun / Moon, Space Wx, Settings).

Satellites

- **Purpose** — browse the catalog, choose favorites, and reach the per-satellite analysis tools.
- **Reached from** — Home → Satellites.
- **Shows** — the satellite list (up to 220 from GP data plus any manual sats). A right-edge mark shows AMSAT activity: a filled dot = heard, filled square = telemetry only, ring = not heard, nothing = no reports. Favorites are flagged.

Figure 9: Satellites list

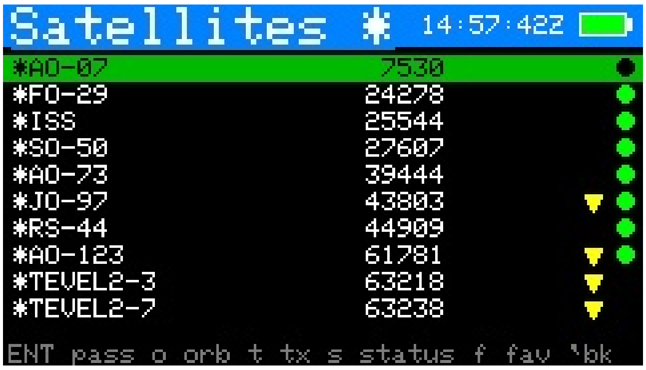


Figure 10: Satellites — favorites filter

- **Keys** — `;`/`.` move (`{`/`}` page); `f` toggle favorite; `v` show favorites only; `/` search all of CelesTrak and add any object as an auto-updating favorite; `n` add a new satellite by hand; `x` delete an added satellite; `o` open **Orbital analysis**; `y` open **Simulation**; `t` open the **Transponder database**; `e` open the **EQX table**; `k` open the **OSCARLOCATOR** view; `2` open the **sat-to-sat visibility finder**; `3` open the **3D Globe**; `d` open the **10-day pass overview**; `i` open the **Illumination** raster; `s` open the **AMSAT status** roster; `L` **share the selected satellite's GP elements over LoRa** to nearby CardSats (needs LoRa enabled); **ENTER** opens **Passes** for the highlighted bird.

Orbital analysis

Figure 11: Orbital analysis — Info

- **Purpose** — an eleven-page deep dive into one satellite's orbit: geometry, dynamics, lighting, and pass planning. Full theory and per-page detail are in §8 → Orbital analy-

Figure 12: Orbital analysis — Live

Figure 13: Orbital analysis — Next pass

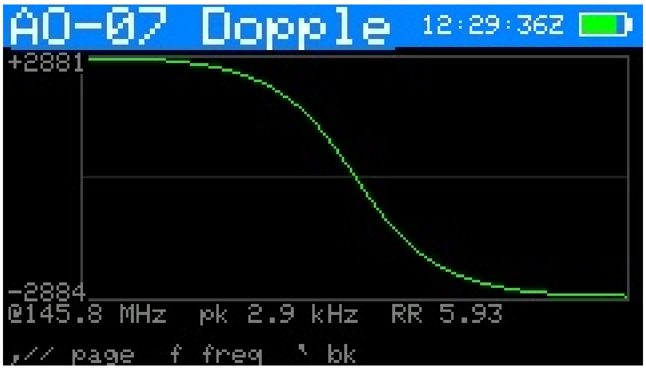


Figure 14: Orbital analysis — Doppler

Figure 15: Orbital analysis — Nodal

Figure 16: Orbital analysis — Sun/Beta

Figure 17: Orbital analysis — Pass outlook

Figure 18: Orbital analysis — Orbit position

sis.

- **Reached from** — Satellites → `o`.
- **Shows** — one page at a time: Info, Live, Next pass, Ground track, Doppler, Nodal, Sun / Beta, Pass outlook, Orbit position, Physical, and **Explore**. The **Explore** page is a sandbox: it seeds apogee, perigee and inclination from the tracked satellite, then lets you edit them (`;/.` pick a row, type a value, `x` reseeds) and watch the derived characteristics recompute live — period, revs/day, eccentricity, apogee/perigee velocity, footprint radius, longest possible pass, nodal drift, and whether the orbit is sun-synchronous. It never alters the real elements, and all quantities stay metric (km, km/s, min).
- **Keys** — `,//` flip pages; `r` recompute; on the **Doppler** page `f` sets the beacon frequency; `` ` `` leaves.

Simulation

Figure 19: Simulation

Figure 20: Simulation — map view

- **Purpose** — a “time machine” that propagates the selected satellite to an arbitrary time so you can preview geometry past or future.
- **Reached from** — Satellites → `y`.
- **Shows** — the sub-point and footprint (and, in world-map view, the track) at the simulated instant, with the offset from now.
- **Keys** — `,//` step the time backward/forward; `;/.` change the step size; `m` toggle the world-map view (sub-point + footprint); `x` reset to now; `` ` `` back.

Transponder database

- **Purpose** — inspect every transponder/beacon entry CardSat holds for a satellite (from SatNOGS plus any you added).

- **Reached from** — Satellites → `t`.
- **Shows** — a scrollable list of entries with description, downlink (and mode), uplink, and tone/inverting flags. The currently selected entry is highlighted with a `>`; entries you added by hand are tagged with a `*`.
- **Keys** — `;/.` select an entry; `x` **delete** the selected entry, *only if it's a manual one* (`*`-tagged) — press `x` once to arm, `x` again to confirm. `` ` `` back. Deleting rewrites that satellite's `/CardSat/mtx_<norad>.json` file; SatNOGS-cached entries can't be deleted here (they'd return on the next Freq update). To **edit** a manual transponder, delete it and re-add it with `n` on the **Passes** screen.

EQX table (OSCARLOCATOR)

- **Purpose** — a table of **equatorial crossing (EQX)** times and longitudes for the selected satellite, for use with a classic **OSCARLOCATOR** plotting board. Each EQX is an **ascending-node** crossing — the moment the satellite's ground track crosses the equator heading **north**. With the EQX time and longitude you rotate the Oscarlocator's track overlay to that longitude and read AOS/LOS and mutual visibility directly off the board, with no live computer at the operating position. A `d` keypress switches the table to the **descending node** (southbound equator crossing) if you'd rather reference that.
- **Reached from** — Satellites → `e`.
- **Shows** — a scrollable, day-grouped table covering the next **3 days**, each row an EQX **date**, **time in UTC**, and **sub-satellite longitude in West-positive** notation (`123.4 W`), matching the convention printed on Oscarlocator dials. Successive crossings step westward by roughly 360°×(period/sidereal-day) per orbit (~28.7° for an AO-7-class orbit).
- **Computed on-device** from the satellite's current GP elements (SGP4) — no network needed — so the longitudes drift as the elements age; run **Update** every week or two to keep them fresh, the same as for tracking.
- **Keys** — `;/.` scroll a page at a time; `d` **toggle ascending ↔ descending** node (recomputes the table — ascending/EQX is the default and the usual OSCARLOCATOR reference, descending gives the southbound equator crossing); `r` recompute (e.g. after the clock is set or new elements are loaded); `` ` `` back. Requires the **UTC clock** to be set (GPS or Location → `c`). The header shows **EQX** for ascending or **DEQX** for descending.

  The table mirrors the output of the **AO-7_OSCARLOCATOR** generator (github.com/prstoetzer/AO-7_OSCARLOCATOR), but works for any satellite in the catalog and runs entirely on the Cardputer.

Next Passes (schedule)

- **Purpose** — the unified upcoming-pass schedule across **all** your favorites, so you see what is next regardless of which bird it is.
- **Reached from** — Home → Next Passes (favs).
- **Shows** — favorites' passes in time order (up to 24 favorites tracked), each with satellite, AOS time/countdown, max elevation and duration. Stale-element sats carry a red `!`.
- **Keys** — `;/.` select; **ENTER** opens **Pass detail**; `m` opens the live **World map**; `t` the **sky-at-a-glance** timeline; `p` the **Rove planner** (below); `w` the **Workable horizon** and `s` the **Target search** (both below); `z` arms **deep sleep** until the next AOS; `` ` `` back.

Three planning tools hang off this screen, and they answer three different questions. The **Rove planner** (`p`) asks *"what would the schedule look like from a place and time I choose?"* — a

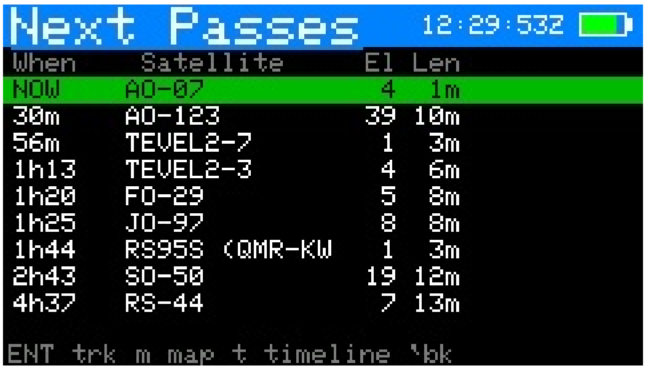


Figure 21: Next Passes schedule

what-if for a trip. The **Workable horizon** (`w`) looks *outward*: over the next ten days, what is the complete set of states, DXCC entities, and grids this station can ever reach? And **Target search** (`s`) looks *inward*: pick one place and find every chance to work it. All three share the same footprint engine underneath; they differ only in the direction of the question.

Rove planner

- **Purpose** — a *from-a-hypothetical-place-and-time* pass survey. Enter a grid square, date, time, and a **± window** (all passes within X hours of that time); for **every favorite** it lists each pass with **AOS, LOS, max elevation**, and the number of **workable US states and DXCC entities** during that pass — so a rover can pick a spot and time and see, at a glance, which passes put the most grid/entity reach under the footprint.
- **Reached from** — Next Passes (favs) → `p`.
- **Shows** — a small input form (grid / date / time / ± hours / **GO**), then a list of passes across all favorites sorted by AOS, each row `name  AOS  maxEl  St Dx` (workable-state and DXCC counts). The survey runs in the background with a progress line so the unit stays responsive.
- **Pass detail (ENTER on a row)** — the **polar plot** for that pass from the entered site, with AOS/LOS/max-el/length and the workable-state, DXCC and grid counts; `s`, `d` and `g` open the full workable **US-state**, **DXCC** and **grid** lists for that pass.
- **Save to text file** — `w` on the results screen writes the whole survey to a formatted `.txt` under `/CardSat/RovePlans/` on the active filesystem: a header (grid, center time, window, pass count) followed by one block per pass with the pass details, the list of workable **states** and **DXCC** entities, and the **count** of workable grids (the grid list itself is not written — only the number).
- **Keys** — in the form: `;`/`.` move between fields, **ENTER** edits a field or starts the survey on **GO**, `+`/`-` bump the window; in the results: `;`/`.` select a pass, **ENTER** opens detail, `w` saves the text file, `g` returns to the form; `` ` `` back.
- **Note** — the workable-entity counts are a property of the satellite's footprint over the pass, so they don't depend on the entered site; the entered grid sets the AOS/LOS and max-elevation geometry. All quantities are UTC.
- **Saved plans (on-device viewer)** — press `l` on the planner to open a browser of the `.txt` plans you've saved with `w`. It lists them newest-first with each plan's date-stamp and size; `ENTER` opens a **read-only, scrolling viewer** (`;`/`.` scroll, `{`/`}` page), `d` deletes (with a confirm), `r` rescans, `` ` `` back. The viewer holds a bounded slice of the file in RAM, so a very large plan shows a "(*truncated — download for full file*)" note; use the web **Files** page (§18) to pull the whole file.

Workable horizon

- **Purpose** — a ten-day *outward* view: over the coming ten days, which US states, DXCC entities, and (optionally) grids will **ever** be workable through **any** of your favorites? Where the Rove planner answers "what does a chosen place and time look like," the Workable horizon answers "over the whole coming week and a half, what is the complete set of places I can reach from here."
- **Reached from** — Next Passes (favs) → `w`.
- **Shows** — a progress bar while the ten-day sweep runs (incrementally, one pass per loop, so the unit stays responsive), with the running counts of workable **states** and **DXCC** growing live as coverage accumulates. The result is the **union** across every favorite and every pass — an entity counts as workable if it falls inside the footprint during *any* pass while the satellite is above your horizon.
- **Grids off by default** — the grid union needs a few kilobytes of working memory, which on this board is better left free for other tasks (notably network uploads). States and DXCC — the common case — use no extra memory. Press `g` on the results screen to re-run the sweep **with grids included** on demand.
- **Drill-down** — `s` opens the full **Workable states** list, `d` the full **Workable DXCC** list (both drawn from the ten-day union, and titled to distinguish them from the Rove planner's per-pass lists).
- **Save to text file** — `w` writes the whole result under `/CardSat/workable/` on the active filesystem.
- **Keys** — `s` states list, `d` DXCC list, `g` re-run including grids, `w` save, ` back.

Target search

- **Purpose** — the *inward* counterpart to the Workable horizon: pick **one** place and find every upcoming chance to work it. This is the direct rover / award-chaser question — *I need this state (or DXCC entity, or grid) — when is my next shot?* — answered for the next ten days across your whole fleet in a single screen.
- **Reached from** — Next Passes (favs) → `s`.
- **Pick a target** — a **US state**, a **DXCC entity**, or a **grid**. Grids are typed into the on-screen editor; states and DXCC are chosen from a filterable list (type a letter to narrow it).
- **Shows** — every pass, across **all** favorites over the next ten days, where the chosen target sits inside the footprint while the satellite is up for you — listed **in time order across the whole fleet** (soonest first, satellites mixed together), each row showing the satellite, the date, the **workable window** (the span of the pass during which the target is actually in the footprint), and the maximum elevation. Up to 40 of the soonest passes are kept. If the target simply can't be reached from your location in the window — for instance a distant DXCC entity no single LEO footprint can cover together with your QTH — the search completes and reports none, rather than leaving you guessing.
- **Polar plot** — press **ENTER** on any row to see that specific pass drawn as a polar plot.
- **Save to text file** — `w` writes the results under `/CardSat/search/`.
- **Keys** — in the picker: type to filter, `;`/`.` select, **ENTER** choose; in the results: `;`/`.` select, **ENTER** polar plot of that pass, `w` save, ` back.

Passes

- **Purpose** — the pass list for one selected satellite, and the jumping-off point to tracking, the visualizers, and the workable lists.
- **Reached from** — Satellites → ENTER, or Home → Passes (sel).

Figure 22: Passes (per-satellite)

- **Shows** — up to twelve upcoming passes for the bird (AOS/LOS, max elevation, duration). A yellow * marks an **optically-visible** pass.
- **Keys** — ;/. select; d **Pass detail**; t or **ENTER** open **Track**; n add a manual transponder; r recompute; x **Mutual windows**; v **10-day overview**; V **visible-pass list** (next 10 days); i **Illumination**; g/w/e workable grids / US states / DXCC **for this pass**; ` back (to Satellites, or Home if opened from the Home menu).

Pass detail

Figure 23: Pass detail (elevation)

- **Purpose** — the full numeric breakdown of one pass plus its polar plot.
- **Reached from** — Passes → d, or Next Passes → ENTER.
- **Shows** — AOS/TCA/LOS times and azimuths, max elevation, duration, and a polar (sky) plot of the arc.
- **Keys** — p toggle to the dedicated **Pass polar** view; ` back.

Pass polar

- **Purpose** — a full-screen polar (azimuth/elevation sky) plot of a single pass's arc.
- **Reached from** — Pass detail → p.
- **Shows** — the pass traced on a polar grid (zenith center, horizon rim), with AOS/LOS marked.
- **Keys** — p toggle back to Pass detail; ` back.

Figure 24: Pass polar plot

Figure 25: Mutual windows

Mutual windows (co-visibility)

- **Purpose** — find the times a satellite is simultaneously visible to you **and** to a second station — the windows in which a contact is geometrically possible.
- **Reached from** — Passes → `x`.
- **Shows** — the other station's grid (editable) and a list of mutual-visibility windows with start/end and the overlap duration.
- **Keys** — `;/.` scroll the windows; `d` open the **DX Doppler table** for the highlighted window; **ENTER** edit the remote grid; ` back.

DX Doppler table

- **Purpose** — for a selected mutual window, the per-30-second RX/TX dial frequencies for **both stations** through the pass, for manual tuning of a coordinated contact.
- **Reached from** — Mutual windows → `d`.
- **Shows** — a scrolling table for the selected transponder, two lines per 30-second step: your dials (green, "me") and the DX station's (cyan, "DX"), each with RX and TX — in one of three modes — **true rule**, **fixed downlink**, or **fixed uplink** — with an anchor dial and (for a linear transponder) a passband operating point. The header shows the passband point **relative to the center of the passband's downlink** (e.g. `ctr`, `+7.5k`, `-12.5k`), so you can see at a glance how far off-center you are working.
- **Keys** — `t` cycle transponder; `m` cycle mode; `a` cycle anchor (my RX/TX, DX RX/TX); `,//` step the passband — in a **fixed** mode they move the **anchored dial** to the next **round 1 kHz** (so you park on, e.g., 145.950 MHz, not 145.9502), and in **true rule** they nudge the operating point by 1 kHz; `;/.` scroll the time steps; ` back to the mutual list.

Sat-to-sat visibility

Figure 26: Sat-to-sat visibility

- **Purpose** — the windows when the selected satellite and a second satellite are both above your horizon at once, over the next few days; detail in §8 → Sat-to-sat visibility.
- **Reached from** — Satellites → `2`.
- **Shows** — first a **pick screen** (choose the second satellite, no calculation yet), then after you start the search a scrolling list of overlap windows with start time (UTC), duration, and the peak elevation of each bird.
- **Keys** — `n`/`p` step the second satellite forward/back **instantly** (no search runs while you cycle); **ENTER** (or `r`) runs the window search for the chosen pair; `;`/`.` scroll the results; ``` ` ``` back to Satellites. Because cycling no longer triggers a multi-day search on every step, choosing a different second satellite is immediate.

10-day pass overview

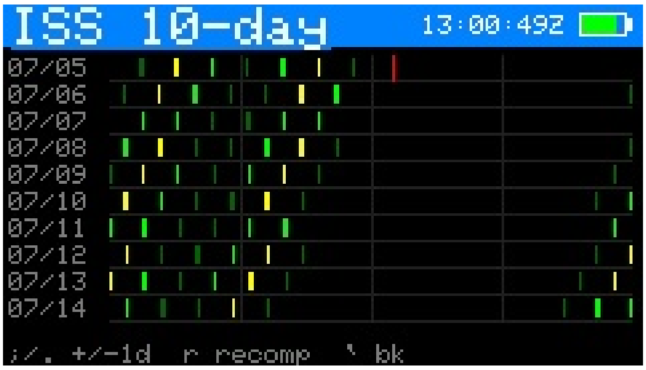


Figure 27: 10-day pass overview

- **Purpose** — an at-a-glance visibility chart of one satellite's passes over ten days, modeled on InstantTrack's multi-day view.
- **Reached from** — Passes → `v` (or Satellites → `d`).
- **Shows** — one row per day (today at top), each a 24-hour UTC timeline; every pass is a bar shaded by peak elevation (dim-green < 15°, green < 40°, yellow above). A red tick marks now.
- **Keys** — `;`/`.` scroll one day (forward indefinitely, not before today); `r` recompute; ``` ` ``` back to Passes.

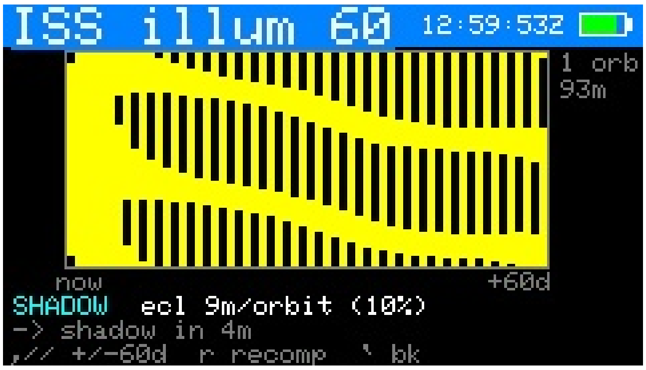


Figure 28: Illumination

Illumination

- **Purpose** — a 60-day solar-illumination raster (DK3WN *illum* style) showing when the satellite is sunlit vs. eclipsed across its orbit.
- **Reached from** — Passes → `i` (or Satellites → `i`).
- **Shows** — horizontal axis date (today → +60 d), vertical axis one orbital period; cells yellow in sunlight, dark in eclipse. A live readout shows current SUN/SHADOW status, eclipse minutes/percent per orbit, and time to the next transition.
- **Keys** — `,//` jump a full 60-day window (forward indefinitely, not before today); `r` recompute; ` back to Passes.

Visible-pass list

- **Purpose** — the optically-visible passes of the selected satellite over the next 10 days: satellite **sunlit**, your **sky dark**, peak elevation over the visibility minimum. Full detail in §8 → Visible-pass list.
- **Reached from** — Passes → `V` (capital; lowercase `v` is the 10-day chart).
- **Shows** — one row per visible pass: AOS time, duration, peak elevation, and the rise compass direction (where on the horizon it comes up).
- **Keys** — `;/.` scroll; **ENTER** open the pass-detail plot (returns here); `r` recompute; ` back to Passes.

Track

Figure 29: Track — live Doppler & CAT

- **Purpose** — the main operating screen: live pointing, Doppler-corrected frequencies,

Figure 30: Track — ground track

transponder selection, calibration, and radio/rotator control. Full detail in §8 → Track, Doppler theory in §9.

- **Reached from** — Passes → t/ENTER, or Home → Track (sel).
- **Shows** — az/el (and GP age), range/range-rate (ECL flag in eclipse), the transponder, DN/RX and UP/TX frequencies, the passband line, the calibration line, the radio status, a rotator status line, and a PL tone line on FM birds that need one.
- **Keys** — m switch TUNE/CAL; d cycle Doppler tune mode (linear birds); t next transponder; n jump to the **beacon**; c set CTCSS/PL tone; k toggle **CW on both legs** (linear birds); N edit the per-satellite **operating note**; r radio output on/off; o rotator on/off; p **Polar**; z large-font readout; f **Manual mode**; l **Log QSO**; i×2 **"I heard it"** AMSAT status report (first tap arms, second sends); v voice memo (SD card); a **point-here arrow**; y tilt tuning on/off (ADV, if IMU); g/w/e workable grids / US states / DXCC now; after LOS, q (60 s prompt) deep-sleeps to the next favorite pass; in TUNE: ,// move spot, s step, x recenter; in CAL: ,// trim downlink, ;/. trim uplink, s step, x zero; **ENTER** save calibration; ` leave (returns to where you came from).
- **Leaving keeps tracking** — pressing ` returns to the previous screen and, if the radio/rotator are engaged, **they keep tracking in the background** (a green **RAD**/**ROT**/**R+R** header tag appears on other screens). Stop with r (radio) or o (rotator, which also parks).

Point-here arrow (a from Track)

- **Purpose** — a big, glanceable compass arrow to the satellite's azimuth plus an elevation bar, for hand-aiming a portable antenna without reading numbers.
- **Reached from** — a on the Track screen.
- **Shows** — a compass rose (N up) with an arrow pointing where to aim (green above the horizon, dim below), an elevation bar (0 at bottom, 90 at top), the numeric az/elevation/range, the rise compass direction, and a status line. The radio and rotator keep running underneath.
- **Keys** — ` back to Track.

Large-font readout

- **Purpose** — a stripped-down, glanceable view of the live session: RX/TX in the largest digits that fit, az/el and the Doppler tune mode beneath, RAD/ROT/ TILT badges. The radio, rotator and Doppler service keep running — it is an alternate view of the same session, not a different mode.
- **Reached from** — Track → z.

- **Shows** — RX and TX frequencies large; az/el; tune mode (FULL/DL/UL/TUNE/CAL); passband position on linear birds; transponder index; status badges.
- **Keys** — mirrors Track in place: `l` **Log QSO**; `v` voice memo; `c` CTCSS tone; `k` CW-on-both-legs toggle; `g`/`w`/`e` workable grids / states / DXCC; `f`/`p` Manual mode / Polar; `z` or `` ` `` back to Track.

Manual mode

- **Purpose** — a no-radio frequency calculator: CardSat shows the Doppler-corrected dial frequencies so you can tune a non-CAT rig by hand.
- **Reached from** — Track → `f`.
- **Shows** — the same RX/TX figures as Track, formatted for manual dialing, with the passband/tune controls but no CAT output.
- **Keys** — mirror Track's tuning set: `m`, `t`, `s`, `x`, the `,`/`/`,`;`/`.` tune/cal keys, `g`/`w`/`e` workable lists, `l` log, `p` polar, `v` voice memo, `z` large-font Manual, `u`/`e`/`f` as on Track; `` ` `` back.

Polar

- **Purpose** — a live full-screen polar sky plot of the satellite you are tracking, for visual aiming.
- **Reached from** — Track → `p`.
- **Shows** — the current az/el as a marker on a polar grid, with the pass arc.
- **Keys** — `p` toggle back to Track; `l` **Log QSO**; `v` voice memo (microSD, ADV); `` ` `` back.

OSCARLOCATOR

- **Purpose** — a live azimuthal-equidistant "plotting board" showing the satellite's sub-point on the Earth in real time, with its footprint; the graphical companion to the tabular **EQX table**.
- **Reached from** — Satellites → `k`.
- **Shows** — a disc centered on your station (QTH mode) or a pole (polar mode), with a coastline, graticule, the satellite marker (sunlit/eclipse color), its ground footprint, an amber **QTH range ring** (footprint radius at mean altitude), the blue **full ground-track arc** across the disc (AOS green, LOS orange, travel arrow), and a sub-point/az/el/range readout. In polar mode the North/South sheet is chosen automatically and flips live as the satellite crosses the equator.
- **Keys** — `m` toggle QTH-centered ↔ polar; `` ` `` back to Satellites.

3D Globe

- **Purpose** — an orthographic 3-D wireframe Earth that auto-follows the selected satellite, for an at-a-glance view of where it (and your other favorites) are on the planet, with the day/night terminator.
- **Reached from** — Satellites → `3`.
- **Shows** — a wireframe globe (near hemisphere only) with graticule, coastline, a yellow **day/night terminator**, your QTH (white cross), all favorites as green dots, the selected satellite centered (sunlit/eclipse color) with its **ground footprint** and a blue **ground-track trail** (one orbit), and a sub-point/az/el/altitude readout. A **second (DX) location** entered by grid (`g`) adds an orange marker and footprint; the overlap with the satellite footprint is the mutual-visibility region. The globe rotates to keep the selected satellite centered.

- **Keys** — arrow keys turn the globe (free-look); **ENTER** re-snaps to auto-follow; `g` set a DX grid, `G` clear it; ` back to Satellites.

Workable grids

Figure 31: Workable grids

- **Purpose** — the Maidenhead grid squares currently (or, from Passes, during the pass) reachable through the satellite's footprint.
- **Reached from** — Track → `g` (live) or Passes → `g` (for the pass).
- **Shows** — the list/scatter of workable grids; from Track it refreshes live while radio and rotator control keep running.
- **Keys** — `;`/`.` scroll; `{`/`}` page; `f` set a prefix filter (e.g. `EM`, `EM2`, `EM21` — upper-cased per the grid rule); `c` clear the filter; ` back.

Workable US states

Figure 32: Workable US states

- **Purpose** — the US states (for WAS chasing) under the footprint now or during the pass.
- **Reached from** — Track → `w` (live) or Passes → `w` (for the pass).
- **Shows** — the workable states list.
- **Keys** — `;`/`.` scroll; ` back.

Workable DXCC

- **Purpose** — the DXCC entities under the footprint now or during the pass.
- **Reached from** — Track → `e` (live) or Passes → `e` (for the pass).

Figure 33: Workable DXCC

- **Shows** — the workable DXCC list (derived from bundled cty.dat geometry).
- **Keys** — `;`/`.` scroll; ` back.

Sun / Moon

Figure 34: Sun / Moon

- **Purpose** — point the rotator at the Sun or Moon for sun-noise/EME work and antenna calibration; full detail in §13 → Sun / Moon.
- **Reached from** — Home → Sun / Moon.
- **Shows** — the selected body's live az/el (and a sky-dome graphic in graphic view). An orange SUN/MOON header tag appears while it is driving the rotator.
- **Keys** — `;`/`.` switch Sun↔Moon; `g` toggle graphic/list view; `o` rotator tracking on/off; `s` open the **Sky sources** plot; `t` open the **Sun/Moon transit finder** (below); `e` open the **EME / moonbounce** screen (below); `x` stop; ` park and back.

Sun / Moon transits

- **Purpose** — find upcoming times the active satellite crosses (or closely approaches) the **Sun or Moon** from your location — the ISS-on-the-disc astrophotography event. Full detail in §13 → Sun / Moon transits.
- **Reached from** — `t` on the **Sun / Moon** screen (predicts for the satellite currently selected).
- **Shows** — an incremental 48-hour scan with a progress bar, then a list of close approaches: body (Sun/Moon), a countdown, minimum angular separation, body elevation, and a **TRANSIT** (crosses the disc) vs **near** (conjunction) label. Requires the clock and your location to be set.

- **Keys** — ; / . move the selection; r rescan; ` back to Sun / Moon. **Never observe a solar transit without proper solar filtering.**

Sky sources

Star layers (0.9.60): the dome carries a live star field — 1,018 stars to magnitude 4.6, constellation lines, and proper names for the sixteen brightest (Sirius through Antares) — recomputed from the clock each frame, exactly like the radio sources. **c** cycles off → stars → +lines → +names. The catalog is compiled into flash (~9 KB) from d3-celestial's BSD-3 data (regenerate with `tools/make_star_tables.py`). Brightness maps to pixel weight, so Orion looks like Orion. Point a yagi at night by starlight.

Figure 35: Sky sources

- **Purpose** — the planets and the strongest cosmic radio sources on a sky dome, for antenna pointing and as an RF-source reference; detail in §13 → Sky sources.
- **Reached from** — Sun / Moon → s.
- **Shows** — a sky dome (zenith center) with planets as cyan dots and radio sources (Cas A, Cyg A, Sgr A*, Tau A, Vir A) as orange crosses; a readout gives the selected object's az/el and type.
- **Keys** — ; / . select an object; ` back to Sun / Moon.

EME / moonbounce

Figure 36: EME / Moon

- **Purpose** — the moonbounce numbers: self-echo Doppler per band, topocentric range/rate, path degradation vs perigee, a galactic sky-noise flag, and a mutual-Moon window vs a DX grid; detail in §13 → EME.

- **Reached from** — Sun / Moon → `e`.
- **Shows** — Moon az/el and up/down; range (km) and range-rate (m/s); degradation (dB, with a near-perigee/apogee note); sky flag (cold/warm/HOT with galactic latitude); self-echo Doppler for 50/144/432/1296/10368 MHz. The `a` analysis page puts the printed per-band table on screen — Doppler, Faraday rotation, sky temperature, and libration spread per band — plus the two-way path loss (+6 dB lunar reflection) and the ground-gain window state. The `m` sub-view lists common Moon windows vs a DX grid over the next 14 days.
- **Keys** — `o` point/stop rotator at the Moon; `x` stop; `m` mutual-Moon window (then `g` change grid, `;/.` select); `p` 30-day planner; `a` per-band analysis page; `w` print the EME report (works from the live and mutual views); ` back (analysis/sub-view → main → Sun/Moon). A red **SUN** flag appears when the Moon is within ~10° of the Sun.

EME 30-day plan

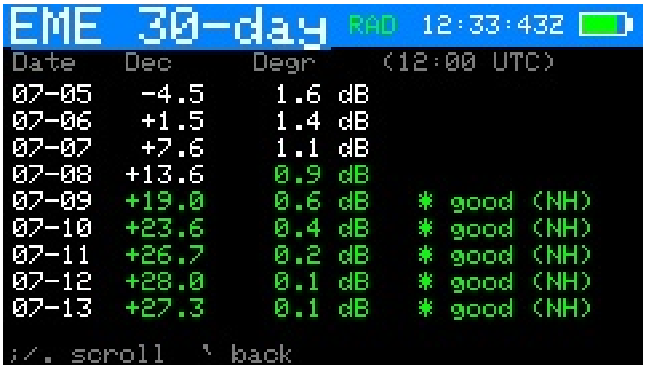


Figure 37: EME 30-day plan

- **Purpose** — pick the good EME days: per-day Moon declination (12:00 UTC) and path degradation for the next 30 days, with a star on high-declination, near-perigee days (the northern-hemisphere heuristic; raw values shown for both hemispheres).
- **Reached from** — EME / moonbounce → `p`.
- **Keys** — `;/.` scroll; `{/}` page; ` back to EME.

Space weather

Figure 38: Space weather

- **Purpose** — solar flux and geomagnetic indices with a plain-language operating outlook; detail in §13 → Space weather.

- **Reached from** — Home → Space Wx.
- **Shows** — F10.7 flux, planetary Kp, running A index, an **aurora likelihood** line derived from Kp (unlikely / possible / likely, with latitude), each labeled and color-coded, an HF/satellite outlook line, and a data-freshness note.
- **Keys** — `p` open the **HF / 6m propagation** guide (below); `r` refresh over WiFi; ` back.

HF / 6m propagation

- **Purpose** — rule-of-thumb band guidance from the solar flux and Kp; detail in §13 → propagation guide.
- **Reached from** — Space Wx → `p`.
- **Shows** — the two indices (color-coded); an HF band summary and a 10/15/20 m open/marginal/shut read; the geomagnetic effect; auroral-VHF likelihood; D-layer absorption; and a rule-of-thumb disclaimer (6 m Es is seasonal).
- **Keys** — `r` refetch in place; ` back to Space Wx.

Weather

Figure 39: Weather

- **Purpose** — terrestrial current conditions and a multi-day forecast for the operating site; detail in §13 → Weather.
- **Reached from** — Home → Weather.
- **Shows** — current temperature, sky, wind and humidity, then per-day condition, high/low and precipitation chance, plus a freshness note. Units per Settings. A second **field conditions** page (`f`) adds feels-like temperature, wind gusts (with a MAST flag), pressure and its 3-hour trend, the UV index, and sunrise/sunset, plus a per-day UV/gust/sun table — for portable and field operation.
- **Keys** — `f` toggles the field-conditions page; `p` prints the weather report; `r` refreshes the fetch; ` backs out (field page → summary → Home).
- **Keys** — `r` refresh over WiFi; ` back.

QRZ callsign lookup

- **Purpose** — resolve a callsign to name, location, grid and license class via the QRZ.com XML API; detail in §13 → QRZ callsign lookup.
- **Reached from** — Home → QRZ Lookup.
- **Shows** — prompts for credentials/WiFi if needed; otherwise the looked-up station's details.
- **Keys** — **ENTER** type a callsign to look up; `;`/`.` scroll a long result; ` back.

Figure 40: QRZ callsign lookup

Grid distance & bearing

- **Purpose** — great-circle distance and beam heading to a Maidenhead grid, for terrestrial VHF/UHF work; detail in §13 → Grid distance & bearing.
- **Reached from** — main menu → **Grid dist/bearing**.
- **Shows** — your grid, the target grid, distance (km/mi), heading (short path), long-path heading and distance, and a rotator-engaged line when pointing.
- **Keys** — `g` enter grid; `q` QRZ → grid lookup; `o` point/stop rotator at the bearing (el 0); `x` stop; ` back to the menu.

QRZ → grid

- **Purpose** — resolve a callsign to its grid (via the QRZ account already set up for QRZ Lookup) and seed the grid calculator with it.
- **Reached from** — Grid dist/bearing → `q`.
- **Shows** — the callsign, the operator's name and grid on success, or the QRZ error.
- **Keys** — `c` enter callsign; **ENTER** use the found grid in the calculator; ` back to the calculator.

AMSAT status

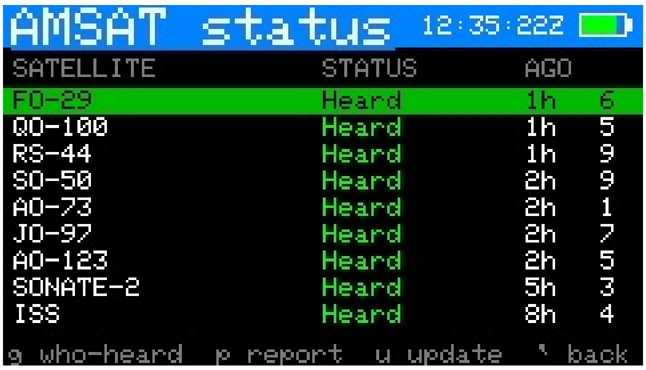


Figure 41: AMSAT status

- **Purpose** — every bird with a recent AMSAT live-status report, sorted most-active-and-most-recent first; detail in §14 → AMSAT status.
- **Reached from** — Satellites → `s`, or the **AMSAT status** main-menu item.
- **Shows** — one row per reported satellite: name, status word (colored), "heard N ago" recency, and how many stations reported, within the configured window.

- **Keys** — `;/.` select; **ENTER** adopt as the active satellite and jump to Track; **g** open the selected bird's **individual reports** (below); **p report this bird's status** (below); `u` re-fetch in place; ` back.

AMSAT reports (per satellite)

Figure 42: AMSAT who-heard

- **Purpose** — the selected satellite's **individual status reports**: who heard it, from which grid, how long ago, and what they reported — headed by a **distinct-grid count** ("7 reports, 5 grids"), a footprint-activity read the summary count can't give.
- **Reached from** — AMSAT status → `g` (fetched on demand from the AMSAT reports API for the configured status window).
- **Shows** — one row per report: callsign, grid, age, and the color-coded status (Heard / Telemetry / Not heard / Crew).
- **Keys** — `;/.` scroll; `{/}` page; `r` refetch; ` back.

Report satellite status

- **Purpose** — submit a **public status report** to amsat.org, exactly like the web form: Heard / Telemetry Only / Not Heard / Crew Active. **Mode-aware**: the status system tracks some birds per mode (`AO-7_[U/v]` vs `AO-7_[V/a]`, `ISS_[FM]` vs `ISS_[SSTV]`, ...), and CardSat targets the entry matching the transponder you are actually working.
- **One-key from Track** — press **i** ("I heard it"): CardSat resolves the mode from the active transponder's uplink/downlink bands, asks once on the status line, and a second `i` within ~3 s sends **Heard**. If the mode can't be resolved (beacon selected on a multi-mode bird), the picker opens instead.
- **Full picker** — `p` on the **AMSAT status** screen (or the ambiguous case above): choose the API name (mode) and any of the four status values, **ENTER** sends. The report is public under your callsign with your grid; the server replaces a repeat report for the same satellite, hour and 15-minute period, so a double-send is harmless.
- **Needs** — your callsign in Settings, the clock set, WiFi, and the satellite present in the AMSAT catalog (the name map refreshes with each elements update and is cached for offline boots).

  Source-independent satellite names (0.9.59): activations, AMSAT status, and LoTW export all resolve satellite names through the same bridge — parenthesised designator first ("FOX-1B (AO-91)" ↔ "AO-91"), then normalized whole-name, then token containment — so they work identically whether your catalog came from AMSAT or CelesTrak. Activations for your **favorites** show green; the LoTW exporter now auto-resolves CelesTrak-named QSOs (a fitting parenthetical designator also beats a truncated long name) instead of prompting.

Activations

Figure 43: Activations

- **Purpose** — list the upcoming satellite activations scheduled on hams.at (roves, grid activations, special operations); detail in §13 → Upcoming activations.
- **Reached from** — Home → Activations.
- **Shows** — a scrollable list of upcoming activations (date, callsign, satellite, grid); entries you've entered yourself are marked with a leading `*` and shown in cyan. ENTER opens a detail view with start/end times (UTC), mode, frequency, the activator's comment (scrollable), and a **footprint note** — whether the satellite is co-visible with the activator near the listed time (±30 min).
- **Keys** — `;/.` move; **ENTER** open the selected activation's detail; `r` refresh the feed; **`n`** add your own activation/sked (works offline — see below); **`e`** edit the selected entry if it's one of your own (feed entries are read-only); in the detail view, `;/.` scroll the comment, **`a`** sets a **sked reminder** (T-60/30/10 countdown beeps and a flash at the start, independent of the favorites AOS alarm), and **`w`** opens the **mutual-window screen** when a footprint exists; when a sked is set, **`c`** on the list clears it; ` back.

Activation mutual window and tailored DX Doppler

- **Purpose** — for an activation with a footprint, show the co-visibility pass and give a DX Doppler view pre-set to the activation's frequency; detail in §13 → Upcoming activations.
- **Reached from** — `w` on an activation's detail (mutual window) → `d` (DX Doppler).
- **Shows** — a small polar plot of the pass (satellite track from your site in cyan and from the DX site in orange) with the date, AOS, LOS, duration and peak elevation for each station; the DX Doppler table then lists RX/TX dial frequencies for both stations across the window, with the transponder and fixed downlink/uplink pre-selected from the activation's frequency when it matches a two-way transponder.
- **Keys** — mutual window: **`d`** DX Doppler, ` back to the detail. DX Doppler: the same controls as the main DX Doppler table (`t` transponder, `m` mode, `a` anchor, `;/.` scroll), ` back to the mutual window.

New / Edit sked (manual activation entry)

- **Purpose** — enter your own activations or personal skeds in the **same format as the hams.at feed**, for operations that aren't posted to that site. Your entries are stored separately and **merged into the Activations list alongside the fetched ones**, surviving feed refreshes and reboots.
- **Reached from** — `n` (new) or `e` (edit a `*`-marked entry) on the Activations screen. Works with no WiFi.
- **Shows** — a field list: Date (YYYY-MM-DD UTC), Call, Sat, Grid, Start/End (HH:MM UTC), Mode, Freq, Comment. A new entry pre-fills today's date and your callsign.
- **Keys** — `;` / `.` move between fields; **ENTER** edit the selected field; `s` save (needs at least a date plus a satellite or callsign); when editing an existing entry, `x` deletes it (press twice); `` ` `` cancel. Saved entries appear immediately in the Activations list marked with `*`, and a sked reminder can then be set on them with `a` like any other activation.

Overhead now

Figure 44: Overhead now

- **Purpose** — a snapshot of every satellite in the loaded catalog that is **above the horizon right now**, sorted by elevation — a quick "what can I work / see this moment" glance.
- **Reached from** — Home → Overhead now.
- **Shows** — each satellite currently up, with its elevation, azimuth, and rise compass direction (high passes in green, near-horizon in yellow), plus a count of how many are up out of how many were scanned. The scan runs once when you open the screen.
- **Keys** — `r` rescan (recompute the snapshot for the current instant); `;` / `.` scroll; `` ` `` back.

Location

QTH presets (0.9.60): press **q** for five named, recallable station sites. **ENTER** recalls a slot straight into the live location (propagated to the predictor immediately) and turns **GPS off** — recalling a named site is an explicit choice the receiver shouldn't quietly override. **s** stores the current position into the slot (prompting for a name if unnamed), **e** renames, **x** clears, **1–5** jump. Stored in config.json.

- **Purpose** — set your station position and clock, and view the GPS sky plot.
- **Reached from** — Home → Location.
- **Shows** — current lat/lon/altitude, grid, time source and clock, and GPS status when enabled.

Figure 45: Location

- **Keys** — `e` edit latitude; `o` edit longitude; `a` edit altitude; `g` enter a Maidenhead grid; `p` enable/disable GPS; `s` pick the GPS source; `c` set the UTC clock by hand; `v` open the **live GPS position** screen; **ENTER** open the **GPS sky plot**; ` back.

Live GPS position

- **Purpose** — a full-precision position readout (DMS, decimal, grid) plus ground speed and course, for rovers and portable ops; detail in §13 → Live GPS position.
- **Reached from** — Location → `v`.
- **Shows** — latitude/longitude in degrees-minutes-seconds to 0.001″ and in decimal degrees, altitude, Maidenhead grid, ground speed (km/h and knots) and course; a status line with fix/sat-count/HDOP.
- **Keys** — ` back to Location.

LoRa Messages

- **Purpose** — CardSat-to-CardSat broadcast text chat over the Cap LoRa (SX1262); detail in §8 → LoRa Messages.
- **Reached from** — Home → Messages.
- **Shows** — a scrolling message history (yours in cyan as `>me`, others in green by callsign) with a status line of frequency / SF / bandwidth / your callsign.
- **Keys** — `n` write + send a message; `;/.` scroll history; `r` retry the radio bring-up if it wasn't ready; ` back to Home.
- **Notes** — needs LoRa enabled in Settings and your callsign set; the LoRa radio is built into the standard binaries (RadioLib required at build time). Untested hardware path.

LoRa RX / hex monitor

- **Purpose** — receive and inspect **any** LoRa signal on the Cap LoRa (SX1262): check whether a transmitter is on the air, view a beacon's raw bytes, peak reception by ear. Not satellite-specific; no network involved. Full detail in §8 → LoRa RX / hex monitor.
- **Reached from** — Messages → `m` (LoRa enabled in Settings; messaging is paused while the monitor holds the radio).
- **Shows** — received frames newest-first (hex + printable ASCII), with RSSI/SNR per frame and the current radio parameters.
- **Keys** — `;/.` scroll; parameter keys as shown in the footer (frequency, SF, bandwidth, etc.); ` back to Messages (releases the radio).

GPS sky plot

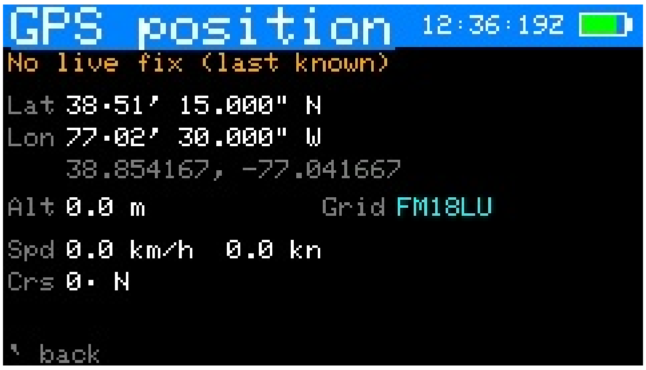


Figure 46: GPS position

Figure 47: GPS sky plot

- **Purpose** — a polar plot of the GNSS satellites currently in view, by signal strength — useful for checking antenna/fix quality.
- **Reached from** — Location → ENTER.
- **Shows** — GNSS satellites placed by azimuth/elevation, colored green (strong) to gray (weak), with fix data.
- **Keys** — ` back. (The plot updates live; no other keys.)

Update

Figure 48: Update

- **Purpose** — refresh everything that comes from the network in one place.
- **Reached from** — Home → Update.

- **Shows** — the last GP age and a note that `k` also refreshes the clock, AMSAT status, space weather and terrestrial weather.
- **Keys** — `k` (or ENTER) download GP + sync clock (NTP) + AMSAT + space wx + weather; `f` fast update (GP + AMSAT + favorites' transponders, skips space wx/weather); `a` fetch and cache **all** transponders for offline use; `w` connect WiFi only; ` back.
- **Primary-catalog courtesy throttle** — CelesTrak asks that the same GP data not be pulled more than about every 2 hours. Both update paths honor that: if the last successful download of the *same source URL* was under 2 h ago (the timestamp is persisted across reboots), the fetch is skipped with a status and the cached catalog reloads instead. Changing the GP source fetches immediately. This applies to any source, AMSAT included — politeness is free.
- **CelesTrak extras** — if you've added satellites with the catalog **search** (Satellites → `/`) that your primary source doesn't carry, both update paths re-fetch each of them from CelesTrak right after the main download, so their elements stay as fresh as everything else. This step is **courtesy-throttled**: at most once per 2 hours (persisted across reboots), 2 s between object fetches, 25 objects per update — the status line reports "CT extras fresh (Nm ago)" when the throttle skips it, and the previously cached elements simply stay in use.

GP source

- **Purpose** — choose where element sets come from.
- **Reached from** — Settings → GP source → ENTER.
- **Shows** — a picker: AMSAT (amateur, default), the CelesTrak category groups (Amateur first, then SatNOGS and the other groups), and Custom URL.
- **Keys** — `;`/`.` move (`{`/`}` page); **ENTER** select; ` back. The choice is saved immediately and used by the next Update.

Settings

Figure 49: Settings menu

- **Purpose** — all configuration, grouped into six submenus (Radio / CAT, Rotator, Passes / alerts, Display / sound, Station / logging, Network / data), each shown with its item count. Each row and its adjust keys are tabulated in §8 → Settings.
- **Reached from** — Home → Settings.
- **Shows** — the submenu list, then the rows within a chosen submenu with their current values.

Figure 50: Settings — Radio / CAT

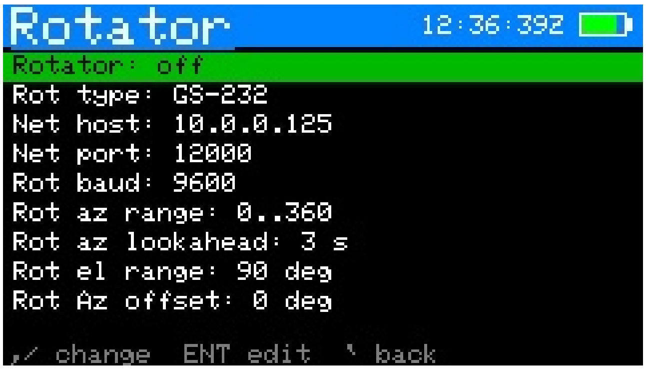


Figure 51: Settings — Rotator

Figure 52: Settings — Station / logging

Figure 53: Settings — Network / data

- **Keys** — `;`/`.` move; **ENTER** open a submenu / edit a text field; `,`/`/` change a value in place; ` back.

WiFi scan

- **Purpose** — pick a network from a live scan instead of typing the SSID.
- **Reached from** — Settings → WiFi SSID row → `s`.
- **Shows** — nearby networks, strongest first, `*` marking secured ones.
- **Keys** — `;`/`.` select; **ENTER** choose (then enter the password unless the network is open); `r` rescan; ` back.

Rotator manual / calibration

- **Purpose** — jog the rotator by hand with live read-back, and — for a **Yaesu (direct)** rotator — calibrate the position ADC. Detail in §17 and ROTOR_INTERFACE.md.
- **Reached from** — Settings → Rotator: manual control → ENTER.
- **Shows** — commanded and actual az/el; the **magnetic target heading** with the local declination (compass hand-pointing needs no arithmetic), flagged `->rotor` when *Rotator → magnetic correction* is also applying it to the hardware; for Yaesu direct, the live ADC counts for each axis.
- **Keys** — `,`/`/` jog azimuth; `;`/`.` jog elevation; `s` step size; `x` stop; (*Yaesu direct only*) `1`/`2`/`3`/`4` capture the ADC counts at az 0 / az full / el 0 / el full; ` back.

CAT serial monitor

- **Purpose** — a live on-device terminal showing the raw CAT traffic between CardSat and the radio, for confirming wiring, model, address, and baud without a PC. Detail in §16 → CAT self-test.
- **Reached from** — Settings → Radio / CAT → *CAT serial monitor*.
- **Shows** — each frame as hex (or ASCII for Kenwood), **TX in cyan, RX in green**, with a decode line. While open it **actively polls the rig** (~every 700 ms) by reading the downlink frequency, so live traffic appears even with no pass running.
- **Keys** — `p` toggle polling on/off (off = passive view); `s` type a raw frame in hex and transmit it; ` back.

CAT self-test

- **Purpose** — exercise the whole CAT link end-to-end in one shot and report what works, before trusting it for a pass. Detail in §16 → CAT self-test.
- **Reached from** — Settings → Radio / CAT → *Run CAT self-test* (radio must be on — start CAT with `r` on Track first).
- **Shows** — a scrolling list of steps (downlink/uplink set+read, modes, band select, sat mode, PTT read, CTCSS), each marked `[PASS]`, `[FAIL]`, or `[INFO]` (the last for capabilities a given radio doesn't expose over CAT).
- **Keys** — `;`/`.` scroll; ` back.

Log menu

- **Purpose** — the QSO logging hub.
- **Reached from** — Home → Log.

Figure 54: Log menu

- **Shows** — New QSO entry, View / edit log, Export to ADIF, Sign & upload to LoTW, Upload to Cloudlog, Voice Memos, Notes. (The list scrolls; a `^`/`v` marker shows when there are more items above or below.)
- **Keys** — `;`/`.` move; **ENTER** open the selected item; ` back.

Sign & upload to LoTW

- **Purpose** — sign un-uploaded satellite QSOs into a `.tq8` and upload them directly to ARRL's Logbook of the World (microSD card + your LoTW key required; full setup in §8 → LoTW upload).
- **Reached from** — Log menu → Sign & upload to LoTW.
- **Shows** — SD-card status, whether the LoTW key/cert are on the card, your station DXCC/CQ/ITU fields, and the count of QSOs not yet uploaded.
- **Keys** — `u` sign & upload (prompts for the key password if encrypted); `a` toggle re-send mode (include QSOs already uploaded); ` back.

Upload to Cloudlog

- **Purpose** — upload satellite QSOs to a self-hosted **Cloudlog** (or **Wavelog**) online logbook over WiFi. Because a Cloudlog instance can itself forward QSOs to LoTW/eQSL, this is usually an alternative to the on-device LoTW upload rather than an addition. Tracked independently of LoTW, so a QSO sent to one isn't marked sent to the other. Full setup in §8 → Cloudlog upload.
- **Reached from** — Log menu → Upload to Cloudlog.
- **Shows** — your configured Cloudlog server, whether the API key and station ID are set, and the count of QSOs not yet sent to Cloudlog.
- **Keys** — `u` upload (connects WiFi if needed); `a` toggle re-send mode (include QSOs already sent to Cloudlog); ` back.

Log entry

- **Purpose** — create or edit one QSO record. Full field list in §8 → Logging QSOs.
- **Reached from** — Track/Polar/Manual → `l`, or Log menu → New QSO entry, or Log list → ENTER on a record.
- **Shows** — every editable field: date, time, sat, mode, DL/UL MHz, call, RST sent/received, grid, notes.
- **Keys** — `;`/`.` move between fields; **ENTER** edit the selected field (the Sat field opens the satellite picker; Mode toggles SSB/CW on linear birds); `s` save; ` cancel.

Figure 55: Log entry

Log list (view / edit)

Figure 56: Log list

- **Purpose** — review and correct stored contacts.
- **Reached from** — Log menu → View / edit log.
- **Shows** — a scrollable list of the most recent 60 QSOs, newest first.
- **Keys** — `;`/`.` move; **ENTER** open a record to edit; (within a record) `x` twice to delete; ` back.

Awards

- **Purpose** — a summary of what you've worked, read straight from the QSO log: all-satellite totals (QSOs, unique grids, distinct satellites, US states, DXCC entities) and a per-satellite breakdown. Full detail in §8 → Awards.
- **Reached from** — Log menu → Awards.
- **Shows** — the totals block, then one row per satellite worked with its QSO and unique-grid counts.
- **Keys** — `g`/`s`/`d` open the full worked **grids / US states / DXCC** lists; `;`/`.` scroll; ` back to the Log menu.

Voice Memos

- **Purpose** — browse, play back, and manage the voice memos on the SD card, and record new standalone memos.
- **Reached from** — Log menu → Voice Memos.
- **Shows** — all memos newest-first: date, time, the satellite recorded on (or `-` for a standalone memo), and length. A `REC` timer replaces the list while recording.

- **Keys** — `;`/`.` move; **ENTER** play the selected memo (any key stops); `n` record a new standalone memo (any key stops & saves); `d` delete (with confirm); `r` refresh; ` back to the Log menu.

Notes (browser)

Figure 57: Notes

- **Purpose** — browse, open, create and delete the plain-text notes stored under `/CardSat/notes/`. Full description in §8 → Notes.
- **Reached from** — Log menu → Notes.
- **Shows** — your notes newest-first, each with the date and time it was last saved (UTC).
- **Keys** — `;`/`.` move; **ENTER** open the selected note; `n` new note; `d` then **ENTER** delete (with confirm); ` back to the Log menu.

Notes (editor)

- **Purpose** — a full multi-line text editor for one note. Because `;` `.` `,` `/` are typed as punctuation, all editor commands use the **Fn** modifier.
- **Reached from** — Notes browser → ENTER on a note, or `n` for a new one.
- **Shows** — the note text with a block cursor; the header shows the note name and a `*` when there are unsaved changes; the footer shows the line number.
- **Keys** — type normally (**ENTER** = new line, **delete** = backspace); **Fn +** **`,`/`/`** cursor left/right; **Fn +** **`;`/`.`** cursor up/down; **Fn +** **`s`** save (prompts for a name the first time); ` exit (unsaved work is saved automatically, or you're prompted for a name).

World map

- **Purpose** — a live equirectangular map of all favorites' footprints with the day/night terminator.
- **Reached from** — Home → World Map, or Next Passes → `m`. Back returns to wherever you entered from.
- **Shows** — every favorite's sub-point and footprint, your station marker, the graticule, and **day/night shading**: the night hemisphere is shaded a dim gray so you can see at a glance which footprints (and which part of your own sky) are in darkness. The shaded region is computed from the sub-solar point and updates live.
- **Keys** — `f` cycle which favorite is highlighted; **`c` recenter the map on your QTH** (press again to return to the classic 0°-centered view); **`n` toggle the night-hemisphere shading** on/off (remembered across reboots); ` back.

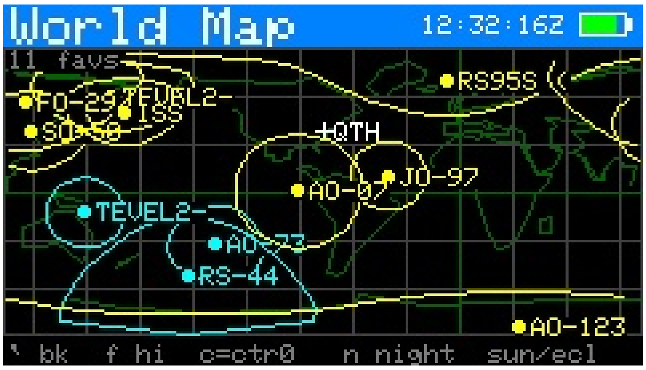


Figure 58: world-map

- **Recentering** — by default the map is centered on 0° longitude. Press `c` to center it on your station's longitude instead, so your QTH sits in the middle and the world wraps around it; this is remembered across reboots. Only the longitude is shifted — north stays up and the equator stays centered. The orbital-analysis ground-track map and the simulation map keep the standard 0°-centered view.

Help

- **Purpose** — an on-device scrollable key reference, available almost anywhere.
- **Reached from** — `h` on most screens.
- **Shows** — per-screen key summaries.
- **Keys** — `;/.` scroll; `g` opens the **Glossary & math**; `m` opens the **User guide**; `s` opens the **Ham satellite history**; `t` opens the **Tech help** guide; `l` opens the **Learn** (radio + orbit theory) screen; `o` opens the **Orbit animations**; `f` opens the **band plan / frequency reference**; `a` opens the **AMSAT Fox anatomy** — an animated, labeled tour of a 1U Fox CubeSat (spin with `;/.`, pause with space, cycle the callouts with `,//`; facts sourced from AMSAT's Fox documentation; `i` inside opens a short **Fox & CubeSats** primer); `c` opens a **CubeSat Simulator** intro — AMSAT's desktop satellite for hands-on learning (kits at the AMSAT Store; cubesatsim.com); ` back.

About

- **Purpose** — build and diagnostic information, and the doorway to the Tools hub, printing, games, readiness and license screens. Full detail in §8 → About.
- **Reached from** — Home → About.
- **Shows** — author credit, firmware version and build date, storage backend, GP catalog size and freshest element age, WiFi/IP, battery, uptime, and the heap line (total free + **largest free block** — the number that matters for TLS uploads on this no-PSRAM board).
- **Keys** — `t` **Tools** hub; `p` **Print a report** submenu; `r` **Station readiness** checklist; `z` **Games menu**; `a` print the **Support AMSAT** page; `c` print the **operator contact card**; `l` **License & credits**; ` or **ENTER** back Home.

Print a report

- **Purpose** — a scrollable list of **every** menu-printable report (twenty-nine), so reports that have no natural home screen can be printed from one place. Reports that need an active satellite guard themselves with a one-line note.

- **Reached from** — About → `p`.
- **Shows** — the report list with the current print destination(s) in the header (printer host, serial, file — or a warning if no output is enabled).
- **Keys** — `;`/`.` move; **ENTER** print the highlighted report (the list **stays open**, with the result on the status line, so several can be run in a row); ` back to About.

Charge / Sleep

- **Purpose** — a minimal low-power mode for charging or parking the device. Full detail in §8 → Charge / Sleep.
- **Reached from** — Home → Charge / Sleep (the last item).
- **Shows** — a blanked screen; any key wakes a brief status view (battery, charge state) before it blanks again.
- **Keys** — any key wakes the status view for a few seconds; `/**DEL** exits to Home, restoring the backlight, the 240 MHz clock, and WiFi (both are reduced/off while parked).

Station readiness

- **Purpose** — one glanceable green/red checklist of everything a working pass depends on: clock, location, GP data and age, WiFi, radio and rotator configuration, SD card, callsign, battery. First-hour onboarding and a pre-pass field check in one screen.
- **Reached from** — About → `r`.
- **Keys** — ` back to About.

Understanding state vectors

Most of CardSat works from **GP mean elements** (the modern successor to the two-line element set, or TLE): a compact set of six orbital numbers — mean motion, eccentricity, inclination, right ascension of the ascending node, argument of perigee, and mean anomaly — plus a drag term, all referenced to an epoch. Those elements are what SGP4 propagates to predict where a satellite will be. But mean elements are not the only way to describe an orbit, and for a brand-new object you often won't have them yet. That's where a **state vector** comes in.

A state vector is the most direct possible description of an orbit: where the object is and how fast it's moving, right now. It is just two 3-D vectors at a single instant (the epoch):

- **Position** `(rx, ry, rz)` — the satellite's location, in kilometers, measured from the center of the Earth along three perpendicular axes.
- **Velocity** `(vx, vy, vz)` — how fast it's moving along each of those axes, in kilometers per second.

Six numbers and a timestamp. Unlike mean elements, a state vector needs no orbital theory to interpret — it is the raw kinematic truth at that moment. Given a state vector and the laws of gravity, you can (in principle) compute the entire orbit forward and backward in time. In fact the two descriptions are interchangeable: a state vector and a set of orbital elements at the same epoch describe the *same* orbit, just in different coordinates — one Cartesian (x/y/z), the other in terms of the ellipse's shape and orientation.

Why the numbers look the way they do. For a satellite in low Earth orbit, the position components are each a few thousand kilometers (the orbital radius is roughly 6,700–7,400 km, so the three components sum in quadrature to that), and the velocity components are each a few kilometers per second (orbital speed at that altitude is about 7.5 km/s). A component can be positive or negative depending on which side of each axis the satellite is on and which way it's heading. There's nothing to "read" from an individual number the way you'd read an inclination; the meaning is in the vectors as a whole.

Frames — the part that trips people up. A state vector is only meaningful together with the **reference frame** its x/y/z axes are defined in. The same physical orbit produces *different* numbers in different frames, because the axes point in different directions. Two frames matter here:

- **TEME** (True Equator, Mean Equinox) — the somewhat idiosyncratic inertial-ish frame that SGP4 itself uses internally. If your source already gives TEME, CardSat uses it directly.
- **J2000** (also called GCRF or, for our purposes, ICRF) — the standard modern inertial frame, fixed to the stars at the J2000.0 epoch. Most launch providers and orbit-determination tools publish in J2000/GCRF. CardSat rotates a J2000 state into TEME on-device (using IAU-76 precession and a truncated IAU-80 nutation model, accurate to about 1–2 meters) before fitting.

Choosing the wrong frame is the single most common mistake: the fit will either fail to converge or produce elements that are subtly wrong, because the axes were misaligned by the small but real rotation between the two frames (tens of meters of position difference at LEO). If your source says "J2000", "GCRF", "ICRF", or "EME2000", pick **J2000**; if it says "TEME" or "true equator", pick **TEME**.

Where you get one. Launch providers, deployment brokers, and rideshare integrators routinely distribute a state vector for a payload — often *before* launch, with an epoch days or weeks in the future — so operators can plan the first passes before the object has a NORAD catalog number and public TLE. Orbit-determination software (from a radar or optical track) also outputs state vectors. Any of these can be typed into CardSat's **State vector → GP** tool.

Why CardSat fits rather than converts. You might expect a state vector to convert *directly* into orbital elements — and mathematically it does, into what are called **osculating** elements (the instantaneous ellipse that exactly matches the position and velocity at that moment). The catch is that SGP4 does **not** expect osculating elements; it expects **mean** elements, which have the short-period gravitational wobbles deliberately averaged out. Feeding osculating elements straight into SGP4 produces errors of many kilometers, because SGP4 then re-adds perturbations that were already baked into the osculating values. So instead of a one-shot conversion, CardSat runs a small **differential-correction fit**: it starts from the osculating elements as a first guess, then repeatedly nudges the six mean elements and re-runs its own SGP4 until the propagator reproduces your entered state vector to within the achievable precision (tens of meters, limited by the TLE number format). The result is a set of mean elements that behaves correctly in every other CardSat prediction. Because a single instant carries no information about atmospheric drag, the drag term `B*` is set to zero — which is fine for the first days but means the elements slowly drift; re-acquire real published elements once the object is cataloged. See **State vector → GP** under *Tools*, below, for the step-by-step operation.

Tools

- **Purpose** — a set of **60 offline** ham-radio and satellite bench tools, all computed locally (no network needed): the **scientific**, **programmer's**, and **graphing** calculators, a **Tiny BASIC** interpreter, a **location converter** (Maidenhead / decimal / DMS / DDM

Figure 59: Tools menu

Figure 60: Scientific calculator

Figure 61: Programmer calculator

Figure 62: Character lookup

Figure 63: DXCC entity lookup

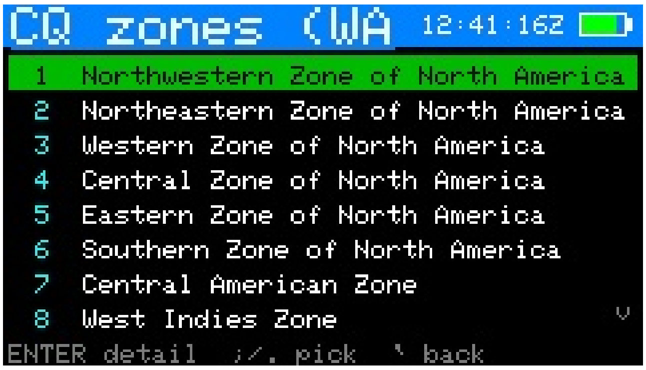


Figure 64: CQ zones

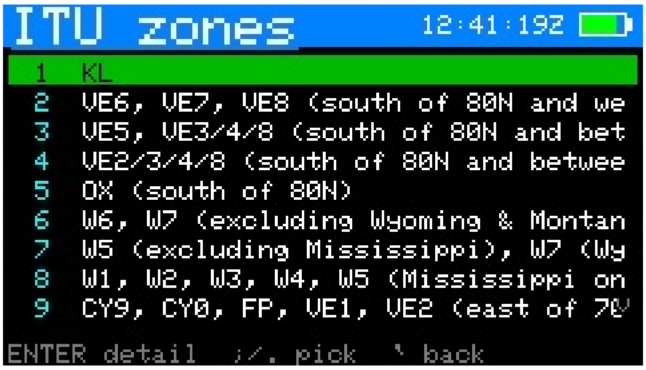


Figure 65: ITU zones

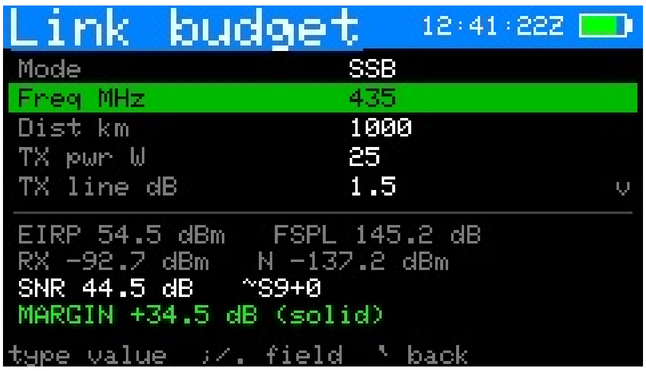


Figure 66: Link budget

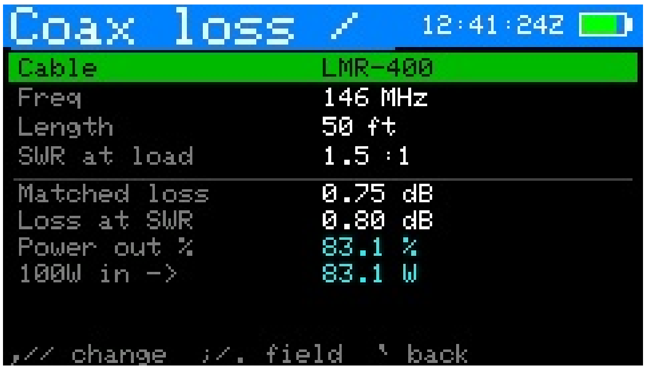


Figure 67: Coax loss / power

Figure 68: Yagi elements

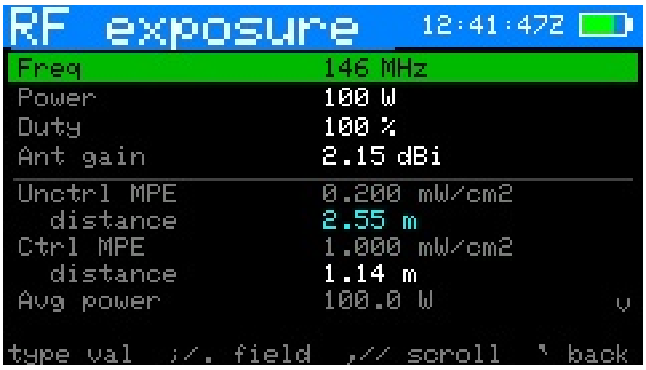


Figure 69: RF exposure (MPE)

Figure 70: Battery runtime

Figure 71: Orbit lifetime (debris)

Figure 72: Cross-section area

/ Plus Code / UTM / MGRS / USNG), the **link budget** chain, **state vector → GP**, the **CubeSatSim C2C** crib, the DXCC / CQ-zone / ITU-zone / CTCSS / operating / radio-math reference cards, **character lookup**, and the live-recalc forms for **coax loss / power**, **dipole**, **vertical / ground plane**, **yagi** and **quad** dimensions, **RF unit** conversions (dBm / W / V), **SWR / return loss**, **free-space path loss**, **RF exposure**, **battery runtime**, **orbit lifetime**, **cross-section area**, **phasing line / stub**, **wavelength / frequency**, **attenuator pad**, **dB chain sum**, the ARRL radio-math calculators (**complex / polar**, **reactance & resonance**, **RC/RL time constant**), and a general **unit converter**, and the **0.9.59 satellite & construction tools**: a **conjunction screener** and **orbital neighborhood** view, a **transponder passband planner**, a **link-margin-vs-elevation** curve, a **debris-group screen**, a **Doppler budget** calculator, **cascade NF & G/T** (Friis), a **sun-noise G/T** measurement helper, a **helix antenna** designer, **L/Pi/T matching networks**, **pointing loss**, an **IMD products** finder, **microstrip / stripline** impedance, **toroid winding**, a **delta-v** mini-set, **thermal equilibrium**, a **polarization / Faraday** estimate, **trace & wire ampacity**, and a **PLL / frequency-plan** helper.

- **Reached from** — About → `t`.
- **Menu layout** — the sixty tools are organized into **six categories**, so Tools opens on a short **category list** rather than one long scroll. Pick a category with `;/.` and ENTER to open its tool list; ` steps back from a tool list to the categories (and again to leave Tools). The categories are **Calculators & programming** (the scientific, graphing, and programmer's calculators, Tiny BASIC, unit and location converters), **Satellite & orbital** (conjunction screener, orbital neighborhood, transponder planner, link-margin curve, debris-group screen, Doppler budget, orbit lifetime, cross-section, delta-v, thermal equilibrium, polarization/Faraday, state vector → GP, CubeSatSim), **Terrestrial**

propagation (link budget, radio horizon, Fresnel clearance, terrestrial path budget, tropo ducting, rain fade, terrain profile, free-space path loss, pointing loss), **Antennas & feedlines** (dipole, vertical, yagi, quad, helix, matching networks, phasing line, coax loss, microstrip/stripline, toroid winding), **RF chain & measurement** (RF units, SWR, wavelength/frequency, dB chain, attenuator, cascade NF & G/T, sun-noise G/T, IMD, RF exposure, PLL/frequency plan), and **Electronics & references** (complex/polar, reactance, RC/RL, ampacity, battery runtime, DXCC, CQ/ITU zones, CTCSS, operating and radio-math cards, character lookup). Each category row shows its tool **count**. A **first-letter jump** (press a letter to hop to the next entry starting with it) works in both the category list and inside a category, and Tools remembers the category you were browsing. The entries below are described grouped roughly by these categories.

- **Calculator** — a traditional tape interface: type an expression (`2+3*sin(45)`), ENTER evaluates, and the expression and its result join a **scrolling tape** above the entry line (`[`/`]` scroll back through history — the arrow keys `;` and `.` stay available as expression characters, `.` being the decimal point). Supports `+ - * / ^ ()`, `pi`, `e`, the previous result as `Ans`, and sin cos tan asin acos atan sqrt ln log exp abs, with trig in **degrees**. It also carries **amateur-radio helpers** (press `'` for the second hint page): the functions `dbm(W)` and `w(dBm)` for power, `db(ratio)` and `undb(dB)`, `wl(MHz)` for wavelength in meters and `fq(m)` for the reverse, `sinh/cosh/tanh`, `floor/ceil/round`, and the constants `c` (speed of light m/s), `kB` (Boltzmann), `Re` (Earth radius km), `mu` (Earth GM) and `g0`. So `wl(146)` gives the 2 m wavelength, `dbm(100)` gives 50 dBm, `db(2)` gives 3.01 dB.

  0.9.59 adds a batch in three flavors. **Math:** two-argument `atan2(y,x)` (degrees out), `hypot(a,b)`, `mod(a,b)`, `min(a,b) / max(a,b)`, `ncr(n,r) / npr(n,r)`, plus `sign`, `log2`, `cbrt`, `fact(n)` ($n \leq 170$), `d2r/r2d`, and `rnd()` for a random 0–1. **RF, chosen to pair with the new tools:** `swr2rl(s) / rl2swr(dB)`, mismatch loss `mml(swr)`, path loss `fspl(MHz,km)`, noise figure ↔ temperature `nf2t(dB) / t2nf(K)` (the cascade tool's little sibling), `dbd(dBi) / dbi(dBd)` (±2.15), and Doppler `dop(MHz, rrKmS)` in Hz straight from a range rate — feed it `SATRR` numbers. **Orbital one-liners** from the audited formulas: `porb(altKm)` circular period in minutes, `vorb(altKm)` km/s, `fpr(altKm)` footprint radius km. The suffix set gains `f` (femto). The function hints stay pinned just above the footer; `'` toggles between the general and ham pages. You can also type **metric-prefix suffixes** on numbers (`100k`, `2.2n`, `47p`, `146M` — case matters, `M`=mega vs `m`=milli), and press `\` to toggle **engineering-notation** output (results shown as a mantissa with a metric prefix, e.g. `4.7 k`). DEL backspaces (and only backspaces — it never exits); `` ` `` returns.

- **Graphing calculator** — plot **y = f(x)** using the same expression parser and function set as the scientific calculator, with the added variable **x**. Press ENTER to type a function (e.g. `sin(x)`, `x^2-4`, `1/x`, `exp(x/50)`); the curve is sampled one point per pixel column and drawn across a pan/zoomable window. **Arrow keys pan**, **`+`/`-` zoom** about the center, **`a`** auto-fits the vertical range to the visible data, and **`r`** resets the window. As with the scientific calculator, **trig is in degrees**, so the default window is $x \in [-180, 180]$. Discontinuities (the poles of `tan(x)`, `1/x` at zero) break the curve rather than drawing a spurious vertical line. All local, no network.

  Since 0.9.59 the plotter is a small instrument:

  - **`2`** sets a **second function Y2** (drawn orange; empty to clear).
  - **`t`** toggles a **trace cursor**: `,`/`/` step it, `{`/`}` jump ×10; the readout shows x, y (and y2, or the numeric **dy/dx** when only Y1 is up).

  - `z` finds the **zeros** of Y1 in the window (bisection-refined, marked with cyan ticks); `Z` finds **Y1 = Y2 intersections** the same way. `x` clears marks and roots.
  - `m` (while tracing) drops **integral marks**: first press sets A, second sets B and the **Simpson ∫** between them appears as `S=` on the plot; third press clears.
  - `b` toggles a **table view** (x / Y1 / Y2 rows; `;/.` shift, `+/-` change the step).
  - `c` toggles **CSV plot mode**: `/CardSat/plot.csv` is streamed and min/max-decimated to the plot columns — a 100,000-row log plots in flat memory. The last comma-separated field of each line is the value, so both `y` and `x,y` files work; `x` re-reads the file, `a` refits the vertical range. Pair it with BASIC's gated `FPRINT` logging or drop a CSV on the card from your computer.

- **Tiny BASIC** — a small line-numbered BASIC interpreter with a built-in editor, for writing and running little programs on the device (4 KB program limit, all local). See **Tiny BASIC reference** below for the full language, editor keys, limits, and worked examples.

- **Location converter** — one position shown at once in every format a satellite or field operator is likely to need. Seeds from your station location; edit the **grid**, **latitude**, or **longitude** (`;/.` pick the field, ENTER to type) and every other format re-derives. Outputs: **Maidenhead** (6- and 8-character), **decimal degrees**, **degrees-minutes-seconds (DMS)**, **degrees-decimal-minutes (DDM)**, **Plus Code** (Open Location Code, 10-digit), **UTM** (zone + easting/northing, WGS84), **MGRS**, and **USNG** (the spaced form of the same military grid). `,//` scroll the derived list; `s` adopts the shown position as your station QTH. All conversions are computed on-device with no network. The UTM/MGRS projection and the Plus Code encoder were validated byte-for-byte against reference implementations across both hemispheres; they are accurate to about a meter, which is well within what these grids represent. (**what3words** is intentionally not offered — it is a proprietary, network-only wordlist lookup, not an offline algorithm, so it doesn't fit an offline tool.)

- **Programmer calc** — a 64-bit value shown at once in **hex / dec / bin / oct**. Type digits in the current entry base; **`;/.` (up/down) move the highlighted base row** and `w` cycles the display width (8/16/32/64 bits, which masks the value). Bitwise `&` `|` `^`, `n` = NOT, `m` = negate, `</>` = shift left/right, plus `+ - * /`, all via a pending-operation model (`=`/ENTER applies). **`x` clears** and DEL drops the low digit (DEL never exits) — `b` and `c` are hex digits, so they can't double as commands. Useful for CI-V byte math and bitmasks.

- **Char lookup** — enter a byte in **hex, dec, bin or oct** (`,//` cycle the entry base, digits type in it, `;/.` browse ±1) and see everything it represents at once: the **ASCII** character or control-code name, its **Morse** pattern, the **Baudot / ITA2** letters- and figures-shift meaning for 5-bit values (US-TTY, as used on RTTY), and the **BCD** reading (CI-V frequency bytes are BCD). Typing any non-hex letter (g–z) looks that character up directly; `x` zeroes, DEL drops the low digit. Press **`=`** for a **browsable full table** — printable ASCII 32–126 with each character's decimal value, Morse, and Baudot code (with L/F letters/figures-shift tag), `;/.` to scroll; `=` returns to the single-character view.

- **DXCC entity lookup** — type a **prefix, partial name, or entity code** and matching DXCC entities list live; ENTER opens a detail card with the **entity code, primary prefix, continent, ITU and CQ zones, name**, status flags (deleted / third-party traffic OK) and any **ARRL footnotes**. The table (current + deleted entities) is embedded from the ARRL DXCC list and works entirely offline; deleted entities are dimmed. `;/.` pick, DEL edits the query, ` back.

- **CQ zones (WAZ)** — all **40 CQ zones** with their names; ENTER opens a detail card with the full prefix/region definition (scroll with `;`/`.` if it runs long). From a **DXCC entity's** detail card, press `z` to jump straight to that entity's CQ-zone definition. Embedded from the CQ WAZ list, fully offline.
- **ITU zones** — the **ITU zone** list (numbers 1–75, 78, 90 — ITU numbering isn't contiguous), each with its prefix/region definition; ENTER opens a scrollable detail card. From a DXCC entity's detail card, `i` jumps to that entity's ITU-zone definition (`z` jumps to the CQ zone). Embedded from the RSGB ITU zones list, offline.
- **Link budget** — the full chain, live-recalc: **TX power → feedline → antenna → free-space path (+ extra losses) → RX antenna → feedline → receiver noise floor → SNR → margin**. Twelve inputs scroll in a window (`;`/`.` move, type to edit); a **Mode** row (`,`/`/`) presets bandwidth + required SNR together (CW 500 Hz, SSB, FM, 1k2/9k6 packet, FT8, JT65/Q65, or Custom). Pinned outputs show EIRP, FSPL, RX power, noise floor (−174 dBm/Hz + 10·log BW + NF), SNR, an **S-meter estimate** (IARU S9 = −93 dBm above 30 MHz, −73 below, 6 dB/S-unit), and the **margin**, color-coded: green ≥ 6 dB (solid), orange 0–6 (thin), red < 0 (no copy). When a satellite is being tracked, the **Distance** field opens pre-filled from its **live slant range** (marked "(live)") and the frequency from the selected transponder downlink; press `p` to re-sync after the satellite moves. Editing Distance by hand drops the "(live)" tag.
- **Forms** — `;`/`.` move between fields; type digits to edit a value (ENTER or moving off the field commits it); on a pick-list field (e.g. the coax type) `,`/`/` cycle the choices; results recompute instantly. **Yagi and quad take an element count** (up to 12 / 8) and list every element; `,`/`/` scroll the output when the list runs past the screen. Each Tools category scrolls when it grows past eight entries. ` back.
- **RF exposure (MPE)** — enter frequency, power, duty cycle and antenna gain; shows the FCC OET-65 **Maximum Permissible Exposure** limits (controlled and uncontrolled, by band) and the estimated **compliance distances** in meters, plus average power and a with-ground-reflection worst case. A far-field estimate for planning — not a substitute for a full station RF-exposure evaluation.
- **Battery runtime** — enter capacity (Ah), RX and TX current draw, TX duty cycle and the usable fraction of capacity; shows the duty-weighted **average current** and the estimated **runtime** (decimal hours and h:m). Handy for sizing a field battery.
- **Orbit lifetime (debris)** — for CubeSat builders assessing post-mission disposal: enter altitude (use **perigee** altitude for an elliptical orbit), mass, cross-sectional area and drag coefficient; shows the **ballistic coefficient** and an estimated **orbital lifetime** (drag decay through an exponential-atmosphere model), with pass/fail against the **25-year** and newer **5-year** debris-mitigation guidelines. This is an order-of-magnitude estimate at nominal solar activity (real lifetime swings several-fold over the solar cycle) — a planning figure, **not** a compliance determination. Use **NASA DAS** for that.
- **Cross-section area** — the satellite's projected (drag) area. Pick a **CubeSat form factor** (`,`/`/` — 0.5U through 16U) to fill the body dimensions, or choose **Custom** and enter any body **X/Y/Z** (cm); add **deployable panel area** (m^2) if fitted. Shows the **end-on** (minimum) and **broadside** faces, the **maximum projected** area at any orientation ($\sqrt{(ab^2+bc^2+ca^2)}$), and the **tumbling average** ((ab+bc+ca)/2 by Cauchy's theorem; a flat panel contributes half its area). The tumbling average is the drag-relevant figure — it's shown as **"→ debris Area"** to drop straight into the Orbit-lifetime tool.

- **Phasing line / stub** — physical cable length for a wanted **electrical** length, the tool a satellite-antenna builder needs for circularly-polarized arrays (turnstiles, eggbeaters, crossed Yagis) and matching stubs. Pick the **coax type** (its velocity factor is applied), enter the **frequency**, and choose a **fraction** (¼/½/3/8/¾/1 λ). Shows the length in ft+in and meters (honors the antenna-units setting) plus the free-space wavelength. **Verify a cut line on an analyzer** before trusting it — real velocity factors vary between cable batches.
- **Wavelength / frequency** — free-space λ = c/f with the common ¼/½/5/8-wave cut lengths, in both unit systems.
- **Attenuator pad** — resistor values for a **pi** and a **T** resistive pad at a target attenuation and system impedance (default 50 Ω), plus the power ratio.
- **dB chain sum** — add a short chain of gains (+) and losses (−) in dB; shows the net dB, the linear power and voltage ratios, and the result on 100 W in.
- **Operating references** — a quick-reference card (no radio needed): common **Q-codes**, the **ITU phonetic alphabet**, and the **RST** system. `,`/`/` switch tab.
- **CTCSS tone reference** — the standard EIA 38-tone list, with the FM satellites' known uplink tones called out (ISS/SO-50/AO-91/92 = 67.0 Hz; SO-50 arms with 74.4 Hz).
- **Radio math reference** — a scrolling cheat sheet distilled from the ARRL *Radio Mathematics* supplement: the decibel table (×2 = +3 dB, etc.), AC voltage factors (Vrms = 0.707 × Vpeak), metric prefixes, useful constants, and the reactance / resonance / SWR / time-constant formulas. `;`/`.` scroll.
- **Complex / polar** — rectangular $a + jb$ to polar (magnitude and angle) and the reciprocal 1/Z, for impedance and phasor work.
- **Reactance & resonance** — enter frequency, L and C; shows **Xl**, **Xc**, the net reactance, and the **LC resonant frequency**.
- **RC/RL time constant** — enter R and C; shows **τ = RC**, the 1/3/5-τ charge percentages (63 / 95 / 99 %), and the corresponding cutoff frequency.

22.0.76.1 0.9.59 satellite & construction tools These were added in 0.9.59. The five **satellite tools** are standalone screens (they read the active satellite and, where noted, the loaded catalog); the rest are live-recalc forms like the ones above. The two screeners that propagate pairs of objects carry an on-screen reminder that **public TLE/GP elements are only kilometre-class accurate — these screens are for situational awareness, not collision avoidance.**

- **Conjunction screener** — pick a second object from the loaded satellites and scan the next **6 hours** for close approaches between it and the active satellite (30 s coarse grid, each candidate minimum refined to 1 s). Lists up to five approaches under 800 km by time, miss distance, and relative velocity; rows turn orange under 100 km and red under 25 km. `;`/`.` choose the second object, ENTER runs the scan.
- **Orbital neighborhood** — a no-propagation view of which loaded objects share the active satellite's altitude shell: everything whose perigee–apogee band comes within 150 km, sorted closest-first, with each object's band, inclination, and the gap (or **OVLP** when the bands overlap). ENTER hands the highlighted object straight to the conjunction screener.

- **Transponder planner** — a satellite-frame crib for a linear transponder: an 11-row table of downlink/uplink dial pairs from 0 – 100 % across the passband (respecting the inversion sense), for coordination and net planning. These are the **on-satellite** frequencies with no Doppler applied — the live tracker adds Doppler on the air. `;`/`.` pick the transponder, `p` prints the table to the configured printer/serial/file.
- **Link margin vs elevation** — plots how link margin changes across a pass from the changing slant range: enter the orbit altitude, frequency, and the margin at the horizon, and the screen draws margin (dB) versus elevation with the extra decibels at TCA called out. `;`/`.` pick a field, type to edit, `x` seeds the altitude from the active satellite.
- **Debris group screen** — fetches a CelesTrak fragmentation-cloud group (Cosmos-2251, Iridium-33, Fengyun-1C, or the last-30-days launches) as **GP JSON** — the same OMM/GP format the rest of CardSat uses, and the one that still lists new objects now that the legacy 5-digit TLE catalog space is full — keeps only the objects in the active satellite's altitude band, and coarse-screens each for closest approach over the next **3 hours**. The download is transient: it is streamed to a temp file, parsed object-by-object (flat memory even for a large cloud), screened, and deleted, so the resident 150-satellite database is never touched. Needs network for the fetch. A very large historical cloud (Fengyun-1C is thousands of objects) can exceed the free-space budget and stop with a storage error — the smaller groups fit comfortably.
- **Doppler budget (orbit)** — max Doppler shift and the peak tuning rate (Hz/s) at TCA for any orbit and frequency, seeded from the active satellite's apogee/perigee. The number that sizes tuning-loop cadence, beacon bandwidth, and digital-mode feasibility on a pass.
- **Cascade NF & G/T** — a Friis noise-figure cascade for the classic sat-station chain (antenna → LNA → coax → radio): system NF, receiver and system noise temperature, and **G/T** — plus the same figure **without** the LNA, and the decibels the mast preamp buys.
- **Sun-noise G/T measure** — turns a measured sun/cold-sky **Y-factor** (off any S-meter or SDR) into a measured **G/T**. Seeds the solar flux from the last-fetched 10.7 cm index (an upper bound below ~2 GHz — enter the flux at your operating frequency for the truest result); the sun's 0.53° disc is a point source for any amateur sat antenna.
- **Helix antenna** — axial-mode helix (Kraus): gain, beamwidth, feed impedance, boom and total wire length, and diameter, from circumference (in wavelengths), turns, and pitch. The gain figure is noted as running about 1 dB optimistic against modern measurements.
- **L/Pi/T match network** — impedance-matching network component values for **L**, **Pi**, and **T** topologies (low-pass form) between two resistive impedances, with the minimum Q flagged for the Pi/T cases. The Pi and T designs were verified by an ABCD/impedance round-trip on the host before shipping.
- **Pointing loss** — antenna pointing loss (dB) from beamwidth and pointing error (parabolic main-lobe approximation), with the ½ dB / 3 dB error angles, seeded from the configured rotator deadband so you can see what the deadband costs.
- **IMD products** — the third- and fifth-order intermodulation products of two carriers, each flagged **IN** or **out** against an entered band (e.g. a transponder passband) — for finding your own products in an inverting passband, or spur planning at the bench.
- **Microstrip/stripline Z0** — characteristic impedance from trace width, dielectric height, and εr (Hammerstad for microstrip; the t→0 approximation for stripline), with effective permittivity and the quarter-wave length. CubeSat RF-board work.
- **Toroid winding** — turns for a target inductance on the common Amidon iron-powder and ferrite cores (built-in AL table), the actual inductance at that integer turn count,

and an approximate wire length.

- **Delta-v (Hohmann/plane)** — the two Hohmann burns and transfer time between two circular altitudes, an optional plane-change cost, and a deorbit-burn estimate — all vis-viva.
- **Thermal equilibrium** — first-order flat-plate equilibrium temperature in sunlight (one- or two-sided radiating) from solar absorptivity and IR emissivity, with handbook presets (white paint, black anodize, solar cell, polished aluminum, gold/MLI, Kapton) and an eclipse note.
- **Polarization / Faraday** — the fixed 3 dB linear↔circular mismatch, the cross-pol penalty, and an order-of-magnitude Faraday-rotation estimate at 146 and 437 MHz from a chosen ionospheric TEC — the "why did the linear Yagi fade" tool.
- **Trace & wire ampacity** — PCB trace width for a current and temperature rise (IPC-2221, external/internal), or AWG wire diameter, resistance, and chassis-wiring ampacity with an OK/OVER check against your load.
- **PLL / frequency plan** — integer-N synthesizer output, comparison and step frequency, and the reference-spur offsets, from reference, R/N dividers, and an output multiplier — for homebrew transverter and LO planning.
- **State vector → GP** — compute **GP mean elements** from an orbital **state vector** (position + velocity) as distributed by a launch provider, and optionally save the result as a **manual satellite**. Rather than converting the state to osculating elements (which SGP4 misinterprets, giving multi-km errors), it runs a **differential-correction fit** that adjusts the mean elements until CardSat's own SGP4 reproduces the state. Enter an epoch (UTC) and `rx ry rz` (km) / `vx vy vz` (km/s), and choose the input **frame**: **TEME** (used directly) or **J2000** (GCRF/ICRF — rotated into TEME on-device via IAU-76 precession and a truncated IAU-80 nutation, accurate to ~1–2 m). Output is the fitted mean motion, eccentricity, inclination, RAAN, argument of perigee and mean anomaly, with derived apogee/perigee and the **fit residual**; `B*` is set to 0 (a single state carries no drag information, so predictions degrade over days — re-acquire real elements once the object is cataloged). `s` saves the elements as a manual satellite.
- **Pre-launch (future) epochs** — launch providers often distribute a state vector before deployment, with an epoch days or weeks ahead of now. That is fully supported: the fit, the saved manual satellite, and all pass/Doppler predictions work with a future epoch (SGP4 propagates correctly on either side of the epoch). While the epoch is still ahead of the clock the satellite's age reads **`pre-lnch`** (in cyan) instead of a "GP *n* d" age, and such a satellite never triggers the stale-element auto-refresh. Passes computed before the epoch are the nominal pre-launch orbit; re-run predictions once the true post-deployment elements are published.

Several tools can **print their result** to the configured print sinks (WiFi/serial/file, set under Settings → Network / data). On tools that don't accept typed text — the **programmer calculator**, **graphing calculator**, **location converter**, and **character lookup**, plus the **Tiny BASIC output console** — press **`p`** to print. On the two tools where you type freely — the **scientific calculator** and the **Tiny BASIC editor** — use **`Fn+p`** instead, so a plain `p` still types normally. The scientific calculator prints the current entry and result; the programmer calculator prints all four bases; the location converter prints every coordinate format; Tiny BASIC prints either the program listing (`Fn+p` in the editor) or the last run's output (`p` on the console).

The Tools menu supports a **first-letter jump** (press a letter to hop to the next tool starting with it) and **remembers the last tool** you used. Form tools **remember their field values** between sessions (your coax, power, gains stay put); press **`x`** in a form tool to reset it to defaults. The **antenna-lengths units** (ft/in or meters) are set in **Settings → Display / sound**. Note that orbital distances, altitudes and satellite sizes are always shown in metric regardless of that setting.

Tiny BASIC reference

CardSat includes a small **line-numbered BASIC interpreter** with a built-in editor, reached from **Tools → Tiny BASIC**. It's meant for short programs — quick calculations, loops, table generation, or just playing with code in the field — and runs entirely on the device with no network. Programs are capped at **4 KB**.

22.0.76.2 The editor The editor opens empty (nothing is pre-populated). Type your program with one statement per **numbered** line, pressing ENTER to start each new line. Plain keys type literally, so `s`, `b`, `h`, and the punctuation keys all insert characters; editor commands are on **`Fn`**:

Key	Action
`Fn+R`	**Run** the program (switches to the output console)
`Fn+S`	**Save** (prompts for a name the first time) to `/CardSat/basic/<name>.bas`
`Fn+L`	**Load** a saved program by name
`Fn+N`	**New** — clear the editor to start a fresh program
`Fn+P`	**Print** the program listing (to the configured print sinks)
`Fn+H`	open **Help** (a plain `h` types the letter)
`Fn+b`	save a **screenshot** (a plain `b` types the letter)
`Fn+,` / `Fn+/`	Move the cursor **left / right**
`Fn+;` / `Fn+.`	Move the cursor **up / down** a line
DEL	Backspace
`	Back to the Tools menu

The header row shows the program name (or `(unsaved)`) and the current size out of 4096 bytes. The footer shows only the most-used keys — press **`Fn+h`** for the full list on the Help screen.

On the **output console**, `;`/`.` scroll, **`p`** prints the output, and ` returns to the editor. If the program stopped with an error, the reason is shown on the last line (in red), prefixed with `?`.

22.0.76.3 Program structure Every statement lives on a numbered line: `10 PRINT "HI"`. Lines may be **entered in any order** — they are sorted by line number before running, so you can insert `15 …` between `10` and `20` later. Line numbers are also the targets for `GOTO`/`GOSUB`/`IF … THEN`.

22.0.76.4 Statements

Statement	Meaning
`LET A = expr`	Assign a variable. The `LET` is optional: `A = 5` works too.
`PRINT …`	Print text and values (see below). `?` is a shorthand for `PRINT`.
`IF cond THEN …`	If the condition is true, either jump to a line number or run the statement after `THEN`.
`GOTO n`	Jump to line `n`.
`GOSUB n` … `RETURN`	Call a subroutine at line `n`; `RETURN` comes back to the line after the `GOSUB`.
`FOR V = a TO b [STEP s]` … `NEXT V`	Loop `V` from `a` to `b` (by `s`, default 1; `s` may be negative).
`REM …`	A comment (ignored).
`END`	Stop the program.
`DIM @(n)`	Allocate the one numeric array, `n` ≤ 256 elements (`@(0)…@(n-1)`), zeroed. Freed when the run ends.
`DATA v, v, …` / `READ X[,@(I),…]` / `RESTORE`	Classic data lists: `DATA` lines hold numbers, `READ` pulls them in order into variables or `@()` slots, `RESTORE` rewinds. Reading past the end halts with `out of DATA`.
`ON expr GOTO n1, n2, …`	Computed jump: `expr`=1 goes to `n1`, 2 to `n2`, …; out of range falls through.
`SATSEL expr`	**Re-snapshot the `SAT*` names** (geometry + elements) for satellite number `expr` in the list (0-based). One SGP4 run per call, budgeted at **2,000 per program**; the pass names (`AOSIN` `LOSIN` `PASSEL` `PASSAOS…`) always refer to the run-start active satellite.
`TXSEL expr`	Snapshot transponder `expr` (0-based) of the **active** satellite into `TXDL` `TXUL` `TXBW` `TXINV` `TXLIN` (`NTX` tells you how many are loaded).
`LPRINT …`	`PRINT`, but the line goes to the **configured report sinks** (printer / serial / file) — same grammar as `PRINT`. Sinks open on the first `LPRINT` and close when the run ends; with no sink configured it halts with `no print output`.

Statement	Meaning
`CLS` / `PSET x,y[,c]` / `LINE x1,y1,x2,y2[,c]` / `CIRCLE x,y,r[,c]` / `TEXT x,y,"s"` / `SHOW`	**Canvas graphics.** Draw on the 240×135 screen (colors 0–9: black, white, red, green, blue, yellow, cyan, orange, gray, dark green; default white), then `SHOW` pushes the frame. A `SHOW`ed frame **stays on screen after the run** until you press a key. `TEXT` also takes an expression instead of a string.
`FOPEN "name"` / `FPRINT …` / `FCLOSE`	**Gated file logging** (off by default — Settings → Station/logging → *BASIC file write*). Appends lines to `/CardSat/basic/<name>` (plain names only, no paths). The file closes automatically at run end.
`FILES`	List the programs and logs in `/CardSat/basic/` to the console.

Still no `INPUT`, by design. A program runs start-to-finish in one keypress and cannot pause for the keyboard — the rule since 0.9.57 is *no interactive programs*, which keeps every run bounded and the watchdog fed. Feed a program its parameters through variables set in the source, `DATA` lines, or the system names.

22.0.76.5 PRINT details

- **Strings** go in double quotes: `PRINT "HELLO"`.
- **Expressions** print their value: `PRINT 3*4` → `12`.
- A **comma** between items inserts spacing (a simple tab): `PRINT A, B`.
- A **semicolon** joins items with no space and, at the end of a line, **suppresses the newline** so the next `PRINT` continues on the same line: `PRINT "X=";X`.
- `PRINT` on its own prints a blank line.

22.0.76.6 Expressions Numbers may be integers or decimals. Operators, in precedence order: `^` (power), then `*` `/` `MOD` (remainder), then `+` `-`, with parentheses `( )` to group. There are **26 numeric variables, `A` through `Z`** (single letters), plus **one numeric array `@( )`** — declare it with `DIM @(n)` (n up to 256), then read and assign `@(I)` anywhere a variable works. Comparisons for `IF`: `=`, `<`, `>`, `<=`, `>=`, `<>` (not equal), combinable with **`AND` `OR` `NOT`** (evaluated left-to-right within each level; both sides are always evaluated).

Functions (in parentheses): `ABS(x)` absolute value, `INT(x)` floor to integer, `SQR(x)` square root, `SGN(x)` sign, `SIN(x)` / `COS(x)` / `TAN(x)` (**x in degrees**, matching CardSat's calculators), `ATN(x)` arctangent in degrees, `LOG(x)` natural log, `EXP(x)`, `MIN(a,b)` / `MAX(a,b)`, and `RND(n)` a random integer from `0` to `n-1`.

22.0.76.7 System data Programs can read the data CardSat already has, as **read-only bare names** (no parentheses). They can be used anywhere a number can:

group	names
Satellite (the active one)	`SATAZ` `SATEL` azimuth/elevation · `SATRNG` slant range km · `SATRR` range rate km/s (+ receding) · `SATLAT` `SATLON` sub-point · `SATALT` altitude km · `SATSUN` 1 = sunlit
Elements	`SATINC` inclination · `SATECC` eccentricity · `SATRAAN` · `SATMM` mean motion · `SATNOR` NORAD number
Next pass	`AOSIN` minutes to AOS (negative = in progress) · `LOSIN` minutes to LOS · `PASSEL` peak elevation · `PASSVIS`
Sun / Moon	`SUNAZ` `SUNEL` `MOONAZ` `MOONEL`
Your station	`MYLAT` `MYLON` `MYALT`
Clock (UTC)	`UTCH` `UTCM` `UTCS` `UTCDAY` `UTCMON` `UTCYR`
Space weather	`SFI` 10.7 cm flux · `KP` · `AINDEX`
Weather	`WXTEMP` `WXWIND` `WXDIR` `WXHUM` (in your configured units)
Device	`BATT` battery % · `GPAGE` element age in days · `NFAV` favorites · `HEAPFREE` free RAM bytes · `UPTIME` seconds since boot · `NSAT` loaded satellites · `NTX` transponders loaded for the active satellite
Availability	`SATOK` `TIMEOK` `POSOK` `WXOK` `SPWXOK` `PASSOK` `GPSOK`
Sidereal	`LSTHR` local sidereal time in hours (needs clock + position)
GPS	`GPSSATS` satellites in view · `GPSSPD` ground speed km/h (both always safe) · `GPSLAT` `GPSLON` `GPSALT` (error without a fix — branch on `GPSOK`)
Pass table	`PASSN` upcoming passes captured (up to 8) · `PASSAOS(k)` / `PASSLOS(k)` minutes to the k-th pass's AOS/LOS · `PASSMAX(k)` its peak elevation, `k = 1..PASSN`
Transponder (after `TXSEL`)	`TXDL` `TXUL` Hz · `TXBW` passband Hz · `TXINV` 1 = inverting · `TXLIN` 1 = linear

Two things to know about how these behave.

They are a snapshot, not a live feed. Every value is read **once, when you press `Fn+R`**, and stays fixed for that run. This is deliberate: a program runs start-to-finish in a few milliseconds, so nothing would visibly move anyway, and computing satellite geometry on *every* statement would let a runaway loop lock up the device. Think of the values as *the sky at the moment you pressed run*.

Missing data stops the program. If you read `WXTEMP` when no weather has been fetched, or `SATAZ` with no satellite selected, the run halts with an error (`no weather data`, `no active satellite`) rather than handing you a number. That is on purpose — a silent `0` for an azimuth means "point due North", which is a perfectly plausible bearing and completely wrong. If you would rather branch than stop, test the matching `…OK` flag first; those never error:

```
10 IF WXOK = 0 THEN 40
20 PRINT "Temp ";WXTEMP
30 END
40 PRINT "No weather cached"
```

A practical example — where do I point, and how long have I got?

```
10 IF SATEL < 0 THEN 60
20 PRINT "AZ ";INT(SATAZ);" EL ";INT(SATEL)
30 PRINT "RANGE ";INT(SATRNG);" KM"
40 PRINT "LOS in ";INT(LOSIN);" min"
50 END
60 PRINT "Below horizon, AOS in ";INT(AOSIN)
```

Estimating Doppler shift on a 145.9 MHz downlink:

```
10 LET F = 145900000
20 LET D = -F * SATRR / 299792
30 PRINT "Shift ";INT(D);" Hz"
```

Nothing here can transmit, move a rotator, touch the network, or write to storage — BASIC reads the system, it does not drive it.

22.0.76.8 Examples

Count to five:

```
10 FOR I = 1 TO 5
20 PRINT I
30 NEXT I
```

A times table using nested loops (the trailing `;` keeps each row on one line; the bare `PRINT` on line 50 ends the row):

```
10 FOR A = 1 TO 3
20 FOR B = 1 TO 3
30 PRINT A*B; " ";
40 NEXT B
50 PRINT
60 NEXT A
```

which prints:

```
1 2 3
2 4 6
3 6 9
```

A subroutine and a conditional loop:

```
10 LET N = 1
20 GOSUB 100
30 LET N = N + 1
40 IF N <= 5 THEN 20
50 END
100 PRINT "SQUARE OF ";N;" IS ";N*N
110 RETURN
```

Roll two dice:

```
10 PRINT "YOU ROLLED ";
20 PRINT RND(6)+1; " AND "; RND(6)+1
```

22.0.76.9 Limits and safety So that nothing you type can lock up the device, execution is **bounded on every axis**:

- Program size: **4 KB**.
- At most **200 numbered lines**.
- `GOSUB` nesting to **16** deep; `FOR` nesting to **8** deep.
- Output capped at **~6 KB** per run.
- A per-run budget of **500,000 statements** — an accidental infinite loop (e.g. `10 GOTO 10`) stops on its own with a `loop?` message instead of hanging.

Common error messages include `no line N` (a `GOTO`/`GOSUB` target that doesn't exist), `RETURN w/o GOSUB`, `NEXT w/o FOR`, `GOSUB too deep`, `FOR too deep`, and `syntax`. Each is prefixed with the line number where it occurred.

22.0.76.10 Notes and limitations This is a deliberately **tiny** BASIC. It has numeric variables only (no strings variables or arrays), no `INPUT` (programs run start to finish without interaction), and no `DATA`/`READ`. Trig is in degrees to match the rest of CardSat. Programs and their state don't persist across a reboot unless you `Fn+S` save them; saved `.bas` files live in `/CardSat/basic/` on the active filesystem and can be copied off over USB like any other file.

Glossary & math

- **Purpose** — concise definitions of the terms CardSat uses (AOS/LOS/TCA, grid, footprint, etc.) and the math behind the orbital parameters and the Doppler shift (period, semi-major axis, apogee/perigee, footprint radius, and the `beta = rr/c` downlink/uplink correction).
- **Reached from** — `g` on the Help screen.
- **Keys** — `;/.` scroll; ` back to Help.

Band plan / frequency reference

- **Purpose** — a scrollable **worldwide amateur band reference**, from **LF to light**. Covers the HF bands with their **ITU Region 1 / 2 / 3** differences, the VHF/UHF/microwave bands with **calling and EME frequencies**, the **satellite subbands**, the IARU **band designators** (H/A/V/U/L/S/C/X/K), and common satellite modes including **QO-100**. A quick on-device answer to "where does this band start" and "what's the calling frequency," with no radio or network needed.
- **Reached from** — `f` on the Help screen.
- **Keys** — `;/.` scroll a line; `` ` `` back to Help.
- **Note** — band edges and especially the microwave/regional segments vary by country within each ITU region; treat this as a quick reference, not a license document.

User guide

- **Purpose** — a concise but reasonably complete on-device manual: first-time setup, making a pass, CAT and rotator control, logging and uploads, the analysis and data screens, and general tips.
- **Reached from** — `m` on the Help screen.
- **Keys** — `;/.` scroll; `` ` `` back to Help.

Ham satellite history

- **Purpose** — a comprehensive history and explanation of amateur radio satellites: what they are, the OSCAR series, the program from OSCAR 1 (1961) through the Phase 2/3/4 orbit classes and the geostationary QO-100, the microsat/CubeSat era and the ISS, how linear vs FM transponders work, why Doppler matters, orbital decay, getting on the air, and a nudge to support AMSAT.
- **Reached from** — `s` on the Help screen.
- **Keys** — `;/.` scroll; `` ` `` back to Help.

Tech help

- **Purpose** — a technical-assistance guide: portable satellite antennas (the Arrow handheld Yagi as the top pick, plus the Elk and tape-measure alternatives), polarization and feedline tips, how to point at a satellite as it arcs overhead, how to work FM and linear birds, and how to get CardSat's CAT and rotator interfaces, logging, and uploads working.
- **Reached from** — `t` on the Help screen.
- **Keys** — `;/.` scroll; `` ` `` back to Help.

Learn

- **Purpose** — an in-depth explainer of the radio and orbital theory behind satellite operating, and how amateur satellites work technically: why orbits work (Newton/Kepler), the LEO/SSO/HEO/GEO orbit classes and what perturbs them (J2, drag), radio bands/modes/modulation, the link budget, antenna gain and polarization, the physics of Doppler, how linear transponders and FM repeater sats work internally, beacons and telemetry, store-and-forward, power and attitude control, and the satellite classes (CubeSat/microsat/ISS/GEO).
- **Reached from** — `l` on the Help screen.

- **Keys** — `;`/`.` scroll; ` back to Help.

Orbit animations

- **Purpose** — an animated explainer of the orbit archetypes. Each shows a satellite tracing its orbit to scale around a drawn Earth (correctly fast at perigee, slow at apogee) with a fading trail, plus its altitude, inclination, period, and what the orbit is used for. Covers LEO, polar / sun-synchronous, MEO, GEO, Molniya / HEO, and the amateur MEO case (GreenCube-like).
- **Reached from** — `o` on the Help screen.
- **Keys** — `,`/`/` cycle orbit type; ` back to Help.

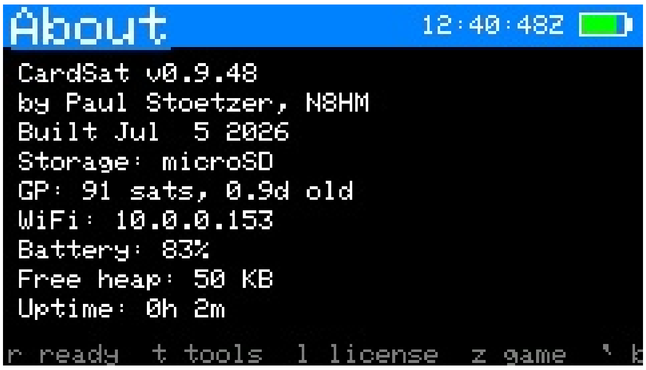


Figure 73: About

- **Purpose** — build and diagnostic information.
- **Reached from** — Home → About.
- **Shows** — firmware version, IP address, free heap and other read-only diagnostics.
- **Keys** — `r` opens the **Station readiness** checklist (below); `l` opens **License & credits**; `z` opens the **Games menu** (seven satellite-themed mini-games — see below); ` back.

Games menu (z from About)

Figure 74: Games menu

Press `z` on the About screen to open a menu of six small games — something to do while waiting on a pass, most of them a light nod to real satellite operating. Use `;`/`.` (or up/down) to move through the list and **ENTER** to launch the highlighted game; ` returns to About. Several of the games can use the **IMU tilt** as well as the keyboard. Game sounds follow the speaker-volume setting and can be silenced with **Game sound: off** in Settings.

Zap the Sats — a Space-Invaders homage where the "invaders" are satellites and your "gun" is a ham operator holding an arrow antenna that fires signals upward. **`T`/`U`** move left/right (`,`/`/` also work, as does tilt), **space** fires, **ENTER** starts a game (and advances a wave or retries after one ends). Clearing a wave speeds the formation up; letting it reach you costs a life.

Doppler Lock — hold your marker on a frequency that drifts along a Doppler-like S-curve, as if tuning a linear transponder through a pass. Nudge left/right with `,`/`/` (or `a`/`l`, or tilt); your score accrues while the cursor stays within the passband around the target.

Catch the Pass — a satellite arcs across a sky dome; press **space** while it's inside the workable elevation window to "log" a QSO. Mistime it and you miss; the windows get tighter as your score climbs. Five misses ends the game.

Rotor Runner — a genuine two-axis game: a satellite drifts around the sky and you slew an antenna crosshair to keep it centered, either by **tilting** the Cardputer (pitch + roll) or with the **arrow keys** (`a`/`l` and up/down). Score accrues while you're on target.

Morse Meteors — letters fall and you clear each by keying its Morse code. Key with two keys: **`t`** (or `.`/`,`) for a dot and **`u`** (or `/`/space) for a dash — a real CW sidetone sounds, with the dash three times the dot length. (T/U are used rather than F/H because `h` is the global Help key; the **Morse swap** setting flips which is dot/dash.)

Grid Chase — a Maidenhead grid-square trainer: a location hint is shown and you pick the correct grid from four options against a countdown. Use `;`/`.` (or up/down) then **ENTER**, or press **`1–4`** to answer directly.

KESSLER (2-player) — the 1991 QBasic classic **GORILLAS.BAS**, altered to a satellite theme: two **lunar ground stations** lob retired CubeSats at each other over a skyline of habitat modules, under **solar wind** and selectable gravity (Moon 1.62 by default; `g` cycles Mars and Earth on the title screen). The physics are GORILLAS' own equations — $x = x_0 + v\cdot\cos(a)\cdot t + \frac{1}{2}(W/5)t^2$, wind drawn from the same `FnRan(10)−5 ± FnRan(10)` distribution, ~5 t-units/second — run unchanged in the original 640×350 virtual space and only scaled to this screen when drawn, so angle 45 / velocity 60 on the Moon *feels* like the game you remember. Type the **angle**, ENTER, the **velocity**, ENTER to fire (`;`/`.` nudge; each player's last shot is prefilled). Blasts carve **craters** in the modules — a crater under a station drops it to the new surface — and a shot through the **Earth** hanging in the black sky earns you its shocked face, exactly where GORILLAS put the sun. Direct hits score; first to N (1–9, title screen) takes the match, and the survivor's dish does the victory dance. Two players, one Cardputer, pass it back and forth. Game state is heap-allocated only while the screen is open (~1 KB) and Two Cardputers can play **over LoRa**. On the title screen, one player presses **`n`** to host — that station becomes P1, picks the terrain seed, gravity, and round count, and beacons an invite on your shared LoRa frequency/SF/BW. Any other CardSat sitting in the game (or reached by the invite) joins as P2 automatically. Only three tiny fixed-layout frames ever cross the air: a hello (seed, gravity, rounds), a fire (angle, velocity, and the shooter's wind), and a round-result sync. Because both radios hold the same seed they build **byte-identical terrain** and run the same physics, so each CubeSat arcs the same on both screens with nothing streamed per frame — the half-duplex-friendly, deterministic-lockstep model. You aim only on your turn (the other station shows "waiting for peer's shot"); the wire is silent between shots. Range is your LoRa link's range — this is the same infrastructure as the messaging screen, on a third frame type (0xC7) that other LoRa traffic ignores. And no, it is **not** written in Tiny BASIC — GORILLAS needs `INPUT`, and the no-interactive-programs rule stands; that is precisely why it lives in firmware.

##**Star & constellation catalog** derived from d3-celestial (BSD-3; BSC5/HYG data).

23

License & credits

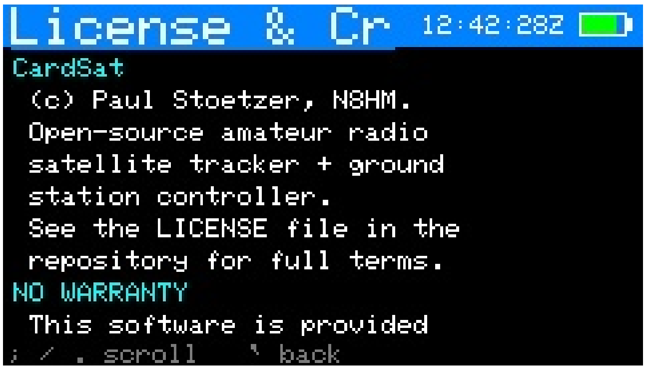


Figure 75: License & credits

- **Purpose** — license pointer and no-warranty/hardware disclaimer, credit for the outside data sources (Celestrak/SpaceTrack, NOAA SWPC, Open-Meteo, hams.at, SatNOGS/AMSAT), acknowledgements, and a recommendation to support AMSAT.
- **Reached from** — `l` on the About screen.
- **Keys** — `;`/`.` scroll; ` back to About.

Edit (text/number entry)

- **Purpose** — the shared on-screen editor used by every field that needs typed input (frequencies, SSID/password, grid, callsign, host/port, clock, etc.).
- **Reached from** — automatically, whenever a field is opened for editing.
- **Shows** — the field title, the current buffer, and the entry footer. Certain fields default to **uppercase** (callsign, grid, SSID and similar) — hold shift for lowercase.
- **Keys** — type to enter; **DEL** backspace; **ENTER** accept; ` cancel.

24

Key reference (cheat sheet)

A **printable version** of this reference is included as `CardSat_CheatCard_4x6.pdf` — a **4×6** landscape index card, front and back (two pages). Print at actual size; the front covers operating and the back covers setup and tools. Regenerate it with `python3 tools_make_cheatcard.py`.

Global: `;` up · `.` down · `,` left · `/` right · ENTER select · `` ` ``/DEL back · `{/}` page · `b` screenshot · `h` help (on screens that use bare letters — the editors, calculator, Tools list, DXCC/char lookups — use **`Fn+b`** / **`Fn+h`**) · **`Fn`+back (`` ` ``/DEL) = emergency stop** (disengages all radio + rotator control from any screen; excluded in the text editors and calculator, where DEL is backspace).

Screen	Keys
Satellites	`f` favorite · `v` favorites-only · `n` new GP sat · `x` delete manual sat (2-press) · `e` EQX table (OSCARLOCATOR) · `k` OSCARLOCATOR · `3` 3D globe · `2` sat-to-sat visibility · `o` orbital analysis · `y` simulation · `t` transponder database · `d` 10-day overview · `i` illumination · `s` AMSAT status roster · `L` share GP over LoRa · ENTER passes · right-edge AMSAT mark: filled dot = heard, filled square = telemetry only, ring = not heard, none = no reports · colored down-arrow = decaying orbit (yellow watch / orange decaying / red imminent)

Screen	Keys
Orbital analysis	`,`/`/` page (Info / Live / Next pass / Ground track / Doppler / Nodal / Sun-Beta / Pass outlook / Orbit position / Phys / Explore) · Info: footprint diameter now/apogee/perigee (= longest possible QSO) + decay estimate & solar-bracket range · Live: az/el/range/Doppler, mean anomaly/phase, sunlit/eclipse + **eclipse depth** (deg; >0 = in shadow) · Next pass: slant ranges + path delay + peak eclipse depth · Doppler: `f` set beacon freq, peak shift + max range-rate · Nodal: J2 node/perigee drift, sun-sync, LTAN, repeat track, longest pass · Sun/Beta: solar beta angle, full-sun vs eclipsed, eclipse %/orbit, next transition · Pass outlook: 7-day pass count/>30° count/longest/avg gap + the best upcoming pass (elevation, when, duration) · Orbit position: mean/true anomaly, argument of latitude, time to perigee/apogee, RAAN, rev number · Phys: orbital velocity now + apo/peri spread, launch year/number + years in orbit (from the COSPAR ID) · Explore: what-if sandbox — `;`/`.` pick apogee/perigee/inclination, type a value, `x` reseeds from the tracked sat; derived orbit recomputes live · `r` recompute · **`p` print report** · ` back
Simulation	`,`/`/` step time · `;`/`.` step size · `m` world-map view (sub-point + footprint at the simulated time) · `x` reset to now · ` back
Next Passes	ENTER track · `m` world map · `t` sky-at-a-glance timeline · `p` rove planner · `w` workable horizon · `s` target search · `r` refresh · `P` print all favorites' passes · `z` deep-sleep until AOS
Passes	`;`/`.` select · `d` detail · `t`/ENTER track · `n` add TX · `r` recompute · `x` mutual · `v` 10-day · `V` visible-pass list · `i` illum · `g` workable grids (this pass) · `w` workable US states (this pass) · `e` workable DXCC (this pass)
Pass detail	`p` polar of this pass · `/ENTER back
Pass polar	`p` back to curve · `/ENTER passes

Screen	Keys
Track	`m` TUNE/CAL · `d` cycle tune mode (FULL/DL/UL/hold) · `t` next TX · `n` jump to beacon · `N` per-sat operating note · `k` CW both legs (linear) · `c` CTCSS tone · `r` radio on/off · `o` rotator on/off · `p` polar · `a` point-here arrow · `z` large-font readout · `y` tilt tuning on/off (if IMU) · `f` Manual mode · `l` log QSO · `v` voice memo (SD card; also drops a **log stub** to finish after LOS) · after LOS: `q` (60 s window) deep-sleeps until the next favorite pass · `i`×2 report Heard to AMSAT (mode-aware) · `g` workable grids now · `w` workable US states now · `e` workable DXCC now (radio/rotator keep running) · ENTER save cal
Large-font readout (`z` from Track)	big RX/TX + az/el + tune mode · `,`/`/` tune · `s`/`x` step/center · `m` TUNE/CAL · `d` mode · `t` next TX · `n` beacon · `k` CW (linear) · `r` radio · `o` rotator · `y` tilt · `l` log · `v` voice memo · `z`/`` ` `` back to Track
Manual large-font (`z` from Manual)	HOLD/TUNE legs in big digits · `u` swap leg · `,`/`/` tune · `s`/`x` · `m` CAL · `t` next TX · `z`/`` ` `` back to Manual
Manual mode (`f` from Track)	no-radio frequency calculator · `u` toggle which leg is fixed (HOLD vs TUNE>) · `,`/`/` move fixed freq in passband (linear) · `s` step · `x` center · `m` CAL · `t` next TX · `l` log · `p` polar · `g` grids (all return here) · ENTER save cal · `` ` ``/`f` back to Track
Workable grids	4-char Maidenhead grids under the footprint (per-pass union or live, refreshing ~3 s; uncapped, works to high orbits) · count shown on a cyan line above the list · `;`/`.` and `{`/`}` scroll · `` ` `` back
Track · TUNE	`,`/`/` tune ∓ · `s` step (100/1k/5k) · `x` recenter
Track · CAL	`,`/`/` downlink ∓ · `;`/`.` uplink ∓ · `s` step (10/100/1k) · `x` zero
Polar	`l` log QSO · `v` voice memo (SD card) · `p`/ENTER/`` ` `` back to track
Log (menu)	`;`/`.` select · ENTER → new QSO / browse / export ADIF / LoTW upload / Cloudlog upload / voice memos / notes / awards / fill grids (QRZ)
Log · list	`;`/`.` scroll · ENTER edit entry · `` ` `` back

Screen	Keys
Log · entry	`;`/`.` field · ENTER edit · `s` save · `x`×2 delete · ` back
Notes (browser)	`;`/`.` select · ENTER open · `n` new · `d`+ENTER delete · ` back · list shows last-saved date/time (UTC), newest first
Notes (editor)	type freely (ENTER = newline, DEL = backspace) · **Fn+`,`/`/`** cursor L/R · **Fn+`;`/`.`** cursor up/down · **Fn+`s`** save (names it the first time) · ` exit (auto-saves)
Mutual	`;`/`.` scroll · `/ENTER back to passes
10-day	`;`/`.` scroll ∓1 day (forward indefinitely, oldest day off the top; not before today) · `r` recompute · **`p` print report** · `/ENTER back
Illum	`,`/`/` jump ∓60-day window (forward indefinitely; not before today) · `r` recompute · **`p` print report** · `/ENTER back
Location	`e`/`o`/`a` lat/lon/alt · `g` grid · `p` GPS on/off · `s` GPS source · `c` set clock · `v` live GPS position · ENTER GPS sky plot
Live GPS position	DMS + decimal lat/lon, altitude, grid, speed, course · ` back
GPS sky plot	live GNSS az/el colored by signal · ` back
Messages	LoRa CardSat-to-CardSat chat · `n` write/send · `;`/`.` scroll · `r` retry radio · ` back
World map	`f` cycle highlighted favorite · `c` recenter on QTH / 0° (sun terminator drawn automatically) · ` back
Rotator (manual)	`,`/`/` az · `;`/`.` el · `s` step · `x` stop · (*Yaesu direct only*) `1`/`2`/`3`/`4` capture ADC at az 0 / az full / el 0 / el full · ` back
Home menu	`;`/`.` move · ENTER open · any letter jumps to the next matching item (repeat cycles)
Help	`;`/`.` scroll · `g` glossary · `m` guide · `s` sat history · `t` tech help · `l` learn · `o` orbit animations · `f` band plan / frequencies · ` back
Update	`k`/ENTER GP (+clock/AMSAT/space-wx/weather) · `f` fast (GP + AMSAT + fav TX) · `a` cache all TX · `w` WiFi only

Screen	Keys
Settings	,// change · ENTER edit/toggle · s scan WiFi (on SSID row) · (Reset = type ERASE)
GP source	pick **AMSAT** / any **CelesTrak** JSON-PP category (Amateur Radio first) / **Custom URL** · ;/. move · {/} page · ENTER select
Sun / Moon	graphical sky-dome view (Sun/Moon glyphs on a polar dome) · g toggle graphic/data list · ;/. pick Sun/Moon · o rotor track on/off (takes the rotator from sat tracking) · s sky sources (radio sources/planets) · t Sun/Moon transit finder · e EME / moonbounce · auto-parks while the body is below the horizon · header shows SUN/MOON tag on other screens · x stop · ` back
Space Wx (main menu)	solar 10.7 cm flux + planetary Kp + running A index, each labeled & color-coded, with a plain-language HF/satellite operating outlook and a data-freshness note · shows cache then auto-fetches on entry (if on WiFi) with an "Updating Space Wx" bar + result · also refreshes with Update · p HF/6m propagation guide · r refresh · ` back
Weather (main menu)	terrestrial current conditions + multi-day forecast for the operating site from Open-Meteo · current temp/sky/wind/humidity then per-day hi/lo & precip chance · shows cache then auto-fetches on entry (if on WiFi) with an "Updating Weather" bar + result · also refreshes with Update · r refresh · cached offline · ` back
QRZ Lookup (main menu)	callsign lookup via QRZ.com XML (needs a QRZ XML subscription + credentials in Settings → Network / data) · ENTER type a callsign · shows name/address/country/grid/class · WiFi required · ` back
EME / moonbounce (Sun/Moon → e)	self-echo Doppler per band (50/144/432/1296/10368, topocentric) · range + rate · degradation vs perigee · galactic sky-noise flag · red SUN flag <10° · p 30-day plan (dec + degr, ;/. scroll) · m mutual-Moon window vs DX grid (g grid, ;/. select) · o point rotator at Moon · x stop · a per-band analysis page · w print · ` back

Screen	Keys
Grid dist/bearing (main menu)	great-circle distance + heading to a Maidenhead grid (short/long path, km/mi) · `g` grid · `q` QRZ→grid lookup · `o` point rotator at bearing (el 0) · ` back
QRZ → grid (Grid dist/bearing → `q`)	resolve a callsign to its grid · `c` callsign · ENTER use grid in the calculator · ` back
HF/6m propagation (Space Wx → `p`)	band guidance from solar flux + Kp: HF conditions, 10/15/20 m open/marg/shut, geomagnetic, auroral VHF, absorption · rule-of-thumb (6 m Es seasonal) · `r` refetch · ` back
Transponder DB (Satellites → `t`)	scrollable list of the selected satellite's transponder/beacon entries (description; **D** downlink + mode; **U** uplink + tone/inv/lin flags) · `;`/`.` scroll · ` back
Edit	type · DEL backspace · ENTER ok · ` cancel
About	build/version, IP, free heap and diagnostics (read-only) · `t` Tools hub · `p` Print submenu · `r` station readiness checklist · `l` license · `z` games
Printing (contextual)	`p` prints the current screen's report on the printable screens (Passes day-sheet, Mutual, DX Doppler, EQX, Target-search, Pass detail/polar, **Orbital analysis, Illumination, 10-day passes, 6-hour timeline**) · `P` prints all-favorite passes from the schedule · `Fn+p` prints the note in the note editor · About → `p` opens a Print submenu listing **every** report; About → `a` Support-AMSAT card; About → `c` operator contact card · **tools:** `p` prints on the programmer calc / graphing calc / location converter / char lookup / BASIC console, and (0.9.59) on **every form tool** (all 34 — inputs + the complete output list) and the **conjunction / neighborhood / debris-group / link-margin** screens; `Fn+p` on the scientific calc and BASIC editor (where plain `p` types)

25

Glossary

- **AOS / LOS** — Acquisition / Loss of Signal: when the satellite rises above and sets below your horizon.
- **TCA** — Time of Closest Approach (maximum elevation).
- **Azimuth / Elevation** — compass bearing (0°=N, clockwise) and angle above the horizon to the satellite.
- **Range / range-rate** — slant distance to the satellite and how fast it's changing (drives Doppler; +ve = receding).
- **Doppler shift** — the frequency change caused by relative motion.
- **Downlink / uplink** — the satellite's transmit frequency (you receive) and receive frequency (you transmit).
- **Linear transponder** — relays a band of frequencies (SSB/CW), as opposed to a single FM channel. May be **inverting** (flips the spectrum) or non-inverting.
- **Passband** — the range of frequencies a linear transponder relays.
- **GP / OMM** — General Perturbations data / Orbit Mean-elements Message: the orbital element set (distributed as JSON) that SGP4 propagates. Replaces the legacy **TLE** (Two-Line Element) text encoding, which CardSat still rebuilds internally to feed the propagator.
- **SGP4** — the standard model that turns an element set into position/velocity over time.
- **Maidenhead grid** — the locator system (e.g. `FM18`) for your station position.
- **Eclipse / sunlit** — whether the satellite is in Earth's shadow or illuminated.
- **CAT** — Computer Aided Transceiver control: the generic term for computer control of a radio. CardSat speaks three CAT dialects — Icom **CI-V**, Yaesu, and Kenwood — behind one abstract rig interface, over a wired serial bus or (for the **IC-9700**) the **Icom LAN** RS-BA1 protocol.
- **CI-V** — Icom's Communications Interface-V serial control bus (Icom's CAT dialect).
- **MAIN / SUB** — the two VFOs/bands of a satellite-capable radio; by default CardSat uses MAIN for uplink and SUB for downlink (the **VFO Type** setting swaps them).
- **One True Rule** — KB5MU's principle: tune so the frequency *at the satellite* stays constant; the computer corrects both legs for Doppler.

26

Supporting AMSAT

CardSat exists because of the work **AMSAT** and its volunteers do — building and keeping amateur satellites flying, publishing the orbital data this app depends on, and advocating for amateur radio in space. **If you find CardSat useful, please consider joining and/or donating to AMSAT at www.amsat.org.** Membership and donations directly fund the next generation of satellites you'll track and work with this very tool.

27

License

CardSat is released under the **MIT License**.

Third-party components keep their own licenses: SGP4 propagation (Hopperpop/Sgp4-Library), GP data from AMSAT, and transponder data from SatNOGS.

CardSat is amateur-radio software. Operate within your license privileges and local band plans. Built on SGP4 (Hopperpop), GP data from AMSAT, and transponder data from SatNOGS.